CLASSIFICATION OF BIPARTITE OBSERVABLES II: NIP OBSERVABLES AND TODORČEVIĆ'S TRICHOTOMY

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Abstract. Two quantum systems, Alice's and Bob's, are prepared independently, and an observable of the joint system, a bipartite observable, is measured. An idealized preparation is a limit of actual preparations. The observable is stable when its expectation value does not depend on which of the two systems is idealized first. An idealized response is the expectation value of the observable for an idealized preparation of Alice's system, regarded as a function of a probe, that is, of a preparation of Bob's system, actual or idealized, with Bob's system idealized first. The observable is NIP for Alice's responses when no sequence of Alice's actual preparations has the independence property of model theory, that is, when no such sequence lets probes produce every finite pattern of high and low expectation values. We show that the observable is stable exactly when its idealized responses are continuous functions of the probe, and NIP for Alice's responses exactly when they are of the first Baire class, the class of pointwise limits of sequences of continuous functions. The idealized responses of a NIP observable then form a Rosenthal compactum: a compact set, for pointwise convergence, of functions of the first Baire class. A separable Rosenthal compactum that is not metrizable is hereditarily separable, or first countable without being hereditarily separable, or not first countable, and Todorčević's trichotomy provides a canonical subspace in each case. Because Alice can mix her preparations, the compactum of all idealized responses of an observable is convex. We prove that, for a NIP observable, this compactum is metrizable as soon as it is hereditarily separable. The NIP observables therefore fall into three classes, according as their compacta of idealized responses are metrizable, first countable but not hereditarily separable, or not first countable. We show that the comparison of photon numbers, the comparison of positions and the causal order of a spacetime with one dimension of space lie in these three classes, respectively, and that the comparison of two phases on a circle lies in the last. The Argyros–Dodos–Kanellopoulos heptachotomy refines the trichotomy: every pointwise bounded family of functions, indexed by the dyadic tree of finite sequences of zeros and ones, whose pointwise closure consists of functions of the first Baire class is, on a suitable dyadic subtree, equivalent to one of seven minimal families. We show that each of the seven minimal families is realized by the responses of a tree of actual preparations of photons, of a particle in a box or of a chain of spins. We show also that the observable by which Bob scans a chain of spins site by site is NIP for Alice's responses but not for Bob's, whereas uniform NIP, a quantitative form of NIP, is symmetric in the two systems. For the Ising model in a transverse field, we prove that the family of magnetizations of finite systems, as functions of an applied field and with an offset field that moves the transition point (the field at which the magnetization changes sign), is stable when the spontaneous magnetization vanishes, and that its closure is first countable but not hereditarily separable when the spontaneous magnetization is positive. In the second case, the two pure phases at a transition point, the equilibrium states magnetized in opposite directions, have idealized magnetizations that differ only at that point. As the transition point varies, the midpoints of these pairs of magnetizations, the idealized magnetizations of the symmetric state (the equal mixture of the two phases), form an uncountable discrete set. The proof rests on a Lee–Yang theorem that we prove for this model: the partition function of each finite system has no zeros off the imaginary axis of the complex longitudinal field.

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1. Introduction

Two quantum systems, Alice's and Bob's, are prepared independently, and a bipartite observable, a physical quantity of the joint system, is measured. Physicists routinely idealize preparations: an oscillator excited to infinite energy, a particle localized at a point, a plane wave of sharp momentum. Such an idealized preparation is a limit of actual ones. Mathematically, an observable of a system is here a bounded self-adjoint operator on the Hilbert space of the system, and an actual preparation is described by a density operator, a positive operator of trace 1 on that space. Every density operator defines a state of the algebra of bounded operators on that space, that is, a positive linear functional with value 1 at the identity; the idealized preparations are described by the remaining states. In Part I [DI26] we called a bipartite observable *stable* when its expectation value does not depend on which of the two systems is idealized first, and we showed that stability is equivalent to a list of properties from analysis, topology, probability and model theory. Many physically natural bipartite observables are stable, among them the swap test, which estimates the overlap of the two prepared states; Hong–Ou–Mandel coincidences, in which two photons meeting at a beam splitter leave it by different ports; and discrete equality tests, such as "equal photon numbers". These observables are even uniformly stable, a quantitative form of stability, with bounds that do not depend on the dimensions of the Hilbert spaces.

Other observables, no less natural, are of a different kind: for a quantity of Alice's system, given by a self-adjoint operator A, and a quantity of Bob's, given by a self-adjoint operator B, the order comparison $[A < B]$ is the orthogonal projection that represents the event "Bob's value of B exceeds Alice's value of A". The comparison $[A < B]$ is unstable whenever A and B have discrete spectra unbounded above, as for the observables "Bob's system has more photons than Alice's" and "Bob's oscillator has the higher energy": if both energies are sent to infinity, the probability that Bob's energy is the higher is 1 when Bob's system is idealized first and 0 when Alice's system is. The order of the idealizations matters because an order relation is built into the observable, and observables of this kind supply the principal examples of this paper.

Although it can be unstable, the comparison $[A < B]$ is tame in a precise sense. Call a sequence of Alice's preparations *independent* for an observable if Bob can read off every finite pattern from it: there are numbers $a < b$ such that, for any two disjoint finite sets of these preparations, a single preparation of Bob's system, a *probe*, gives expectation values at most a on the first set and at least b on the second. An observable with an independent sequence has the *independence property*, and an observable without one is *NIP*, an acronym for "no independence property". In Section 7 we show that $[A < B]$ is NIP, and even uniformly NIP, for all self-adjoint operators A and B: for each gap $b - a$, the finite sets of preparations

from which probes can read off every pattern have a size bounded by a number that depends only on the gap. By contrast, the fringe observable, which asks whether a photon lands on a bright fringe of an interference pattern, has an independent sequence, as we prove in Proposition 7.13: if Alice's preparations produce fringes of wavenumbers $8, 8^2, 8^3, \dots$ on Bob's screen, then, for any two disjoint finite sets of these preparations, some position of Bob's detector lies on a bright fringe of every preparation in the first set and on a dark fringe of every preparation in the second. Uniform NIP governs learning: by a theorem of Alon, Ben-David, Cesa-Bianchi and Haussler [ABDCBH97], a class of functions with values in $[0, 1]$ is learnable from a number of samples that does not depend on their distribution exactly when its fat-shattering dimension, the size that uniform NIP bounds, is finite for every gap. Uniform NIP implies NIP, but the converse fails (see Corollary 8.7).

The division of first-order theories, sets of axioms in a formal language, by properties such as stability and NIP, which model theorists call dividing lines, is the subject of Shelah's classification theory [She90]. Part I and the present paper apply it to bipartite observables: in Part I we treated the first dividing line, stability, and in this paper we treat a weaker one, NIP. In model theory, a linear order is the prototype of an unstable NIP structure, that is, of a set equipped with relations whose theory is unstable and NIP. The rational order $(\mathbb{Q}, <)$ is unstable, yet it is NIP: over a finite set of parameters it has only polynomially many types in the size of the set, a type being a complete description of a new element in terms of the parameters. The ordered field of real numbers is likewise unstable and NIP, and so, more generally, is every o-minimal structure, that is, every ordered structure in which every set of elements defined by a formula with parameters (a definable set) is a finite union of points and intervals; the o-minimal structures are the best-known unstable NIP structures.[1]

These structures, like most structures to which NIP has been applied, are two-valued: their formulas take the values true and false, even when, as for the real field, the underlying set is a continuum. In this paper the formulas are real-valued, since their values are expectation values, and the examples come from physics rather than from algebra or geometry. Real-valued forms of NIP have met quantum theory before, in finite dimensions, as the fat-shattering dimension that governs the learnability of quantum states and measurements [Aar07, CHY15]. To our knowledge, however, the structure theory of NIP, and with it the theory of Rosenthal compacta, the compact sets, for the topology of pointwise convergence, of pointwise limits of sequences of continuous functions on a Polish space (a separable, completely metrizable space), has not previously been brought to bear on physical observables.

The dictionary. Fix an observable, and regard its expectation value for an idealized preparation of Alice's system as a function of Bob's probe, idealized probes included, with Bob's system idealized first; we call this function an *idealized response*. In Section 5 we show that the idealized responses, with the topology of pointwise convergence, form a compact set of functions on a compact metrizable space of probes, and in Theorem 6.1 that Shelah's dividing lines correspond to degrees of regularity of these functions:

- the observable is stable exactly when every idealized response is continuous;
- it is NIP exactly when every idealized response is of the first Baire class, that is, a pointwise limit of a sequence of continuous functions; the idealized responses then form a *Rosenthal*

[1] Other standard examples are the ordered group of integers $(\mathbb{Z}, +, <)$, whose theory is Presburger arithmetic, and, more generally, ordered abelian groups [GS84], and the fields of p-adic numbers, for every prime p, and algebraically closed valued fields (see [Sim15a, Chapter 2]). For o-minimal structures see [PS86, KPS86]; the real field with the exponential function is o-minimal by a theorem of Wilkie [Wil96].

compactum, a compact set of functions of the first Baire class, and every one of them is the limit of a sequence of responses of actual preparations (see [BFT78, Theorem 3F]);
- otherwise, some sequence of Alice's actual preparations is independent, and the pointwise closure of the responses of a subsequence is a copy of $\beta\mathbb{N}$, the space of ultrafilters on the natural numbers (see Theorem 6.1(c)).

The fine classification. The compactum of idealized responses is separable (see Lemma 6.3), and separable Rosenthal compacta form a vast class; a priori there is no reason why the failure of such a compactum to be metrizable should be witnessed by anything simple. Todorčević's trichotomy [Tod99] shows that a simple witness always exists: a non-metrizable separable Rosenthal compactum is either hereditarily separable (every subspace has a countable dense subset), or first countable (every point has a countable base of neighborhoods) but not hereditarily separable, or not first countable, and in these three cases it contains, respectively, a copy of the split interval, a copy of the Alexandroff duplicate of the Cantor set or of its analogue for the split Cantor set, and a copy of the one-point compactification of a discrete set of size continuum. The split interval is the unit interval with each point doubled into two adjacent points, in the order topology, and the split Cantor set is the Cantor set doubled in the same way; the Alexandroff duplicate of the Cantor set adds to the Cantor set an isolated twin of each of its points, in such a way that the twins of a sequence of points converging to another point converge to that point as well. In this the trichotomy resembles Rosenthal's ℓ_1 theorem [Ros74]: a bounded sequence in a real Banach space has either a weakly Cauchy subsequence, one whose values under each continuous linear functional form a Cauchy sequence, or a subsequence equivalent to the unit vector basis of ℓ_1. The Argyros–Dodos–Kanellopoulos heptachotomy [ADK08], which they extracted from the methods of Todorčević's paper with the help of Ramsey theory for trees, refines the trichotomy to the level of families. Index a pointwise bounded family of functions on a Polish space by the dyadic tree, the set of finite sequences of zeros and ones, and suppose that its pointwise closure consists of functions of the first Baire class. Then, for some dyadic subtree, a subtree in which every node has exactly two immediate successors, the subfamily indexed by it is equivalent to one of exactly seven minimal families; two families indexed by the same set are *equivalent* when the correspondence between them extends to a homeomorphism of their pointwise closures, and a family is *minimal* when it is equivalent to its restriction to every dyadic subtree. For a NIP observable, such a family is given by the responses of a tree of Alice's preparations, for instance a tree of successively sharper localizations of a particle, and the heptachotomy shows that, up to passing to a subtree, there are seven ways, and only seven, in which the responses of such a tree can approach their idealized limits. Through the connection among NIP, the Vapnik–Chervonenkis dimension (the two-valued form of the fat-shattering dimension) and Rosenthal compacta, the trichotomy also classifies NIP formulas whose parameters range over a compact metrizable space [DIMW+26, Section 4.2].

For observables, the classification is coarser, for a reason that comes from probability. Alice can mix two preparations, that is, make one or the other at random with given probabilities, and the response of the mixture is the corresponding convex combination of the two responses; hence the idealized responses form a convex set (see the proof of Corollary 6.8). In Theorem 6.6 we give a direct proof that a separable convex Rosenthal compactum that is hereditarily separable is metrizable, a special case of a known result on locally connected Rosenthal compacta. Consequently, the compactum of idealized responses of a NIP observable is metrizable, or first countable but not hereditarily separable, or not first countable (see Corollary 6.8 and Theorem 6.9), and NIP observables fall into three Todorčević classes. As

we show in Section 7, three physical order relations lie in these three classes, respectively: the comparison of photon numbers, the comparison of positions or of quadratures of light (the two field amplitudes of a mode, which behave like a position and a momentum), and the causal order of a spacetime with one dimension of space. The compactum of idealized responses of the causal order fails to be first countable because an event that occurs at the time of Bob's detection can influence a detection only at its own position. The comparison of two phases on a circle, we prove in Theorem 7.12, lies with the causal order in the last class: on a circle, for each pair of antipodal points there is an idealized response that differs from the constant response of the uniform distribution only on probes that give weight to one of the two points. Trees of preparations, nevertheless, realize all seven minimal families, as we show in Theorem 8.5 (see Figure 2): the closure of the responses of a tree is a closed subset of the compactum of idealized responses, and it need not be convex.

The classifications are theorems of ZFC, the Zermelo–Fraenkel axioms of set theory with the axiom of choice, but Todorčević's trichotomy is proved by the method of forcing [Tod99]. The required configuration is first found in a generic extension, a larger universe of sets built from the given one, and the conclusion is then brought back to the given universe by absoluteness: it concerns countable codes for the objects involved, and statements of this kind have the same truth value in both universes. To our knowledge, no proof of the trichotomy that avoids forcing is known [DIMW$^+$26]. The results of this paper that depend on the trichotomy, namely Theorems 4.7, 6.6 and 6.9, Corollary 6.8 and the canonical subfamilies in Theorems 7.8, 7.10 and 7.12, therefore rest on forcing as well; the location of each example in the classification is proved directly.

Orders of limits and the quantum Ising model. Many central notions of physics are defined by an order of limits. Below the critical temperature of a ferromagnet, a spontaneous magnetization appears when the volume becomes infinite before the external field is switched off, and it disappears in the other order (see the proof of Theorem 9.2). The classification applies to such phenomena, because its definitions concern families of responses, continuous functions on a compact metrizable space of probes, whether or not the family comes from a bipartite observable; for a ferromagnet, the responses are the magnetizations of finite samples, as functions of the applied field, and the probes are the values of the field. A NIP family with separable closure is NIP_1, NIP_2 or NIP_3 when this closure is, respectively, first countable, hereditarily separable or metrizable (see Section 4), and each level has a reading in terms of probes; for instance, such a family is NIP_3 exactly when countably many probes tell all its idealized responses apart (see Propositions 4.5 and 4.6). In Theorem 9.2 we show that the family of magnetizations of finite samples is stable exactly when the spontaneous magnetization vanishes. When the spontaneous magnetization is positive, the family enlarged by an offset field that moves the transition point (the field at which the magnetization changes sign) is NIP_1 but not NIP_2, the class of the comparison of positions in a box: at each transition point, the idealized responses of the two pure phases, the equilibrium states magnetized in opposite directions, form a split pair, two functions that differ only at that point, and the idealized response of the symmetric state, the equal mixture of the two phases, is their midpoint, as in the proof of Theorem 6.6. For the Ising model in a transverse field, a lattice of spins whose neighbors interact through their components along one axis, in a field perpendicular to that axis, these results are, to our knowledge, new (see Theorem 9.6). Their proof needs only one argument specific to this model: the differentiability, away from zero field, of the pressure, the limit per site of the logarithm of the partition function (the normalizing constant of the thermal equilibrium state). We derive it, together with the real analyticity of the pressure there, from a Lee–Yang theorem for the transverse-field model: the partition

function of each finite system, as a function of the complex longitudinal field (the field along the axis of the couplings), has no zeros off the imaginary axis (see Propositions 9.4 and 9.5). The Lee–Yang theorems of Asano [Asa70] and of Suzuki and Fisher [SF71] for quantum spin systems, in the form given by Fröhlich and Rodriguez [FR12], exclude zeros, in the presence of a transverse field, only in a smaller half-plane (see the remarks after Proposition 9.4). We have not found the theorem for the whole half-plane in the literature, and we include a proof, which combines standard tools. In Section 10 we show that two further results of physics that rest on an order of limits are statements about stability (see Theorems 10.3 and 10.4): the Banks–Casher relation [BC80], which ties the chiral condensate of quarks, the vacuum expectation value whose nonvanishing signals the spontaneous breaking of chiral symmetry, to the density of near-zero eigenvalues of the Dirac operator (the operator that governs the quarks in a given gluon field), and the Scalapino–White–Zhang criterion [SWZ93], which tells superconductors, metals and insulators apart by their current responses.

Contributions of the paper. For mathematicians, the news is that Shelah's dividing lines in model theory and Todorčević's classification of Rosenthal compacta, with its refinement by Argyros, Dodos and Kanellopoulos, although developed for purely mathematical reasons, are realized by physical observables. In Sections 7 and 8 we show that each class of Todorčević's trichotomy, in the form it takes for observables, is realized by an observable of a physical system, and each of the seven minimal families of the heptachotomy by the responses of a tree of preparations for such an observable; the systems are photons, particles in a box, planar rotors (particles on a ring, whose positions are angles), events in spacetime and chains of spins. Each realization displays what distinguishes its class or family from the others. For physicists, the news is a scale of levels on which orders of limits can be placed and compared, each level with an operational meaning, and a theorem on the Ising model in a transverse field, a model whose Hamiltonian, the operator that represents its energy, has terms that do not commute: with an offset field, the family of its magnetizations is stable when the spontaneous magnetization vanishes, and NIP_1 but not NIP_2 when the spontaneous magnetization is positive (see Theorem 9.6).

The logical setting. Expectation values are real numbers, known only up to the finite resolution of a measurement, and the formulas of our logic are therefore real-valued; the resolution appears as the gap $b - a$ in the definition of the independence property (see Definition 3.3). We work in Keisler's model theory of real-valued structures [Kei23], in which no metric is given a priori and each formula carries its own bound, rather than in the continuous logic of Ben Yaacov and Usvyatsov [BYU10], in which each sort, that is, each kind of object, carries a metric given in advance and all formulas take values in $[0, 1]$. For the bounded observables of this paper either setting would serve, but physics requires unbounded quantities: the position Q and the momentum P of a particle satisfy the canonical commutation relation $QP - PQ = i\hbar I$, where $\hbar$ is the reduced Planck constant and I is the identity operator, and no pair of bounded operators satisfies this relation [Win47, Wie49]. For comparisons, unboundedness is harmless, because $[A < B]$ is a projection, unchanged when A and B are replaced by the same strictly increasing function of themselves (see Proposition 7.9). Unboundedness is not harmless for the expectation value of an unbounded quantity, such as the energy $\rho \mapsto \mathrm{Tr}(\rho H)$, where ρ ranges over density operators, H is the Hamiltonian, an unbounded self-adjoint operator, and Tr is the trace. Such a formula can be brought into $[0, 1]$ only by a nonlinear rescaling or by adjoining a point at infinity [Ben08]; expectations are affine in the state, whereas a rescaled energy is not, and the rescaled energy of a mixture of states is no longer the corresponding mixture of rescaled energies. In Keisler's setting each formula φ takes its values in an interval $[-r_\varphi, r_\varphi]$, whose radius r_φ depends on the formula.

An unbounded quantity therefore enters only through its restrictions to confinements, sets of states on each of which it is bounded, such as the states of a particle in a box, the states of a cavity with a frequency cutoff, the states of a detector with a finite range, or the states that satisfy an energy constraint $\mathrm{Tr}(\rho H) \leq E$ of quantum information, for a real number E. The confinements play the role of the shards of our joint work with Matos-Wiederhold, Salvetti and Tall [DIMW$^+$26], the parts of a state space on which every predicate stays within prescribed bounds. A confinement of interest is compact for the weak* topology, the topology of convergence of all expectation values, but not for the trace norm, the norm that measures how well two states can be distinguished: on a set of Bob's states that is compact in the trace norm, every bounded observable is stable, because its expectation value is Lipschitz in Bob's state, uniformly in Alice's.

Related work. Model theory has met operator algebras and quantum theory along other routes. Farah, Hart and Sherman developed the continuous model theory of C*-algebras and von Neumann algebras, the algebras of bounded operators on a Hilbert space that are closed under adjoints and closed in the norm and in the weak operator topology (the topology of convergence of all matrix elements), respectively, and studied stability in that setting [FHS13]; there the structures are the algebras themselves, and stability is a property of their theories, whereas here the object is a single observable together with its idealized responses. Zilber gave a model-theoretic semantics of the canonical commutation relation [Zil16]. On the side of real-valued NIP, Ben Yaacov showed that NIP is preserved under randomization, the passage from a theory to a theory of random variables with values in its models [Ben09], whence averages of NIP relations over random parameters are again NIP; Proposition 7.3 is an elementary, quantitative instance of this phenomenon for relations between quantities measured by Alice and by Bob. Hrushovski and Pillay studied invariant measures in NIP theories [HP11], Simon connected NIP with Rosenthal compacta [Sim15b], and Glasner and Megrelishvili used Todorčević's trichotomy to build a hierarchy of tame dynamical systems [GM22]. To our knowledge, none of these works considers the orders of limits of a bipartite observable, the compactum of its idealized responses, or the place of physical observables in Todorčević's trichotomy and in the Argyros–Dodos–Kanellopoulos heptachotomy; Part I concerns the first of these topics, and the present paper concerns all three.

The paper is organized as follows. We state the main results in Section 2. The facts about Rosenthal compacta that we need, and a classification of countable NIP families of continuous functions that we derive from them (see Theorem 4.7), occupy Sections 3 and 4. In Sections 5 and 6 we attach to each observable its compactum of idealized responses, prove Theorem 6.1, and show that, for the compactum of idealized responses of a NIP observable, hereditary separability implies metrizability (see Theorem 6.6 and Corollary 6.8). The examples follow in Sections 7 and 8; spontaneous magnetization, classical and quantum, is the subject of Section 9, and two further orders of limits are treated in Section 10. Open questions are collected in Section 11. Standard arguments are cited, and sketched where the adaptation is not routine.

2. Main results

The classification of observables and of families of responses is drawn in Figure 1, with the physical systems that inhabit each class and the minimal families that each class contains, and in Figure 2 each of the seven minimal families of the heptachotomy is followed to its idealized limits. The classes of Todorčević's trichotomy are listed in Table 1 and the minimal

families of the heptachotomy in Table 2, each with a physically natural example and with the feature of the example that the class or family isolates.

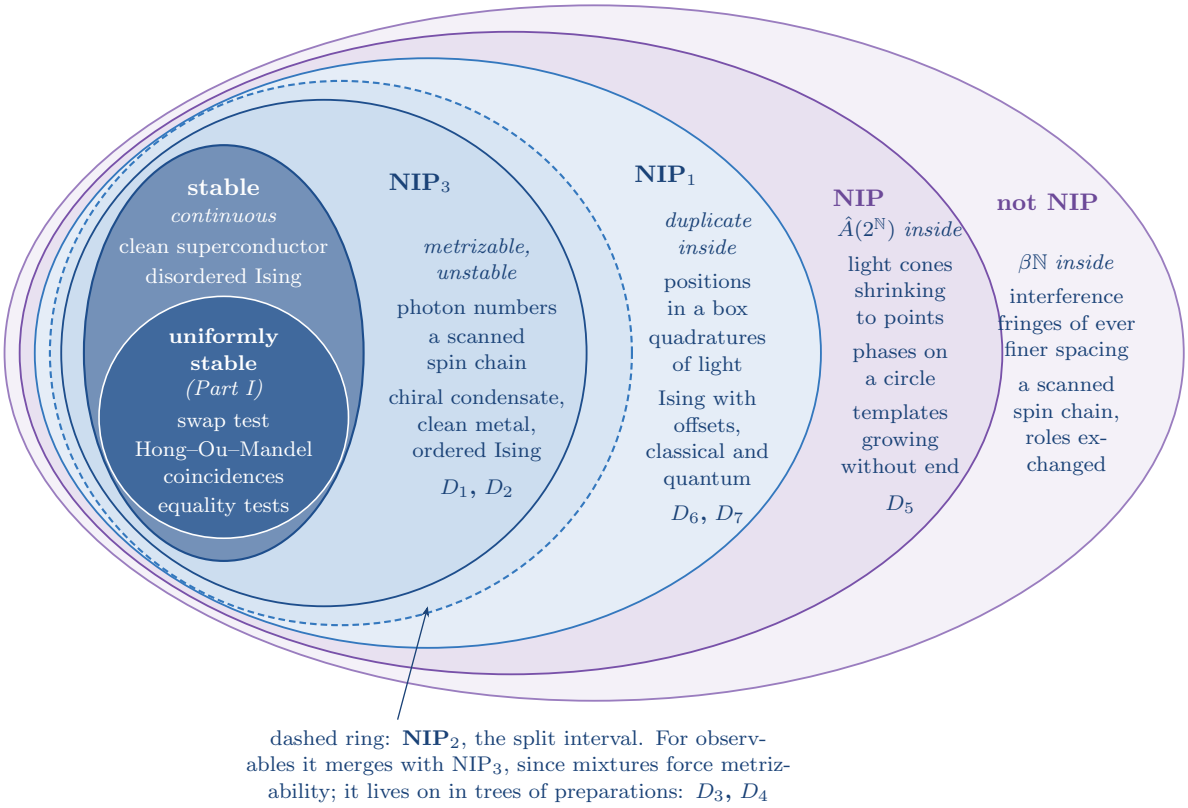


FIGURE 1. The classes of observables and of families of responses, with their physical examples and the minimal families of the heptachotomy that they contain. The regions are nested: uniformly stable $\subseteq$ stable $\subseteq$ NIP$_3$ $\subseteq$ NIP$_2$ $\subseteq$ NIP$_1$ $\subseteq$ NIP. The innermost examples are those of Part I [DI26].

Throughout, "Alice's responses" are the expectation values of Alice's actual preparations as functions of Bob's probes, and her idealized responses, the corresponding functions for her idealized preparations, with Bob's system idealized first, are the functions in the pointwise closure of her responses. The classes refer to this closure: a NIP observable, or a NIP family with separable closure, is NIP$_1$, NIP$_2$ or NIP$_3$ when the closure is, respectively, first countable, hereditarily separable or metrizable.

The results behind the tables are the following:

(A) *The dictionary.* An observable is stable exactly when its idealized responses are continuous, and NIP exactly when they are of the first Baire class; when it is not NIP, some sequence of responses of Alice's actual preparations has a closure homeomorphic to $\beta\mathbb{N}$ (see Theorem 6.1).

(B) *Mixtures force metrizability.* A separable convex Rosenthal compactum that is hereditarily separable is metrizable (see Theorem 6.6); hence NIP$_2$ observables are NIP$_3$ (see Corollary 6.8), and the fine classification of NIP observables has three classes of Todorčević's trichotomy (see Theorem 6.9).

Class	Witness	The picture	The physics	
stable	continuous	idealizations commute	swap test; superconductor	10.4
NIP_3	metrizable	countably many probes suffice	photons; spin chains	7.8
$\mathrm{NIP}_2 \setminus \mathrm{NIP}_3$	split interval	mixtures supply the midpoints	merges with NIP_3	6.6
$\mathrm{NIP}_1 \setminus \mathrm{NIP}_2$	duplicate	approach from either side	positions; quadratures; Ising	9.6
$\mathrm{NIP} \setminus \mathrm{NIP}_1$	$\hat{A}(2^{\mathbb{N}})$	a light cone shrinks to a point	causal order; phases on a circle	7.10, 7.12
not NIP	$\beta\mathbb{N}$	ever finer oscillation	interference fringes	7.13

TABLE 1. The classes of Todorčević's trichotomy. The witness of a class is a property of the idealized responses or of their closure, or a canonical compactum of which the closure contains a copy (see Section 4): the split interval, the Alexandroff duplicate of the Cantor set or of the split Cantor set, the compactum $\hat{A}(2^{\mathbb{N}})$, which extends the one-point compactification of $2^{\mathbb{N}}$, the set of infinite sequences of zeros and ones, regarded as a discrete space, or the space $\beta\mathbb{N}$ of ultrafilters on $\mathbb{N}$. The last column names the main result.

Family	Closure	The picture	The physics
D_1	$A(2^{<\mathbb{N}})$	escape to infinity	photon number states
D_2	$2^{\leq\mathbb{N}}$	infinite configurations	a spin chain read site by site
D_3	$\hat{S}_+(2^{\mathbb{N}})$	approach from the right	a particle in a box
D_4	$\hat{S}_-(2^{\mathbb{N}})$	approach from the left	a particle in a box
D_6	$\hat{D}(2^{\mathbb{N}})$	twins over a Cantor set	a spin chain, sites and templates
D_7	$\hat{D}(S(2^{\mathbb{N}}))$	approach from both sides	a particle in a box, half on each side
D_5	$\hat{A}(2^{\mathbb{N}})$	a point-supported limit	light cones; templates

TABLE 2. The seven minimal families $D_1, \ldots, D_7$ of the heptachotomy and their closures, the canonical compacta of Section 4, grouped by class of Todorčević's trichotomy as in Figure 2 (see Theorem 8.5). Here $2^{<\mathbb{N}}$ is the dyadic tree of finite sequences of zeros and ones, $2^{\mathbb{N}}$ is the Cantor set of infinite such sequences, and $2^{\leq\mathbb{N}}$ is the compact space of all such sequences, finite and infinite, in which the finite ones are isolated.

(C) *Natural examples.* The comparison of photon numbers is NIP_3, the comparison of positions or of quadratures of light (the two field amplitudes of a mode) is NIP_1 but not NIP_2, and the causal order of a spacetime with one dimension of space and the comparison of two phases are NIP but not NIP_1 (see Theorems 7.8, 7.10 and 7.12 and Proposition 7.9); each minimal family is realized by a tree of preparations of one of these observables or of an observable that probes a chain of spins, by reading it site by site, by matching it with a template, or both (see Theorem 8.5). The scanning observable, which reads the

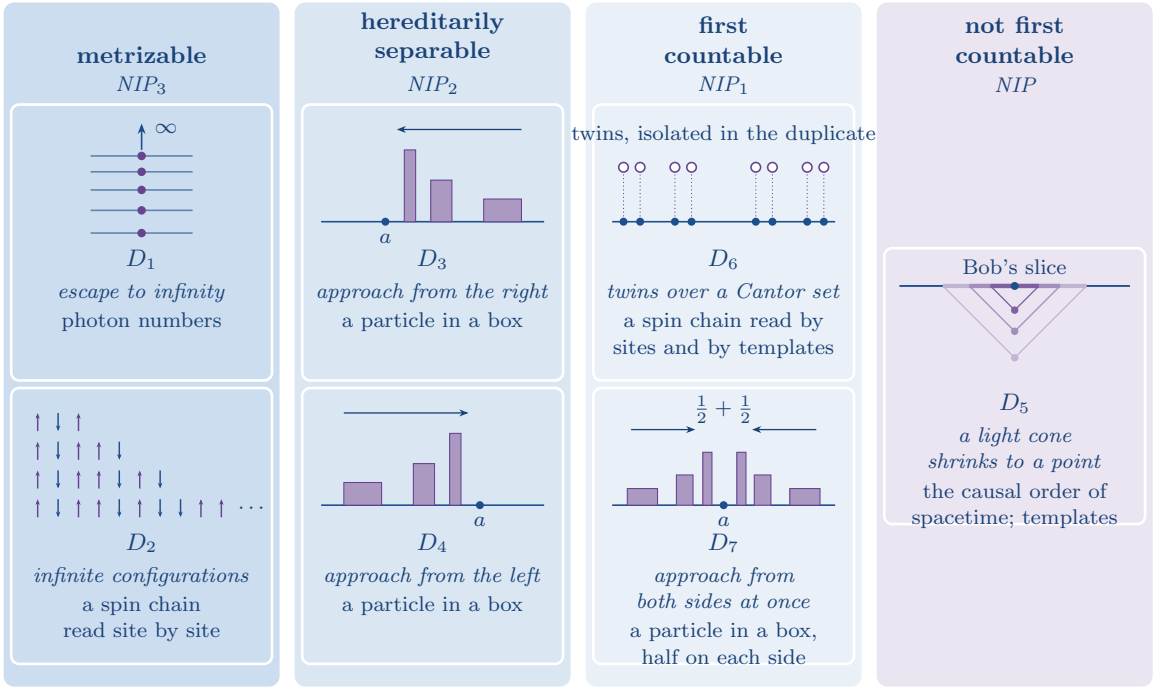


FIGURE 2. The seven minimal families of the heptachotomy, arranged by the class of Todorčević's trichotomy to which their closures belong, each with the physical picture of its realization (see Theorem 8.5): how a tree of preparations approaches its idealized limits. In the panels of D_3, D_4 and D_7, the bumps, the distributions of successive preparations of the particle, approach the point a; in that of D_7, each preparation is split into halves, one on each side of a, and in that of D_6, the duplicate is the Alexandroff duplicate of the Cantor set.

chain site by site, is NIP for Alice's responses but not for Bob's, so NIP is not symmetric for observables (see Corollary 8.7), although uniform NIP is (see Remark 3.8).

(D) *The quantum Ising model.* For the Ising model in a transverse field, the family of magnetizations of finite systems, as functions of the applied field and with an offset field that moves the transition point, is stable, with a metrizable closure, when the spontaneous magnetization vanishes, and it is NIP_1 but not NIP_2 when that magnetization is positive (see Theorem 9.6). The proof uses the real analyticity of the pressure (the limit per site of the logarithm of the partition function) away from zero field, which we derive from a Lee–Yang theorem for this model: the partition function of each finite system has no zeros off the imaginary axis of the complex longitudinal field, the field along the axis of the couplings (see Propositions 9.4 and 9.5).

(E) *Orders of limits.* Spontaneous magnetization, the Banks–Casher relation and the Scalapino–White–Zhang criterion are statements of stability (see Theorems 9.2, 10.3 and 10.4).

Theorem 6.6 in (B) is a special case of a known result (see Remark 6.7). To our knowledge, its consequences for observables in (B), result (D), the realizations in (C) and the asymmetry of nonuniform NIP are new.

3. A family of continuous functions is NIP exactly when its closure consists of Baire-1 functions

We begin with the topological dichotomy on which the paper rests: a pointwise bounded family of continuous functions on a compact metrizable space either has the independence property or has its pointwise closure inside the functions of the first Baire class. Throughout the paper the functions will be responses of Alice's preparations and the points will be Bob's probes.

3.1. **Functions of the first Baire class.** For a topological space Y, let $C_p(Y)$ be the set of continuous functions $Y \to \mathbb{R}$ with the topology of pointwise convergence. A function $f\colon Y \to \mathbb{R}$ is of the *first Baire class*, or *Baire-1*, if it is the pointwise limit of a sequence of continuous functions, and we write $B_1(Y)$ for the set of these functions, again with the topology of pointwise convergence. For $\mathcal{R} \subseteq \mathbb{R}^Y$, we write $\overline{\mathcal{R}}$ for the closure of $\mathcal{R}$ in $\mathbb{R}^Y$, and we call $\mathcal{R}$ *pointwise bounded* if $\sup_{r\in\mathcal{R}} |r(y)| < \infty$ for every $y \in Y$.

We use Baire's characterization of Baire-1 functions on Polish spaces, the separable, completely metrizable spaces; its second condition concerns F_σ sets, the countable unions of closed sets.

Fact 3.1 (see [Tod97, Section 10]**).** *Let Y be a Polish space and $f\colon Y \to \mathbb{R}$ a function. The following are equivalent:*

1. *the function f is Baire-1;*
2. *for every open $U \subseteq \mathbb{R}$, the set $f^{-1}[U]$ is F_σ;*
3. *for every closed $F \subseteq Y$, the restriction $f|_F$ has a point of continuity.*

Moreover, if f is not Baire-1, then there are countable sets $D_0, D_1 \subseteq Y$ with $\overline{D_0} = \overline{D_1}$ and reals $a < b$ such that $f \le a$ on D_0 and $f \ge b$ on D_1.

The last clause of Fact 3.1 lets us test membership in $B_1(Y)$ on countable sets, which is what makes pointwise closures manageable.

Lemma 3.2. *Let Y be a Polish space and $\mathcal{R} \subseteq B_1(Y)$ a pointwise bounded family. The following are equivalent:*

(i) *the family $\mathcal{R}$ is relatively compact in $B_1(Y)$;*
(ii) *every sequence in $\mathcal{R}$ has a cluster point in $B_1(Y)$;*
(iii) *we have $\overline{\mathcal{R}} \subseteq B_1(Y)$.*

Sketch. This is the Bourgain–Fremlin–Talagrand argument [BFT78]. The implication (i)⇒(ii) is clear, and (iii)⇒(i) is Tychonoff's theorem. For (ii)⇒(iii), suppose that some $f \in \overline{\mathcal{R}}$ is not Baire-1, and take D_0, D_1, a and b as in Fact 3.1. A sequence in $\mathcal{R}$ converging to f on the countable set $D_0 \cup D_1$ has no cluster point in $B_1(Y)$, because such a point would agree with f on $D_0 \cup D_1$ and would therefore have no point of continuity on $\overline{D_0} = \overline{D_1}$. □

3.2. **The independence property.** In Theorem 3.6 below, we show that the obstruction to Baire-1 closures is combinatorial: a sequence of functions whose sublevel and superlevel sets meet in every finite pattern.

Definition 3.3. Let Y be a set. A family $\mathcal{R} \subseteq \mathbb{R}^Y$ has the *independence property* if there are a sequence $(r_n)_{n\in\mathbb{N}}$ in $\mathcal{R}$ and reals $a < b$ such that, for all disjoint finite sets $E, F \subseteq \mathbb{N}$, there is $y \in Y$ with $r_n(y) \le a$ for $n \in E$ and $r_n(y) \ge b$ for $n \in F$. Otherwise, the family $\mathcal{R}$ is *NIP*.

On the Cantor space $2^{\mathbb{N}}$ of infinite sequences of zeros and ones, the family of coordinate functions p_n, given by $p_n(y) = y(n)$ for $n \in \mathbb{N}$, has the independence property, and the closure of $\{p_n : n \in \mathbb{N}\}$ in $\mathbb{R}^{2^{\mathbb{N}}}$ is a copy of $\beta\mathbb{N}$, the space of ultrafilters on $\mathbb{N}$. The next two facts say that this example is the only obstruction.

Fact 3.4 (Rosenthal's dichotomy; see [Ros74, Tod97]**).** *Let Y be a Polish space and $(r_n)_{n\in\mathbb{N}}$ a pointwise bounded sequence in $C_p(Y)$. Then the sequence (r_n) has either a pointwise convergent subsequence or a subsequence whose closure in $\mathbb{R}^Y$ is homeomorphic to $\beta\mathbb{N}$.*

Fact 3.5 (Bourgain–Fremlin–Talagrand; see [BFT78, Theorem 4G]**).** *Let Y be a Polish space and $\mathcal{R} \subseteq C_p(Y)$ a pointwise bounded family. Then $\overline{\mathcal{R}} \subseteq B_1(Y)$ if and only if, for every sequence $(r_n)_{n\in\mathbb{N}}$ in $\mathcal{R}$ and all reals $a < b$, there is $I \subseteq \mathbb{N}$ such that no $y \in Y$ satisfies $r_n(y) \leq a$ for all $n \in I$ and $r_n(y) \geq b$ for all $n \notin I$.*

When the space Y is compact, the condition in Fact 3.5 reduces to a statement about finite patterns, and we obtain the dichotomy announced at the beginning of this section.

Theorem 3.6. *Let Y be a compact metrizable space and $\mathcal{R} \subseteq C_p(Y)$ a pointwise bounded family. Then exactly one of the following holds:*

(i) *we have $\overline{\mathcal{R}} \subseteq B_1(Y)$;*
(ii) *the family $\mathcal{R}$ has the independence property.*

The following proof is the classical passage from infinite to finite patterns by compactness (see [DIMW+26, Section 1]).

Proof. For a sequence (r_n) in $\mathcal{R}$, reals $a < b$ and $I \subseteq \mathbb{N}$, the sets $\{y : r_n(y) \leq a\}$ for $n \in I$ and $\{y : r_n(y) \geq b\}$ for $n \notin I$ are closed, because each function r_n is continuous. By compactness of Y, these closed sets have empty intersection if and only if finitely many of them do. Hence the condition in Fact 3.5 fails for a given sequence (r_n) and given reals $a < b$ exactly when every finite pattern is realized, that is, exactly when (r_n), a and b witness the independence property. By Fact 3.5, alternative (i) therefore fails exactly when alternative (ii) holds. □

We will also need to replace a space of probes by a dense subset; in Lemma 3.7 below, we show that the replacement does not affect the independence property.

Lemma 3.7. *Let Y be a topological space, $\mathcal{R} \subseteq C_p(Y)$ a family and $L \subseteq Y$ a dense set. Then the family $\mathcal{R}$ is NIP if and only if the family $\{r|_L : r \in \mathcal{R}\}$ of restrictions is NIP.*

Sketch. If the restrictions have the independence property, then so does $\mathcal{R}$. Conversely, if (r_n) and $a < b$ witness the independence property of $\mathcal{R}$, and $a < a' < b' < b$, then each finite pattern holds strictly, with a' and b', on a nonempty open set, which meets L; hence the restrictions witness the independence property with the reals $a' < b'$. □

Remark 3.8 (Symmetry). For a formula of a first-order theory, NIP is symmetric in the two variables: the formula $\varphi(x, y)$ is NIP if and only if the formula $\varphi^{\mathrm{opp}}(y, x) = \varphi(x, y)$ is NIP (see [She90, Chapter II, Section 4] and [Sim15a, Lemma 2.5]); if φ has Vapnik–Chervonenkis (VC) dimension d, that is, if d is the largest size of a finite set of elements every subset of which is cut out by some instance $\varphi(x, b)$ of the formula, with a fixed parameter b, then the formula φ^{opp} has VC dimension less than 2^{d+1} [Ass83]. Assouad's argument is finite and combinatorial, and it applies, with the reals $a < b$ fixed, to real-valued functions. For uniform NIP, which bounds, for each gap $b - a$, the size of the finite sets of functions on which every pattern of values at most a and at least b is realized at some point (see Remark 6.2 below), the roles of Alice and Bob are therefore interchangeable. The notion of Definition 3.3, which asks for an infinite sequence inside a single family, is not symmetric: the relation $n \in u$ between numbers n and finite sets $u \subseteq \mathbb{N}$ has the independence property in the variable n but not in the variable u.

3.3. **Rosenthal compacta.** A *Rosenthal compactum* is a compact space homeomorphic to a compact subset of $B_1(Y)$, with the topology of pointwise convergence, for some Polish space Y. By Theorem 3.6 and Lemma 3.2, if Y is compact metrizable and $\mathcal{R} \subseteq C_p(Y)$ is pointwise bounded and NIP, then the closure $\overline{\mathcal{R}}$ is a Rosenthal compactum. The classification of Rosenthal compacta (see Section 4) will therefore classify the NIP families that arise below.

3.4. **Historical remarks.** Rosenthal's dichotomy grew out of his ℓ_1 theorem [Ros74]; Bourgain, Fremlin and Talagrand proved Fact 3.5 in 1978 [BFT78], and Godefroy named Rosenthal compacta [God80] (see also [Ai92, Tka11]). The independence property was isolated by Shelah [She71, She90]; the polynomial bound behind its $\{0,1\}$-valued case, the Sauer–Shelah lemma, was found independently by Sauer [Sau72] and Shelah [She72], and by Vapnik and Chervonenkis [VC71, VC74] in learning theory [Val84]. The connection between NIP and Rosenthal compacta is due to Simon [Sim15b], and its use for limits of computations goes back to our joint work with Matos-Wiederhold, Salvetti and Tall [DIMW+26].

4. Separable Rosenthal compacta fall into finitely many canonical classes

We now state Todorčević's trichotomy and the Argyros–Dodos–Kanellopoulos heptachotomy, and derive from them a classification of countable NIP families of continuous functions on a compact metrizable space. By Section 3, the pointwise closure of such a family is a separable Rosenthal compactum; hence the classification of these compacta, which describes their failure to be metrizable by finitely many canonical examples, applies to it. We follow the notation of Argyros, Dodos and Kanellopoulos [ADK08].

4.1. **Three topological invariants.** Recall that a space is hereditarily separable if each of its subspaces has a countable dense subset, and first countable if each of its points has a countable base of neighborhoods. Every compact metrizable space is hereditarily separable, and by a theorem of Pol every hereditarily separable Rosenthal compactum is first countable (see [Deb13, Section 10]). For a Rosenthal compactum K we therefore have:

$$K \text{ metrizable} \implies K \text{ hereditarily separable} \implies K \text{ first countable.}$$

For separable Rosenthal compacta, the failure of each converse is witnessed by a canonical subspace, and these canonical subspaces are pointwise closures of families indexed by the dyadic tree, the set of finite sequences of zeros and ones (see Fact 4.3 below).

4.2. **The canonical compacta.** Let $2^{<\mathbb{N}}$ be the set of finite binary sequences, with $|t|$ the length of a sequence $t \in 2^{<\mathbb{N}}$, and let $2^{\mathbb{N}}$ be the Cantor space. For $t \in 2^{<\mathbb{N}}$, let v_t be the characteristic function of the clopen set $V_t = \{\sigma \in 2^{\mathbb{N}} : t \sqsubset \sigma\}$ of extensions of t, and let $t^\frown 0^\infty$ and $t^\frown 1^\infty$ be the sequences that extend t by zeros and by ones. Let $\prec$ be the lexicographic order of $2^{\mathbb{N}}$. For $\sigma \in 2^{\mathbb{N}}$, let x_σ^+ and x_σ^- be the characteristic functions of $\{\tau \in 2^{\mathbb{N}} : \sigma \preceq \tau\}$ and $\{\tau \in 2^{\mathbb{N}} : \sigma \prec \tau\}$. For functions f and g on spaces X_1 and X_2, let (f,g) be the function on the disjoint sum $X_1 \oplus X_2$ that is f on X_1 and g on X_2; a point of $2^{\mathbb{N}}$ is regarded as a function on $\mathbb{N}$.

The following compacta appear in the classification:

- the *one-point compactification* $A(2^{\mathbb{N}})$ of a discrete set of size continuum, realized as $\{\delta_\sigma : \sigma \in 2^{\mathbb{N}}\} \cup \{0\}$, where δ_σ is the characteristic function of $\{\sigma\}$, and its separable extension $\hat{A}(2^{\mathbb{N}})$, the pointwise closure of $\{v_t : t \in 2^{<\mathbb{N}}\}$, which equals $\{v_t : t \in 2^{<\mathbb{N}}\} \cup A(2^{\mathbb{N}})$;

- the *split Cantor set* $S(2^{\mathbb{N}}) = \{x^+_\sigma, x^-_\sigma : \sigma \in 2^{\mathbb{N}}\}$, the *split interval* $S(I)$, that is, $[0,1]\times\{0,1\}$ ordered lexicographically, with the order topology, and the extended split Cantor sets $\hat{S}_+(2^{\mathbb{N}})$ and $\hat{S}_-(2^{\mathbb{N}})$ defined below; the space $\hat{S}_+(2^{\mathbb{N}})$ is homeomorphic to a closed subspace of $S(I)$;
- the *Alexandroff duplicate* $D(2^{\mathbb{N}})$ of the Cantor set and its separable extension $\hat{D}(2^{\mathbb{N}})$, and the extended duplicate $\hat{D}(S(2^{\mathbb{N}}))$ of the split Cantor set, the last two being defined below; the duplicate $D(2^{\mathbb{N}})$ is the set $2^{\mathbb{N}}\times\{0,1\}$ in which the points $(\sigma,1)$ are isolated and the basic neighborhoods of a point $(\sigma,0)$ are the sets $(U\times\{0,1\})\setminus\{(\sigma,1)\}$, for open neighborhoods U of σ in $2^{\mathbb{N}}$.

Among these, the space $S(2^{\mathbb{N}})$ is hereditarily separable but not metrizable, the space $D(2^{\mathbb{N}})$ is first countable but not separable, and the space $A(2^{\mathbb{N}})$ is not first countable (see [ADK08]).

4.3. **The heptachotomy.** A *dyadic subtree* of $2^{<\mathbb{N}}$ is a subtree in which every node has exactly two immediate successors; we write it as $(s_t)_{t\in 2^{<\mathbb{N}}}$, using its canonical enumeration, which preserves length and the lexicographic order (see [ADK08, Section 2]). A dyadic subtree is *regular* if each of its levels is contained in a level of $2^{<\mathbb{N}}$. Fix a regular dyadic subtree $\Theta = (s_t)_{t\in 2^{<\mathbb{N}}}$ such that the sequences $s^\frown 0^\infty$, for $s\in\Theta$, are pairwise distinct, and so are the sequences $s^\frown 1^\infty$.

Definition 4.1. Let X be a Polish space.

(1) Two pointwise bounded families $\{f_i : i\in I\}$ and $\{g_i : i\in I\}$ of real functions, indexed by the same countable set I, are *equivalent* if the map $f_i\mapsto g_i$ extends to a homeomorphism between their pointwise closures.
(2) A family $\{f_t : t\in 2^{<\mathbb{N}}\}$ that is relatively compact in $B_1(X)$ is *minimal* if it is equivalent to $\{f_{s_t} : t\in 2^{<\mathbb{N}}\}$ for every dyadic subtree (s_t) of $2^{<\mathbb{N}}$.

The seven minimal families are the following, and the pointwise closure of each of them is a separable Rosenthal compactum that contains the family as a discrete set (see [ADK08]):

(1) $D_1 = \{\frac{1}{|t|+1}v_t : t\in 2^{<\mathbb{N}}\}$, whose closure $A(2^{<\mathbb{N}})$ is the one-point compactification of a countable discrete set;
(2) $D_2 = \{s^\frown 0^\infty : s\in\Theta\}$, a family of points of $2^{\mathbb{N}}\subseteq\mathbb{R}^{\mathbb{N}}$, whose closure is denoted $2^{\leq\mathbb{N}}$;
(3) $D_3 = \{x^+_{s^\frown 0^\infty} : s\in\Theta\}$, whose closure is $\hat{S}_+(2^{\mathbb{N}})$;
(4) $D_4 = \{x^-_{s^\frown 1^\infty} : s\in\Theta\}$, whose closure is $\hat{S}_-(2^{\mathbb{N}})$;
(5) $D_5 = \{v_t : t\in 2^{<\mathbb{N}}\}$, whose closure is $\hat{A}(2^{\mathbb{N}})$;
(6) $D_6 = \{(v_t, t^\frown 0^\infty) : t\in 2^{<\mathbb{N}}\}$, whose closure is $\hat{D}(2^{\mathbb{N}})$;
(7) $D_7 = \{(v_s, x^+_{s^\frown 0^\infty}) : s\in\Theta\}$, whose closure is $\hat{D}(S(2^{\mathbb{N}}))$.

The closures in (1) and (2) are metrizable, those in (3) and (4) are hereditarily separable but not metrizable, those in (6) and (7) are first countable but not hereditarily separable, and the closure in (5) is not first countable (see [ADK08]).

Fact 4.2 (Heptachotomy; see [ADK08, Theorem 2]**).**

(a) *Up to equivalence, there are exactly seven minimal families, namely* $D_1,\dots,D_7$*, and they are pairwise non-equivalent.*
(b) *Let* X *be a Polish space and* $\{f_t : t\in 2^{<\mathbb{N}}\}$ *a family that is relatively compact in* $B_1(X)$*. Then there is a regular dyadic subtree* (s_t) *of* $2^{<\mathbb{N}}$ *such that* $\{f_{s_t} : t\in 2^{<\mathbb{N}}\}$ *is equivalent to one of* $D_1,\dots,D_7$*.*

4.4. **Todorčević's trichotomy.** Let K be a separable Rosenthal compactum and $\{f_n : n\in\mathbb{N}\}$ a countable dense subset of K. A minimal family D_i *canonically embeds* into K (with respect to $\{f_n\}$) if there is an injection $\iota\colon 2^{<\mathbb{N}}\to\mathbb{N}$, increasing for the order of $2^{<\mathbb{N}}$ by length

and, within each length, by the lexicographic order, such that $\{f_{\iota(t)} : t \in 2^{<\mathbb{N}}\}$ is equivalent to D_i. In that case the closure of D_i is homeomorphic to a closed subspace of K.

Fact 4.3 (Todorčević's trichotomy in canonical form; see [Tod99] and [ADK08, Theorem 3]). *Let K be a separable Rosenthal compactum and $\{f_n : n \in \mathbb{N}\}$ a countable dense subset of K.*

(a) *If K consists of bounded functions, is metrizable and is not separable in the supremum norm, then the family D_2 canonically embeds into K.*
(b) *If K is hereditarily separable but not metrizable, then the family D_3 or the family D_4 canonically embeds into K.*
(c) *If K is first countable but not hereditarily separable, then the family D_6 or the family D_7 canonically embeds into K.*
(d) *If K is not first countable, then the family D_5 canonically embeds into K.*

In particular, a non-metrizable separable Rosenthal compactum contains a copy of $\hat{S}_+(2^{\mathbb{N}})$ or $\hat{S}_-(2^{\mathbb{N}})$, of $\hat{D}(2^{\mathbb{N}})$ or $\hat{D}(S(2^{\mathbb{N}}))$, or of $\hat{A}(2^{\mathbb{N}})$, according as it is hereditarily separable, first countable but not hereditarily separable, or not first countable.

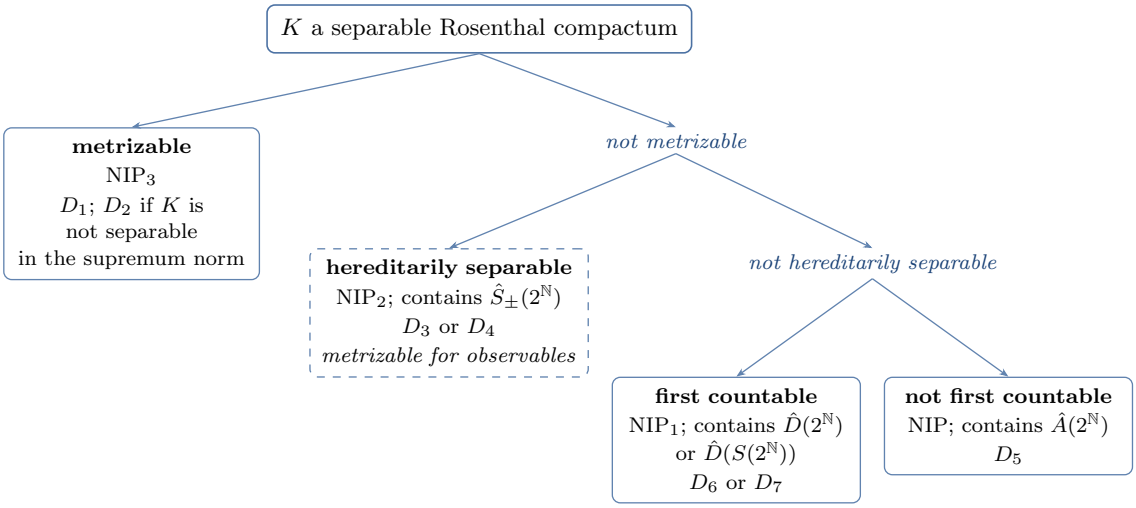


FIGURE 3. Todorčević's trichotomy as a decision tree, with the canonical compacta and the minimal families of the heptachotomy at its leaves (compare [DIMW+26, Section 4.2]). For observables the dashed leaf merges with the metrizable one (see Theorem 6.6).

4.5. **Levels of NIP.** We now transfer the classification to NIP families, as in our work on deep computations [DIMW+26, Section 4.2].

Definition 4.4. Let Y be a compact metrizable space and $\mathcal{R} \subseteq C_p(Y)$ a pointwise bounded NIP family whose closure $\overline{\mathcal{R}}$ is separable. The family $\mathcal{R}$ is *NIP$_1$*, *NIP$_2$* or *NIP$_3$* if $\overline{\mathcal{R}}$ is, respectively, first countable, hereditarily separable or metrizable.

By the implications among the three invariants, we have NIP$_3$ $\Rightarrow$NIP$_2$ $\Rightarrow$NIP$_1$ $\Rightarrow$NIP.

The top level has a concrete meaning. A set $D \subseteq Y$ *separates* a set K of functions on Y if any two distinct members of K differ at some point of D.

Proposition 4.5. *Let Y be a set and $K \subseteq \mathbb{R}^Y$ a set that is compact in the topology of pointwise convergence. Then the space K is metrizable if and only if some countable set $D \subseteq Y$ separates K.*

Proof. Suppose that a countable set D separates K. Restriction to D is a continuous injection of K into $\mathbb{R}^D$, which is metrizable because D is countable. A continuous injection of a compact space into a Hausdorff space is a homeomorphism onto its image, so the space K is metrizable.

Conversely, suppose that K is metrizable. Being compact, it is second countable, and so is $K \times K$. Hence the open set $W = \{(f, g) \in K \times K : f \neq g\}$ is Lindelöf. For $y \in Y$, the set $U_y = \{(f, g) \in K \times K : f(y) \neq g(y)\}$ is open, and these sets cover W. If $\{U_y : y \in D\}$ is a countable subcover, then the set D separates K. □

Thus a NIP family with separable closure is NIP_3 exactly when countably many points of Y distinguish all functions in its closure. When the functions are responses and the points are probes, this says that countably many measurements tell all idealized preparations apart. The other two levels have readings of the same kind.

Proposition 4.6. *Let Y be a compact metrizable space and $\mathcal{R} \subseteq C_p(Y)$ a pointwise bounded NIP family whose closure $K = \overline{\mathcal{R}}$ is separable.*

(a) *The family is NIP_3 if and only if some countable set $D \subseteq Y$ separates K.*
(b) *The family is NIP_1 if and only if every $f \in K$ is* countably determined*: there is a countable set $D_f \subseteq Y$ such that f is the only member of K that agrees with f on D_f.*
(c) *The family is NIP_2 if and only if there is no set $E \subseteq K$ of cardinality $2^{\aleph_0}$ whose members are* finitely separated*: for each $f \in E$ there are a finite set $F_f \subseteq Y$ and $\varepsilon_f > 0$ such that every $g \in E \setminus \{f\}$ satisfies $|g(y) - f(y)| \geq \varepsilon_f$ for some $y \in F_f$.*

Proof. Part (a) is Proposition 4.5. (b) If f is countably determined, then the set $\{f\}$ is the intersection of the countably many open sets $\{g \in K : |g(y) - f(y)| < \frac{1}{m}\}$, for $y \in D_f$ and $m \geq 1$, and in a compact Hausdorff space a point that is a countable intersection of open sets has a countable base of neighborhoods. Conversely, if $(U_k)_{k \in \mathbb{N}}$ is a countable base at f and each U_k contains a basic neighborhood defined by a finite set $F_k \subseteq Y$, then the set $D_f = \bigcup_k F_k$ works: a member of K that agrees with f on D_f lies in every U_k. (c) A set is finitely separated exactly when it is a discrete subspace, and a separable Rosenthal compactum is hereditarily separable if and only if it has no discrete subspace of cardinality $2^{\aleph_0}$, by a theorem of Todorčević [Tod99] (see also [GM22, Theorem 0.4]). □

The three levels thus measure how many points of Y are needed to determine a limit: one countable set for all of them (NIP_3), a countable set for each of them (NIP_1), or no continuum of limits that finitely many points tell apart one by one (NIP_2). When limits are taken as types and points as parameters, the levels say how many instances of a formula determine a type. This partly answers a question, raised in our work on deep computations [DIMW+26], that asks for a non-topological characterization of the levels; a finitary characterization, in the spirit of the fat-shattering dimension (the largest size, at a given gap, of a finite set of functions on which every pattern of low and high values is realized at some point), remains open (see Question (5)).

Theorem 4.7. *Let Y be a compact metrizable space and $\{r_n : n \in \mathbb{N}\} \subseteq C_p(Y)$ a pointwise bounded NIP family. Then exactly one of the following holds:*

(i) *the family is NIP$_3$;*
(ii) *the family is NIP$_2$ but not NIP$_3$, and there is an increasing injection $\iota\colon 2^{<\mathbb{N}} \to \mathbb{N}$ such that $\{r_{\iota(t)} : t \in 2^{<\mathbb{N}}\}$ is equivalent to D_3 or to D_4;*
(iii) *the family is NIP$_1$ but not NIP$_2$, and there is an increasing injection $\iota\colon 2^{<\mathbb{N}} \to \mathbb{N}$ such that $\{r_{\iota(t)} : t \in 2^{<\mathbb{N}}\}$ is equivalent to D_6 or to D_7;*
(iv) *the family is not NIP$_1$, and there is an increasing injection $\iota\colon 2^{<\mathbb{N}} \to \mathbb{N}$ such that $\{r_{\iota(t)} : t \in 2^{<\mathbb{N}}\}$ is equivalent to D_5.*

The following proof merely combines the facts above.

Proof. By Theorem 3.6 and Lemma 3.2, the closure K of the family is a Rosenthal compactum, and the family is a countable dense subset of it. The four cases are mutually exclusive by the definitions and cover all possibilities. In cases (ii)–(iv), the required injection is provided by Fact 4.3(b)–(d), applied to K and the dense subset $\{r_n\}$. □

Thus a non-metrizable NIP closure is always explained by a subfamily of the original functions, arranged along the dyadic tree and equivalent to one of five canonical families. Applied to realized types, the complete descriptions of elements of the structure in terms of the parameters, the classification of Theorem 4.7 yields that of NIP formulas described in the introduction (see [DIMW+26, Section 4.2]). There the realized types are viewed as continuous functions on a compact metrizable space of parameters, and this proviso matters: the space of types of a formula over a countable set of parameters, with no topology on the parameters, is always metrizable, since the countable set of parameters separates the types (see Proposition 4.5).

4.6. **Historical remarks.** Todorčević proved his trichotomy in 1999 [Tod99], using forcing and absoluteness; Argyros, Dodos and Kanellopoulos extracted the canonical form and the heptachotomy from the methods of his paper in 2008 [ADK08], using Ramsey theory for trees [HL66, Mil81, LSV93]. Glasner and Megrelishvili used the trichotomy to build a hierarchy of tame dynamical systems [GM22], and the levels NIP$_1$, NIP$_2$ and NIP$_3$ were introduced for limits of computations in our work on deep computations [DIMW+26].

The hierarchy obtained in this paper differs from these. In the setting of Glasner and Megrelishvili [GM22] and in that of deep computations [DIMW+26], Todorčević's trichotomy is applied to families of functions that carry no linear structure, such as enveloping semigroups (pointwise closures of the transformations of dynamical systems) or sets of computations, and all its classes can occur. By contrast, the idealized responses of an observable (its expectation values for Alice's idealized preparations, as functions of Bob's probes, with Bob's system idealized first) form a convex set (see the proof of Corollary 6.8), and convexity collapses NIP$_2$ onto NIP$_3$ (see Theorem 6.6 and Remark 6.7), whence the hierarchy has three levels. At the level of families indexed by the dyadic tree, all seven minimal families occur (see Theorem 8.5). The difference is a feature of quantum, or more generally of probabilistic, response spaces.

5. The idealized responses of an observable form a compact set of functions

Here we place the responses of an observable in the setting of Section 3: we attach to each bipartite observable a compact metrizable space of probes, on which Alice's responses are

continuous, and we identify the pointwise closure of these responses with the responses of Alice's idealized preparations.

5.1. **Observables and responses.** We work in the setting of Part I [DI26] and explain its operator theory briefly. Alice's and Bob's systems are described by separable complex Hilbert spaces H_1 and H_2. A *density operator* on a Hilbert space H is a positive trace-class operator of trace 1; density operators are the *actual states* of the system, and we write $\mathfrak{D}(H)$ for the set they form. The trace-class operators, those bounded operators x with $\operatorname{Tr}|x| < \infty$, where $|x| = (x^*x)^{1/2}$, form a Banach space $\mathfrak{T}(H)$ under the trace norm $\|x\|_1 = \operatorname{Tr}|x|$, and we write $B(H)$ for the algebra of bounded operators on H. A bounded self-adjoint operator C on $H_1 \otimes H_2$ is an *observable*, and its expectation when the two systems are prepared independently in actual states $\rho \in \mathfrak{D}(H_1)$ and $\sigma \in \mathfrak{D}(H_2)$ is

$$\varphi_C(\rho, \sigma) = \operatorname{Tr}\big((\rho \otimes \sigma)C\big).$$

Fixing Alice's state ρ produces an observable on Bob's system alone, the *effective observable* $T_C(\rho) = \operatorname{Tr}_1((\rho \otimes I)C)$, obtained by tracing out Alice's system. It is characterized by $\varphi_C(\rho, \sigma) = \operatorname{Tr}(\sigma\, T_C(\rho))$ for all $\sigma \in \mathfrak{D}(H_2)$, and it extends linearly to a bounded operator $T_C \colon \mathfrak{T}(H_1) \to B(H_2)$ with $\|T_C\| \le \|C\|$, the *slice map*.

Idealized states are described through C*-algebras. A *C*-algebra* is a norm-closed algebra of bounded operators closed under adjoints; we use only unital ones, which contain the identity. A *state* of a unital C*-algebra $\mathfrak{A}$ is a linear functional ω on $\mathfrak{A}$ that is positive, in the sense that $\omega(a^*a) \ge 0$ for all $a \in \mathfrak{A}$, and satisfies $\omega(I) = 1$. The set $\mathcal{S}(\mathfrak{A})$ of states, with the weak* topology (pointwise convergence on $\mathfrak{A}$), is compact; it is metrizable when $\mathfrak{A}$ is norm-separable. Every density operator ρ defines a state $a \mapsto \operatorname{Tr}(\rho a)$; the remaining states of $B(H)$, which no density operator represents, are the *idealized* states, and each of them is a limit of actual states.

5.2. **The space of probes.** The responses of Alice's preparations involve Bob's system only through the effective observables $T_C(\rho)$. The natural probes are therefore the states of the smallest C*-algebra containing them.

Definition 5.1. Let C be an observable on $H_1 \otimes H_2$. The *probe algebra* $\mathfrak{A}_C \subseteq B(H_2)$ is the unital C*-algebra generated by $T_C(\mathfrak{T}(H_1))$, and the *probe space* is $Y_C = \mathcal{S}(\mathfrak{A}_C)$. For $\rho \in \mathfrak{D}(H_1)$, the *response* of ρ is the function $r_\rho \colon Y_C \to \mathbb{R}$ given by $r_\rho(\tau) = \tau(T_C(\rho))$, and the *response family* is $\mathcal{R}_C = \{r_\rho : \rho \in \mathfrak{D}(H_1)\}$.

Lemma 5.2. *Let C be an observable on $H_1 \otimes H_2$.*

(a) *The probe space Y_C is compact and metrizable.*
(b) *Every response is continuous on Y_C and bounded by $\|C\|$, so the response family $\mathcal{R}_C$ is a pointwise bounded subset of $C_p(Y_C)$.*
(c) *The restrictions to $\mathfrak{A}_C$ of the actual states of Bob's system form a dense subset of Y_C, and for $\sigma \in \mathfrak{D}(H_2)$ we have $r_\rho(\sigma|_{\mathfrak{A}_C}) = \varphi_C(\rho, \sigma)$.*

Proof. (a) Because H_1 is separable, so is $\mathfrak{T}(H_1)$; its image under the norm-continuous slice map is therefore separable, and so is the C*-algebra $\mathfrak{A}_C$ that this image generates. The compact space Y_C is thus metrizable.

(b) The response r_ρ is evaluation at the element $T_C(\rho)$ of $\mathfrak{A}_C$, and evaluation at a fixed element is weak* continuous. Since $\rho \ge 0$ and C is self-adjoint, the operator $T_C(\rho)$ is self-adjoint, whence r_ρ is real-valued; moreover, $|r_\rho(\tau)| \le \|T_C(\rho)\| \le \|C\|$.

(c) By the Hahn–Banach theorem, every state of $\mathfrak{A}_C$ extends to a state of $B(H_2)$ (see [KR83, Theorem 4.3.13]); restriction to $\mathfrak{A}_C$ is thus a weak*-continuous map of the state

space of $B(H_2)$ onto Y_C. The states given by density operators are weak* dense in the state space of $B(H_2)$ (see [DI26, Lemma 2.2]), and their restrictions are therefore dense in Y_C. The formula restates the definition of the effective observable. □

By Lemma 5.2(c) and Lemma 3.7, Bob's actual probes alone decide whether $\mathcal{R}_C$ has the independence property. The family $\mathcal{R}_C$ has it exactly when there are actual states $\rho_1, \rho_2, \ldots$ of Alice's system and reals $a < b$ with the following property: for all disjoint finite $E, F \subseteq \mathbb{N}$, some actual state σ of Bob's system satisfies $\varphi_C(\rho_n, \sigma) \leq a$ for $n \in E$ and $\varphi_C(\rho_n, \sigma) \geq b$ for $n \in F$. A single probe then lets Bob read any finite pattern off Alice's sequence.

Definition 5.3. An observable C on $H_1 \otimes H_2$ is *NIP for Alice's responses* if the response family $\mathcal{R}_C$ is NIP.

5.3. **Idealized responses.** To describe the closure of $\mathcal{R}_C$, we use the bitranspose of the slice map, which extends it from actual to idealized states. Every bounded operator $T\colon E \to F$ between Banach spaces has a *transpose* $T^*\colon F^* \to E^*$, given by $T^*(\lambda) = \lambda \circ T$, and a *bitranspose* $T^{**}\colon E^{**} \to F^{**}$, which extends T and is continuous for the weak* topologies. Here we regard the slice map as a map $T_C\colon \mathcal{T}(H_1) \to \mathfrak{A}_C$. The dual of $\mathcal{T}(H_1)$ is $B(H_1)$; the bidual of $\mathcal{T}(H_1)$ is therefore $B(H_1)^*$, which contains all states of $B(H_1)$, idealized ones included.

For an observable C on $H_1 \otimes H_2$ and a state ω of $B(H_1)$, let $r_\omega\colon Y_C \to \mathbb{R}$ be given by $r_\omega(\tau) = \langle T_C^{**}(\omega), \tau \rangle$. We also recall from Part I [DI26] the two products of a state ω of $B(H_1)$ and a state ω' of $B(H_2)$, evaluated at the observable C: if (ρ_i) and (σ_j) are nets of actual states, indexed by directed sets, with weak* limits ω and ω', then $(\omega \otimes_{\mathsf{A}} \omega')(C) = \lim_j \lim_i \varphi_C(\rho_i, \sigma_j)$ and $(\omega \otimes_{\mathsf{B}} \omega')(C) = \lim_i \lim_j \varphi_C(\rho_i, \sigma_j)$, the subscript naming the system that is idealized first.

Lemma 5.4. *Let C be an observable on $H_1 \otimes H_2$. Then*

$$\overline{\mathcal{R}_C} = \{r_\omega : \omega \in \mathcal{S}(B(H_1))\}.$$

Moreover, if $\tau \in Y_C$ and $\tilde{\tau}$ is a state of $B(H_2)$ extending τ, then $r_\omega(\tau) = (\omega \otimes_{\mathsf{B}} \tilde{\tau})(C)$.

Proof. The map $\omega \mapsto T_C^{**}(\omega)$ is weak* continuous, and on bounded subsets of $\mathfrak{A}_C^{**}$ the weak* topology is the topology of pointwise convergence on Y_C, because states span $\mathfrak{A}_C^*$. The map $\omega \mapsto r_\omega$ is therefore a continuous map of the compact space $\mathcal{S}(B(H_1))$ into $\mathbb{R}^{Y_C}$, and its image K is compact. As T_C^{**} extends T_C, the set K contains $\mathcal{R}_C$, and, being closed, it contains $\overline{\mathcal{R}_C}$. Conversely, actual states are dense in $\mathcal{S}(B(H_1))$ (see [DI26, Lemma 2.2]), whence, by continuity, $K \subseteq \overline{\mathcal{R}_C}$. The last formula is [DI26, Lemma 2.4]; the pairing $\langle T_C^{**}(\omega), \tilde{\tau} \rangle$ depends only on the restriction of $\tilde{\tau}$ to $\mathfrak{A}_C$. □

The functions r_ω are the *idealized responses*: the responses of Alice's idealized preparations to Bob's probes, idealized probes included, with Bob's system idealized first. On actual probes the order of idealization does not matter (see [DI26, Corollary 2.5]), and the idealized response satisfies $r_\omega(\sigma|_{\mathfrak{A}_C}) = (\omega \otimes_{\mathsf{B}} \sigma)(C) = (\omega \otimes_{\mathsf{A}} \sigma)(C)$.

6. Stability, NIP and the independence property are continuity, Baire class 1 and $\beta\mathbb{N}$

We come to the main result: the three levels of tameness of an observable are three levels of regularity of its idealized responses, the expectation values of Alice's idealized preparations as functions of Bob's probes, with Bob's system idealized first. In Theorem 6.1 below, we show that stability corresponds to continuity, through weak compactness, and NIP to the first Baire class, through Theorem 3.6. Recall from Part I [DI26] that an observable is stable when

its expectation value does not depend on which of the two systems is idealized first. As in Section 5, an observable C is a bounded self-adjoint operator on the tensor product $H_1 \otimes H_2$ of the Hilbert spaces of Alice's and Bob's systems, its probe space Y_C is the compact metrizable state space of its probe algebra (see Definition 5.1 and Lemma 5.2), and its response family $\mathcal{R}_C$ consists of the responses of Alice's actual preparations; the closure $\overline{\mathcal{R}_C}$ of this family is the set of idealized responses (see Lemma 5.4).

For stability we need one more fact from operator theory, Kadison's function representation: on the state space of a unital C*-algebra $\mathfrak{A}$, the continuous affine real functions are exactly the evaluations $\tau \mapsto \tau(a)$ at self-adjoint elements $a \in \mathfrak{A}$ (see [Kad51]).

Theorem 6.1. *Let C be an observable on $H_1 \otimes H_2$.*

(a) *The observable C is stable if and only if every idealized response is continuous, that is, $\overline{\mathcal{R}_C} \subseteq C_p(Y_C)$.*

(b) *The observable C is NIP for Alice's responses if and only if every idealized response is Baire-1, that is, $\overline{\mathcal{R}_C} \subseteq B_1(Y_C)$. In that case, the set $\overline{\mathcal{R}_C}$ of idealized responses is a Rosenthal compactum.*

(c) *If the observable C is not NIP for Alice's responses, then there is a sequence $(\rho_n)_{n\in\mathbb{N}}$ of actual states of Alice's system such that the closure of $\{r_{\rho_n}\}$ is homeomorphic to $\beta\mathbb{N}$.*

Proof. Part (a) rests on the characterization of stability by weak compactness (see [DI26, Theorem 5.1]), and parts (b) and (c) rest on the dichotomy of Theorem 3.6.

For part (a), recall that a bounded operator between Banach spaces is *weakly compact* if it maps bounded sets to relatively weakly compact sets, and that, by Gantmacher's theorem, this holds exactly when its bitranspose takes values in the original target space. If C is stable, the slice map is weakly compact as a map into $B(H_2)$, and therefore also as a map into the closed subspace $\mathfrak{A}_C$, whose weak topology is the restriction of that of $B(H_2)$. The bitranspose T_C^{**} then maps $B(H_1)^*$ into $\mathfrak{A}_C$, whence every idealized response r_ω is evaluation at an element of $\mathfrak{A}_C$ and is continuous on Y_C.

Conversely, suppose that every idealized response is continuous, and let ω be a state of $B(H_1)$. The element $T_C^{**}(\omega)$ of $\mathfrak{A}_C^{**}$ is self-adjoint, so the idealized response r_ω is a continuous affine real function on Y_C. By Kadison's function representation, there is a self-adjoint $a \in \mathfrak{A}_C$ with $r_\omega(\tau) = \tau(a)$ for all $\tau \in Y_C$. The elements $T_C^{**}(\omega)$ and a of $\mathfrak{A}_C^{**}$ thus agree on all states, and since states span $\mathfrak{A}_C^*$, they are equal; in particular, the element $T_C^{**}(\omega)$ lies in $B(H_2)$. As ω was arbitrary, the observable C is stable (see [DI26, Theorem 5.1]).

By Lemma 5.2, Theorem 3.6 applies to the family $\mathcal{R}_C \subseteq C_p(Y_C)$, and part (b) follows from it together with Lemma 3.2. For part (c), let $(r_n)_{n\in\mathbb{N}}$ be an independent sequence in $\mathcal{R}_C$, with reals $a < b$. By compactness of Y_C (see the proof of Theorem 3.6), for every $I \subseteq \mathbb{N}$ some probe τ satisfies $r_n(\tau) \le a$ for $n \in I$ and $r_n(\tau) \ge b$ for $n \notin I$. No subsequence of (r_n) can therefore converge pointwise, and Fact 3.4 yields a subsequence whose closure is homeomorphic to $\beta\mathbb{N}$. □

In terms of preparations and probes, Theorem 6.1 describes how the responses of Alice's preparations depend on Bob's probe at each level:

- for a stable observable, the response of each idealized preparation of Alice's system depends continuously on Bob's probe, idealized probes included;

- for an observable that is NIP but unstable, such as the comparison of photon numbers (see Section 7), these responses can be discontinuous, but each is a pointwise limit of a sequence of continuous functions, and together they form a Rosenthal compactum;
- for an observable with the independence property, a sequence of Alice's actual preparations lets Bob read every finite pattern with a single probe, and, by Fact 3.4, any such sequence without a convergent subsequence has a subsequence whose closure is a copy of $\beta\mathbb{N}$.

Remark 6.2. In Part I [DI26] we distinguished stability (no infinite ladder) from uniform stability (bounded ladders at every margin, equivalently stability on ultrapowers). A ladder is a pair of sequences of Alice's and of Bob's preparations, of the same length, finite or infinite, whose expectation values are high when Alice's index does not exceed Bob's and low otherwise, or the reverse, and it has margin δ when every high value exceeds every low value by at least δ. The same distinction appears here. The notion of NIP in Definition 3.3 forbids an infinite independent sequence, whereas *uniform NIP* bounds, at each margin, the size of finite independent sets, that is, the fat-shattering dimension. Uniform NIP implies NIP. Comparisons $[A < B]$, the projections that represent the event "Bob's value of B exceeds Alice's value of A" for quantities A of Alice's system and B of Bob's, are uniformly NIP, with bounds independent of the Hilbert spaces (see Corollary 7.4), and they are unstable when both quantities have discrete spectra unbounded above, by the ladder argument of Section 7. By Theorem 6.1, their idealized responses form a Rosenthal compactum of Baire-1 functions, not all continuous. As in Part I, the uniform notion is the one that survives ultrapowers: a family is uniformly NIP if and only if its extension to every ultrapower is NIP. Recall that, for an ultrafilter $\mathcal{U}$ on $\mathbb{N}$, every bounded sequence $(x_n)_{n\in\mathbb{N}}$ of reals has a unique ultralimit $\lim_{n,\mathcal{U}} x_n$, the real x such that $\{n : |x_n - x| < \varepsilon\} \in \mathcal{U}$ for every $\varepsilon > 0$. Here the extension of a family to an ultrapower is the family of ultralimits of sequences of responses, evaluated at ultralimits of sequences of probes, and a finite set of responses is *δ-shattered* if it is independent, in the sense of Definition 3.3, with a gap $b - a \geq \delta$. By Łoś's theorem, which says that the value of a formula at ultralimits is the ultralimit of its values at the terms of the sequences, a δ-shattered set of size n in an ultrapower yields, for every $\delta' < \delta$, a δ'-shattered set of size n in the original family. Hence, if the original family has finite fat-shattering dimension at every margin, so does its extension, which then has no infinite independent sequence; conversely, if the fat-shattering dimension at some margin is infinite, then in an ultrapower by a countably incomplete ultrafilter (for instance, a nonprincipal ultrafilter on $\mathbb{N}$) the finite shattered sets combine, by $\aleph_1$-saturation (every countable set of conditions, each finite subset of which is realized in the ultrapower, is realized in it), into an infinite independent sequence.

6.1. **The finer classification of NIP observables.** For NIP observables, the classification of Section 4 applies to the idealized responses. We first check that they form a separable compactum.

Lemma 6.3. *Let C be an observable on $H_1 \otimes H_2$ and $(\rho_n)_{n\in\mathbb{N}}$ a sequence of density operators that is dense in $\mathfrak{D}(H_1)$ for the trace norm. Then the responses r_{ρ_n} are dense in $\overline{\mathcal{R}_C}$.*

Proof. For density operators ρ and ρ', we have $\sup_{\tau\in Y_C} |r_\rho(\tau) - r_{\rho'}(\tau)| \leq \|T_C(\rho - \rho')\| \leq \|C\| \, \|\rho - \rho'\|_1$. Every response is therefore a uniform, and a fortiori a pointwise, limit of responses r_{ρ_n}; the closure of $\{r_{\rho_n}\}$ thus contains $\mathcal{R}_C$, and with it $\overline{\mathcal{R}_C}$. Sequences such as (ρ_n) exist, because $\mathfrak{T}(H_1)$ is separable. □

Definition 6.4. An observable C that is NIP for Alice's responses is *NIP_1*, *NIP_2* or *NIP_3* if the response family $\mathcal{R}_C$ is, respectively, NIP_1, NIP_2 or NIP_3 (see Definition 4.4).

Proposition 6.5. *Every stable observable is NIP$_3$: its idealized responses form a compact metrizable space.*

Proof. By Theorem 6.1(a), every idealized response is continuous on the compact metrizable space Y_C, so a countable dense set of probes separates them, and Proposition 4.5 applies. □

6.2. **Mixtures force metrizability.** The idealized responses of an observable form a convex set, because responses depend affinely on Alice's state (see the proof of Corollary 6.8). In Theorem 6.6 below, we show that for separable convex Rosenthal compacta hereditary separability implies metrizability. This is a special case of a known result on locally connected Rosenthal compacta (see Remark 6.7); we give a direct proof. The mechanism is visible in the split interval $S(I)$ of Section 4, which is homeomorphic to the set of indicator functions $\mathbf{1}_{(a,1]}$ and $\mathbf{1}_{[a,1]}$ on $[0,1]$, for $0 \le a \le 1$. For each such a, the two indicators form a split pair, two functions that differ only at a, and a convex set of functions that contains both of them contains their midpoint, which takes the value $\frac{1}{2}$ at a. These midpoints form an uncountable discrete set: the midpoint for a point $b \neq a$ takes the value 0 or 1 at a.

Theorem 6.6 (see [Tod99, Theorems 5 and 7]**,** [Gru88, Lemma 2.4] **and Remark 6.7).** *Let Y be a set, and let $K \subseteq \mathbb{R}^Y$ be a separable compact convex set which, with the topology of pointwise convergence, is a Rosenthal compactum. If K is hereditarily separable, then K is metrizable.*

Proof. The idea is the one described before the theorem: if K were hereditarily separable but not metrizable, it would contain a copy of a split interval, and the midpoints of the split pairs of that copy would form a subspace of K that is not separable.

Suppose, then, that K is hereditarily separable but not metrizable. By Fact 4.3(b), there is a homeomorphism h from $\hat{S}_+(2^{\mathbb{N}})$ or $\hat{S}_-(2^{\mathbb{N}})$, the closures of the minimal families D_3 and D_4 of Section 4, respectively, onto a subspace of K. We treat the case $\hat{S} = \hat{S}_+(2^{\mathbb{N}})$. The other case is symmetric: the map c that exchanges the digits 0 and 1 reverses the lexicographic order $\prec$ of $2^{\mathbb{N}}$, and the map $f \mapsto 1 - f \circ c$ carries D_4 to a family of the form of D_3 for the tree $c(\Theta)$, the image under c of the regular dyadic subtree Θ fixed in Section 4. Write $Q = \{s^\frown 0^\infty : s \in \Theta\}$ for the countable set of points indexing D_3, and Σ for the closure of Q in $2^{\mathbb{N}}$. Recall that, for $\sigma \in 2^{\mathbb{N}}$, the functions x^+_σ and x^-_σ are the characteristic functions of $\{\tau \in 2^{\mathbb{N}} : \sigma \preceq \tau\}$ and $\{\tau \in 2^{\mathbb{N}} : \sigma \prec \tau\}$; thus D_3 consists of the functions x^+_q for $q \in Q$.

We first record the structure of $\hat{S}$. If $(\sigma_i)_{i\in\mathbb{N}}$ is a strictly increasing sequence in $2^{\mathbb{N}}$ converging to σ, then $x^\pm_{\sigma_i} \to x^+_\sigma$ pointwise; if it is strictly decreasing, then $x^\pm_{\sigma_i} \to x^-_\sigma$. Indeed, for $\tau \in 2^{\mathbb{N}}$, the value of $x^\pm_{\sigma_i}$ at τ is eventually 1 if $\tau \succ \sigma$, eventually 0 if $\tau \prec \sigma$, and at $\tau = \sigma$ it is eventually 1 in the increasing case and 0 in the decreasing case. Since $\hat{S}$ is Fréchet–Urysohn (every point in the closure of a subset is the limit of a sequence from the subset; see [BFT78, Theorem 3F]) and every sequence of distinct points of $2^{\mathbb{N}}$ has a strictly monotone subsequence, every point of $\hat{S} \setminus D_3$ has the form x^+_ρ or x^-_ρ with $\rho \in \Sigma$; and if $x^{e_i}_{\sigma_i} \in \hat{S}$ for a strictly monotone sequence (σ_i) and signs $e_i \in \{+,-\}$, then the limit computed above lies in $\hat{S}$, because $\hat{S}$ is closed. Note also that $x^+_\sigma = x^-_\rho$ exactly when σ is the immediate successor of ρ in $\prec$, and that two of these functions with the same sign are equal only when their indices are.

We use the following fact repeatedly: if $A \subseteq 2^{\mathbb{N}}$ is uncountable, then all but countably many points of A are limits of strictly increasing and of strictly decreasing sequences from A. Indeed, if $\sigma \in A$ is not a limit of a strictly increasing sequence from A, there is $\tau_\sigma \prec \sigma$ with

no point of A strictly between them, and the intervals $(\tau_\sigma, \sigma]$ for distinct such σ are disjoint; each of them either contains a basic clopen set or makes τ_σ the immediate predecessor of σ. The first alternative occurs for only countably many σ, because the intervals are disjoint and the basic clopen sets are countable, and so does the second, because only countably many points of $2^{\mathbb{N}}$ have an immediate predecessor. The decreasing case is the same. Let P be the set of $\sigma \in \Sigma$ that are limits of strictly increasing and of strictly decreasing sequences from Q. The set Σ is uncountable, because distinct branches of the dyadic tree Θ have distinct limits. By the same argument applied to Q, all but countably many points of Σ belong to P (an open interval that misses Q also misses Σ); hence P is uncountable. For $\sigma \in P$, both x_σ^+ and x_σ^- belong to $\hat{S}$, they differ at σ, and neither equals $x_\rho^\pm$ for $\rho \neq \sigma$, because σ has neither an immediate predecessor nor an immediate successor in $\prec$.

The split pairs indexed by P now supply the midpoints. For $\sigma \in P$, put $F_\sigma^\pm = h(x_\sigma^\pm)$ and $M_\sigma = \frac{1}{2}(F_\sigma^+ + F_\sigma^-)$, which lies in K by convexity. Averaging is continuous for the pointwise topology, and so is h; consequently, if (σ_i) is a strictly increasing (respectively, decreasing) sequence in P converging to σ, then $M_{\sigma_i} \to h(x_\sigma^+)$ (respectively, $h(x_\sigma^-)$). The proof that the midpoints form a non-separable subspace takes three steps.

Step 1: each value is taken by at most countably many midpoints. Let $m \in K$ and suppose that $A = \{\sigma \in P : M_\sigma = m\}$ is uncountable. Choose $\sigma \in A$ that is a limit of strictly increasing and of strictly decreasing sequences from A. Then $h(x_\sigma^+) = m = h(x_\sigma^-)$, and since h is injective, $x_\sigma^+ = x_\sigma^-$, a contradiction.

Step 2: only countably many midpoints lie on the copy of $\hat{S}$. Let $E = \{\sigma \in P : M_\sigma \in h(\hat{S})\}$. By Step 1 and the countability of D_3, all but countably many $\sigma \in E$ satisfy $M_\sigma = h(x_{\rho(\sigma)}^{e(\sigma)})$ for some sign $e(\sigma)$ and some $\rho(\sigma) \in \Sigma$, and $\rho(\sigma) \neq \sigma$, since a midpoint of two distinct points differs from both. Suppose that uncountably many such σ exist. Fix a metric d on $2^{\mathbb{N}}$; then for some positive integer n the set E_n of these σ with $d(\sigma, \rho(\sigma)) \geq 1/n$ is uncountable. Choose $\sigma \in E_n$ and a strictly monotone sequence (σ_i) in E_n converging to σ, say increasing. Passing to a subsequence, we may assume that the signs $e(\sigma_i)$ are constant and that the points $\rho(\sigma_i)$ are either constant or strictly monotone, converging to some ρ^*. Then the midpoints M_{σ_i} converge both to $h(x_\sigma^+)$ and to $h(x_{\rho^*}^\pm)$. Since h is injective, $x_\sigma^+ = x_{\rho^*}^\pm$, and since $\sigma \in P$ has no immediate predecessor, $\rho^* = \sigma$. Thus $d(\sigma_i, \rho(\sigma_i)) \to 0$, contrary to $\sigma_i \in E_n$. Consequently, the set E is countable.

Step 3: the midpoints form a non-separable subspace. Let $P' = P \setminus E$, an uncountable set, and let $L = \{M_\sigma : \sigma \in P'\}$. Let $D \subseteq L$ be countable. By Step 1, there is $\sigma \in P'$ with $M_\sigma \notin D$. If M_σ were in the closure of D, then, since Rosenthal compacta are Fréchet–Urysohn (see [BFT78, Theorem 3F]), there would be a sequence (σ_i) in P' with $M_{\sigma_i} \in D$ and $M_{\sigma_i} \to M_\sigma$. As $M_\sigma \notin D$, no point occurs infinitely often in (σ_i), so a subsequence is strictly monotone and converges to some $\rho \in \Sigma$, and then the midpoints M_{σ_i} converge to $h(x_\rho^+)$ or $h(x_\rho^-)$. Thus $M_\sigma \in h(\hat{S})$, contrary to $\sigma \notin E$. The subspace L is therefore not separable, and K is not hereditarily separable, a contradiction. □

Remark 6.7. The proof uses convexity only to know that midpoints of points of K lie in K. The same midpoints settle a question about the split interval $S = [0,1] \times \{0,1\}$, ordered lexicographically, with the order topology. In a remark following his Proposition 10 [God80], Godefroy states that the space $P(S)$ of Borel probability measures on S, with the weak* topology, is hereditarily separable. This is not the case: for $0 < a < 1$, let $a^- = (a, 0)$ and $a^+ = (a, 1)$, and let m_a be the midpoint $\frac{1}{2}(\delta_{a^-} + \delta_{a^+})$ of the point masses at a^- and a^+. The set $\{s \in S : s \leq a^-\}$ is clopen, so its indicator f_a is continuous, and $m_a(f_a) = \frac{1}{2}$,

while $m_b(f_a) \in \{0,1\}$ for $b \neq a$. Hence the set $\{m_a : 0 < a < 1\}$ is an uncountable discrete subspace of $P(S)$, exactly as in the proof of Theorem 6.6. Todorčević has pointed out to us that Theorem 6.6 is known, and that it holds in the larger class of locally connected Rosenthal compacta, which contains the convex ones: it follows from his Theorems 5 and 7 [Tod99] and from an argument of Gruenhage [Gru88, Lemma 2.4]. The proof above is direct and uses only the midpoints, the same midpoints that refute Godefroy's remark. The theorem answers, for convex compacta, the question of which classes of Todorčević's trichotomy occur, in the spirit of the Glasner–Megrelishvili hierarchy of tame systems [GM22].

Corollary 6.8. *Every observable that is NIP$_2$ is NIP$_3$.*

Proof. Let C be NIP$_2$. We show that the set $\overline{\mathcal{R}_C}$ of idealized responses satisfies the hypotheses of Theorem 6.6. Since the slice map and evaluation at a probe are linear and $\mathfrak{D}(H_1)$ is convex, the identity $r_{t\rho+(1-t)\rho'} = t\, r_\rho + (1-t)\, r_{\rho'}$, for $\rho, \rho' \in \mathfrak{D}(H_1)$ and $0 \le t \le 1$, shows that the family $\mathcal{R}_C$ is convex. For fixed t, the map $(f,g) \mapsto tf + (1-t)g$ is continuous on $\mathbb{R}^{Y_C} \times \mathbb{R}^{Y_C}$ for the pointwise topology and sends $\mathcal{R}_C \times \mathcal{R}_C$ into $\mathcal{R}_C$; it therefore sends $\overline{\mathcal{R}_C} \times \overline{\mathcal{R}_C}$, the closure of $\mathcal{R}_C \times \mathcal{R}_C$, into $\overline{\mathcal{R}_C}$, which is thus convex. By Lemma 6.3 and Theorem 6.1(b), the set $\overline{\mathcal{R}_C}$ is a separable Rosenthal compactum, and it is hereditarily separable because C is NIP$_2$. By Theorem 6.6, it is metrizable, that is, the observable C is NIP$_3$. □

For observables, then, only three of the four classes of Todorčević's trichotomy occur. For families of functions all four occur: the minimal family D_3 of Section 4, formed by the characteristic functions of the sets $\{\tau \in 2^{\mathbb{N}} : \tau \succeq s^\frown 0^\infty\}$ for s in the regular dyadic subtree Θ, consists of continuous functions on the Cantor set, since each of these sets is clopen, and its closure $\hat{S}_+(2^{\mathbb{N}})$ is hereditarily separable but not metrizable, so the family D_3 is NIP$_2$ but not NIP$_3$. This example comes from the lexicographic order of the Cantor set, which is isomorphic to the order that the real line induces on the middle-thirds Cantor set; the members of D_3 are realized types of this order, and they take only the values 0 and 1. Expectations bring in mixtures, and the midpoints of the split pairs turn the split interval into a space with an uncountable discrete subset; for the order of positions, the resulting space is the Helly space, the space of nondecreasing functions from $[0,1]$ to $[0,1]$ that take the value 0 at 0 and 1 at 1 (see Section 7). This is why, for averages over Alice's mixed preparations, hereditary separability forces metrizability (see Theorem 6.6).

Theorem 6.9. *Let C be an observable on $H_1 \otimes H_2$ that is NIP for Alice's responses, and let $(\rho_n)_{n\in\mathbb{N}}$ be a trace-norm dense sequence in $\mathfrak{D}(H_1)$. Then exactly one of the following holds:*

- (i) *the observable C is NIP$_3$;*
- (ii) *the observable C is NIP$_1$ but not NIP$_2$, and there is an increasing injection $\iota\colon 2^{<\mathbb{N}} \to \mathbb{N}$ such that the responses $\{r_{\rho_{\iota(t)}} : t \in 2^{<\mathbb{N}}\}$ are equivalent to D_6 or to D_7;*
- (iii) *the observable C is not NIP$_1$, and there is an increasing injection $\iota\colon 2^{<\mathbb{N}} \to \mathbb{N}$ such that the responses $\{r_{\rho_{\iota(t)}} : t \in 2^{<\mathbb{N}}\}$ are equivalent to D_5.*

Proof. By Lemma 5.2, the family $\{r_{\rho_n}\}$ is a pointwise bounded family of continuous functions on the compact metrizable space Y_C, and it is NIP because it is contained in $\mathcal{R}_C$. By Lemma 6.3, its closure is $\overline{\mathcal{R}_C}$. Hence Theorem 4.7 applies, and its case (ii) is excluded by Corollary 6.8. □

Physically, when the idealized responses of a NIP observable do not form a metrizable space, the reason is always visible in Alice's actual preparations: a family of them, arranged along the dyadic tree, has idealized limits that reproduce one of three canonical compacta, the closures of the minimal families D_5, D_6 and D_7. In the next section we locate the comparisons $[A < B]$ in this classification.

6.3. **Historical remarks.** Kadison proved the function representation used in Theorem 6.1 in 1951 [Kad51]. Grothendieck's theorem [Gro52], on which the characterizations of stability in Part I [DI26] rest, lies behind the characterization of stability by the continuity of the idealized responses in Theorem 6.1(a); the NIP version of this characterization in Theorem 6.1(b), with Baire-1 functions in place of continuous ones, follows our work on deep computations [DIMW$^+$26].

7. Three physical orders realize the three classes of Todorčević's trichotomy that occur for observables

Each of the three classes of Todorčević's trichotomy in Theorem 6.9, we show here, contains a physically natural observable; all three observables are orders, and all three are unstable and NIP. The comparison of two discrete quantities, such as photon numbers, has a metrizable compactum of idealized responses and is thus NIP_3 (see Theorem 7.8). The comparison of two continuous quantities, such as the positions of two particles in a box or two quadratures of light (the two field amplitudes of a mode, which behave like a position and a momentum), has as its compactum of idealized responses a copy of the Helly space, the space of nondecreasing functions from $[0,1]$ to $[0,1]$ that take the value 0 at 0 and 1 at 1; this space is first countable but not hereditarily separable, and the comparison is therefore NIP_1 but not NIP_2 (see Theorem 7.8 and Proposition 7.9). The causal order of a spacetime with one dimension of space, "Bob's detection lies in the causal future of Alice's event", and the comparison of two phases, "Bob's angle is ahead of Alice's by less than half a turn", are NIP, but their compacta of idealized responses are not first countable, and they are therefore not NIP_1 (see Theorems 7.10 and 7.12): the causal order because a light cone can shrink to a point, and the phases because on a circle the jumps at the two ends of a half-turn can cancel.

7.1. **Relations between classical quantities.** All three orders relate a classical quantity measured by Alice to one measured by Bob, and their responses can be computed from classical probability distributions. Let (X_1, m_1) and (X_2, m_2) be σ-finite measure spaces such that the Hilbert spaces $H_1 = L^2(X_1, m_1)$ and $H_2 = L^2(X_2, m_2)$ are separable, fix an integer $p \geq 1$, let $\alpha\colon X_1 \to \mathbb{R}^p$ and $\beta\colon X_2 \to \mathbb{R}$ be measurable, and let $R \subseteq \mathbb{R}^p \times \mathbb{R}$ be a Borel set. The *relation observable* C_R is multiplication on $L^2(X_1 \times X_2, m_1 \otimes m_2) = H_1 \otimes H_2$ by the indicator of $\{(s,t) : (\alpha(s), \beta(t)) \in R\}$; it represents the event "Alice's value and Bob's value are related by R". In this section, the letters ρ and σ denote actual states of Alice's and of Bob's systems, that is, density operators on H_1 and on H_2. For a state ρ of Alice's system, the *distribution* of Alice's value in ρ is the Borel probability measure on $\mathbb{R}^p$ given by $\mu_\rho(U) = \mathrm{Tr}\big(\rho\, M_{\mathbf{1}_{\alpha^{-1}(U)}}\big)$ for Borel sets $U \subseteq \mathbb{R}^p$, where M_f denotes multiplication by f, and the distribution μ_σ of Bob's value in a state σ of Bob's system is defined in the same way. For $y \in \mathbb{R}$, let $R^y = \{x \in \mathbb{R}^p : (x,y) \in R\}$.

By the spectral theorem, every self-adjoint operator on a separable Hilbert space is unitarily equivalent to multiplication by a real measurable function on a space $L^2(X, m)$ of this kind. Thus, for self-adjoint operators A on H_1 and B on H_2, realized as multiplication by α and by β, the *comparison* $[A < B]$ is the relation observable C_R for $R = \{(x,y) : x < y\}$; it is the

projection $(E_A \otimes E_B)(R)$, where E_A and E_B are the spectral measures, so it does not depend on the realization. When A and B are bounded, the operator $I \otimes B - A \otimes I$ is multiplication by $\beta(t) - \alpha(s)$, and $[A < B]$ is its spectral projection for the interval $(0, \infty)$, or equivalently the spectral projection of $A \otimes I - I \otimes B$ for the interval $(-\infty, 0)$.

Lemma 7.1. *Let C_R be a relation observable. For states ρ of Alice's system and σ of Bob's, the expectation value of C_R when the two systems are prepared independently in these states is*

$$\varphi_{C_R}(\rho, \sigma) = (\mu_\rho \otimes \mu_\sigma)(R) = \int g_\rho \, d\mu_\sigma, \qquad g_\rho(y) = \mu_\rho(R^y),$$

and the effective observable $T_{C_R}(\rho)$, the observable on Bob's system obtained by fixing Alice's state ρ, is multiplication by $g_\rho \circ \beta$.

Proof. For unit vectors $\psi \in H_1$ and $\chi \in H_2$, with distributions μ_ψ and μ_χ in the corresponding pure states, Tonelli's theorem and a change of variables give

$$\langle \psi \otimes \chi, C_R(\psi \otimes \chi) \rangle = \iint \mathbf{1}_R\big(\alpha(s), \beta(t)\big) \, |\psi(s)|^2 |\chi(t)|^2 \, dm_1(s) \, dm_2(t) = (\mu_\psi \otimes \mu_\chi)(R).$$

A density operator is a countable convex combination of pure states, $\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|$, the series converging in trace norm; here j ranges over a countable set, the weights $p_j \geq 0$ have sum 1, and $|\psi_j\rangle\langle\psi_j|$ is the orthogonal projection onto the line spanned by a unit vector $\psi_j \in H_1$. Accordingly, the distribution of Alice's value in ρ is $\mu_\rho = \sum_j p_j \mu_{\psi_j}$. Both sides of the first identity are affine in each variable and preserve such combinations, so the first identity holds for all states; the second is Tonelli's theorem, and g_ρ is measurable by the same theorem. Finally, for every unit vector $\chi \in H_2$,

$$\langle \chi, T_{C_R}(\rho) \chi \rangle = \varphi_{C_R}(\rho, |\chi\rangle\langle\chi|) = \int g_\rho\big(\beta(t)\big) |\chi(t)|^2 \, dm_2(t),$$

and a bounded operator on a complex Hilbert space is determined by these values. □

In Lemma 7.2 and Proposition 7.3 below, we show that when the functions g_ρ are monotone, or differences of monotone functions, the responses cannot shatter large sets. Recall from Remark 6.2 that a finite set of responses is δ-shattered if every pattern on it is realized with reals $a < b$ such that $b - a \geq \delta$.

Lemma 7.2. *Let $\delta > 0$ and $m = \lceil 9/\delta \rceil$. Let $h_1, \dots, h_n \colon \mathbb{R} \to \mathbb{R}$ be functions of the form $h_i = u_i - v_i$, where $u_i, v_i \colon \mathbb{R} \to [0,1]$ are nondecreasing, and let $a < b$ be reals with $b - a \geq \delta$. Suppose that for every $S \subseteq \{1, \dots, n\}$ there is a Borel probability measure ν_S on $\mathbb{R}$ with $\int h_i \, d\nu_S \geq b$ for $i \in S$ and $\int h_i \, d\nu_S \leq a$ for $i \notin S$. Then $2^n \leq (2nm + 1)^m$. In particular, $n \leq N(\delta)$, where $N(\delta)$, the largest integer n satisfying this inequality, is finite and depends only on δ.*

Proof. We first approximate a measure by m equally weighted points. Let ν be a Borel probability measure on $\mathbb{R}$, with distribution function $F(t) = \nu((-\infty, t])$. For $k = 1, \dots, m$, let $y_k = \min\{t : F(t) \geq (k - \frac{1}{2})/m\}$; the minimum exists because F is nondecreasing and right continuous, with limits 0 and 1 at $-\infty$ and $+\infty$, and $0 < (k - \frac{1}{2})/m < 1$. Let $\nu' = \frac{1}{m} \sum_{k=1}^m \delta_{y_k}$ be the average of the point masses δ_{y_k} at the points y_k. Since $y_k \leq t$ exactly when $F(t) \geq (k - \frac{1}{2})/m$, we have $\nu'((-\infty, t]) = \lfloor mF(t) + \frac{1}{2} \rfloor / m$, whence $|\nu((-\infty, t]) - \nu'((-\infty, t])| \leq 1/m$ for every t; by letting s increase to t, we obtain the same bound for the half-lines $(-\infty, t)$. Hence $|\nu(J) - \nu'(J)| \leq 1/m$ for every set J of the form $\emptyset$, $\mathbb{R}$, $[t, \infty)$ or (t, ∞). For a nondecreasing

$u\colon \mathbb{R} \to [0,1]$, every set $\{u > s\}$ has this form, and $\int u\,d\nu = \int_0^1 \nu(\{u > s\})\,ds$; therefore

$$\left|\int u\,d\nu - \int u\,d\nu'\right| \le \frac{1}{m}. \tag{1}$$

Next we discretize the functions. Let $\tilde{u}_i = \lfloor m u_i \rfloor / m$ and $\tilde{v}_i = \lfloor m v_i \rfloor / m$. These functions are nondecreasing, take values in $\{0, \frac{1}{m}, \dots, 1\}$, and differ from u_i and v_i by at most $\frac{1}{m}$. Each of the $2n$ functions $\tilde{u}_i, \tilde{v}_i$ changes value at most m times along $\mathbb{R}$, so the map $Q(y) = (\tilde{u}_1(y), \dots, \tilde{u}_n(y), \tilde{v}_1(y), \dots, \tilde{v}_n(y))$ takes at most $2nm+1$ values.

For each S, let $y_1^S, \dots, y_m^S$ be the points constructed from ν_S as above, and let $w(S) = (Q(y_1^S), \dots, Q(y_m^S))$. By (1), applied to u_i and to v_i, and by the bound on the discretization,

$$\left|\int h_i\,d\nu_S - \frac{1}{m}\sum_{k=1}^{m}\big(\tilde{u}_i(y_k^S) - \tilde{v}_i(y_k^S)\big)\right| \le \frac{4}{m} \le \frac{4\delta}{9},$$

and the average on the left is determined by $w(S)$. If $S \neq S'$, choose i in one of the two sets but not the other, say $i \in S \setminus S'$. Then $\int h_i\,d\nu_S - \int h_i\,d\nu_{S'} \ge \delta > \frac{8\delta}{9}$, so the two averages differ, and $w(S) \neq w(S')$. Thus the map $S \mapsto w(S)$ is injective, and $2^n \le (2nm+1)^m$. □

Proposition 7.3. *Let C_R be a relation observable and $c \ge 1$ a constant such that, for every state ρ of Alice's system, $g_\rho = c\,(u_\rho - v_\rho)$ for two nondecreasing functions $u_\rho, v_\rho\colon \mathbb{R} \to [0,1]$. Then, for every $\delta > 0$, no set of more than $N(\delta/(2c))$ responses of Alice's states is δ-shattered by points of the probe space. Hence the observable C_R is uniformly NIP, with bounds that do not depend on the Hilbert spaces, and in particular it is NIP.*

Proof. Suppose that the responses $r_{\rho_1}, \dots, r_{\rho_n}$ are δ-shattered with reals $a < b$, the pattern S being realized by a probe τ_S in the probe space Y_{C_R} of C_R. Let $a' = a + \delta/4$ and $b' = b - \delta/4$. The set of probes τ with $r_{\rho_i}(\tau) > b'$ for $i \in S$ and $r_{\rho_i}(\tau) < a'$ for $i \notin S$ is open, by Lemma 5.2(b), and contains τ_S, so, by Lemma 5.2(c), it contains the restriction of an actual state σ_S of Bob's system. By Lemma 7.1, the value of r_{ρ_i} there is $\int g_{\rho_i}\,d\mu_{\sigma_S}$, and Lemma 7.2, applied with $h_i = g_{\rho_i}/c = u_{\rho_i} - v_{\rho_i}$, $\nu_S = \mu_{\sigma_S}$ and the reals $a'/c < b'/c$, yields $n \le N(\delta/(2c))$. An infinite independent sequence with reals $a < b$ would contain $(b-a)$-shattered sets of every finite size, so the observable C_R is NIP. □

Corollary 7.4. *For all self-adjoint operators A on H_1 and B on H_2, the comparison $[A < B]$ is uniformly NIP, with bounds that depend only on the margin, a lower bound for the gap $b - a$ in Definition 3.3, and in particular it is NIP.*

Proof. Here $R^y = (-\infty, y)$, so the function $g_\rho(y) = \mu_\rho((-\infty, y))$ is nondecreasing, with values in $[0,1]$. Proposition 7.3 therefore applies, with $c = 1$, $u_\rho = g_\rho$ and $v_\rho = 0$. □

Remark 7.5 (Commuting observables and the general framework). In the orders of this section, as in the label-controlled observables of Section 8, in which a label held by Alice selects the test applied to Bob's system (see Definition 8.1), the probe algebra is commutative, and the responses are integrals against classical probability distributions. Let us notice that this does not reduce the paper to classical probability, for four reasons. First, the definitions of Section 5 apply, and Theorems 6.1, 6.9 and 6.6 hold, for every observable on $H_1 \otimes H_2$, with no commutativity assumption, including observables such as the swap test, whose probe algebra contains all compact operators. The examples show that every class of the classification is inhabited, and the fact that commuting observables already suffice is information in itself, just as in model theory the dividing lines are first exhibited by the simplest structures, such as the random graph and the dense linear order. Second, the framework imposes restrictions that

hold for all observables: the idealized responses always form a compact convex set, because they are the affine image of the state space of $B(H_1)$ (see Lemma 5.4), and by Corollary 6.8 convexity alone removes a class of Todorčević's trichotomy. Third, noncommutativity enters where the physics requires it: in Section 9 the responses are magnetizations of the Ising model in a transverse field, a Hamiltonian whose terms do not commute, and Theorem 9.6 shows that the classification detects its phase transition. Finally, whenever an observable is a relation between commuting quantities measured by Alice and by Bob, Lemma 7.1 reduces its classification to classical probability, and Proposition 7.3 shows how that reduction yields bounds that do not depend on the Hilbert spaces. In Question (3) we ask for phenomena that noncommutative probe algebras produce and commuting observables do not.

7.2. **Order observables.** We now fix two model comparisons, one of discrete and one of continuous quantities, whose values lie in a compact set $\Xi \subseteq [0,1]$ containing 0 and 1. The two cases are the following:

- *discrete*: the set Ξ is $\{1 - \frac{1}{l+1} : l \in \mathbb{N}\} \cup \{1\}$, the Hilbert spaces are $H_1 = H_2 = \ell^2(\mathbb{N})$, and the operator $A = B$ is the diagonal operator with eigenvalue $1 - \frac{1}{l+1}$ on the l-th basis vector; this operator is a bounded relabeling of the number operator, which counts photons and has eigenvalue l on the l-th basis vector, and the order of its eigenvalues is that of the photon numbers;
- *continuous*: the set Ξ is $[0,1]$, the Hilbert spaces are $H_1 = H_2 = L^2[0,1]$, and the operator $A = B$ is multiplication by x, the position of a particle confined to the interval $[0,1]$.

In both cases, let $C = [A < B]$ be the spectral projection of $I \otimes B - A \otimes I$ for the interval $(0,\infty)$, which represents the event "Bob's value exceeds Alice's". For a state ρ, let μ_ρ be the distribution of A in ρ, a Borel probability measure on Ξ. In the discrete case every probability measure on $\Xi \setminus \{1\}$ occurs, and in the continuous case the measures that occur are exactly the absolutely continuous ones (for a probability density p, take the pure state given by the unit vector $\psi = \sqrt{p}$). By Lemma 7.1, the expectation value of C for Alice's state ρ and Bob's state σ is

$$\varphi_C(\rho,\sigma) = (\mu_\rho \otimes \mu_\sigma)\big(\{(x,y) : x < y\}\big) = \int g_\rho \, d\mu_\sigma, \qquad g_\rho(y) = \mu_\rho\big([0,y)\big).$$

The function $g_\rho \colon \Xi \to [0,1]$ is nondecreasing, with $g_\rho(0) = 0$ and $g_\rho(1) = 1$ in both cases, since μ_ρ assigns no mass to the point 1; in the continuous case it is also continuous. Accordingly, the effective observable is $T_C(\rho) = g_\rho(B)$. By Corollary 7.4, the observable C is uniformly NIP, and in particular NIP. It is unstable: in the discrete case by the interleaving argument of Part I [DI26, Proposition 4.6(3)], which uses only the order of the eigenvalues, and in the continuous case by the same ladder argument, with Alice's positions a_k and Bob's positions b_l, for $k, l \geq 1$, interleaved as $a_1 < b_1 < a_2 < b_2 < \cdots$.

7.3. **The probe space and the idealized responses.** For both comparisons of the previous subsection, the discrete and the continuous one, and for every state ρ of Alice's system, the function g_ρ, whose value at a point $y \in \Xi$ is the probability that Alice's value lies below y, is nondecreasing. In Proposition 7.7 below, we show that the pointwise limits of these functions describe all idealized responses. Let M_Ξ be the set of nondecreasing functions $g \colon \Xi \to [0,1]$ with $g(0) = 0$ and $g(1) = 1$, with the topology of pointwise convergence. It is a closed subset of $[0,1]^\Xi$, hence compact. Although M_Ξ contains discontinuous functions, integration against a fixed measure is continuous on it, by the following lemma.

Lemma 7.6. *Let ν be a Borel probability measure on Ξ, and let (g_i) be a net in M_Ξ, indexed by a directed set, that converges pointwise to a function $g \in M_\Xi$. Then $\int g_i \, d\nu \to \int g \, d\nu$.*

Proof. Let $\varepsilon > 0$ be given. Since ν has only finitely many atoms of mass at least ε, there are points $0 = p_0 < p_1 < \cdots < p_m = 1$ of Ξ such that each open cell $c_k = \Xi \cap (p_{k-1}, p_k)$ has $\nu(c_k) < \varepsilon$. Put $\eta_i = \max_k |g_i(p_k) - g(p_k)|$, which tends to 0. On c_k both g_i and g take values between their values at p_{k-1} and at p_k, because they are nondecreasing. Hence

$$\Big|\int_{c_k} g_i \, d\nu - \int_{c_k} g \, d\nu\Big| \le \nu(c_k)\big(g(p_k) - g(p_{k-1}) + 2\eta_i\big),$$

and summing over k, together with the contribution of the points p_k, shows that $|\int g_i \, d\nu - \int g \, d\nu| \le \varepsilon + 3\eta_i$, since the differences $g(p_k) - g(p_{k-1})$ add up to at most 1. □

Proposition 7.7. *In both cases, the probe algebra $\mathfrak{A}_C$, the unital C^*-algebra generated by the effective observables, is $\{f(B) : f \in C(\Xi)\}$, so the probe space Y_C, its state space, is the space $P(\Xi)$ of Borel probability measures on Ξ with the weak* topology. The map $\Phi\colon r \mapsto \big(r(\delta_y)\big)_{y\in\Xi}$, where δ_y is the point mass at y, is a homeomorphism from the compactum $\overline{\mathcal{R}_C}$ of idealized responses onto M_Ξ, and every idealized response is given by $r(\nu) = \int \Phi(r) \, d\nu$.*

Proof. The functions g_ρ for states ρ whose distribution has full support in Ξ separate the points of Ξ, and the map $f \mapsto f(B)$ is an isometric $*$-isomorphism from $C(\Xi)$ onto its image, because the spectrum of B is Ξ. By the Stone–Weierstrass theorem, the unital C*-algebra generated by the operators $g_\rho(B)$ is $\{f(B) : f \in C(\Xi)\}$, and the slice map takes values in it. By the Riesz representation theorem, its state space is $P(\Xi)$, and the response of ρ is $r_\rho(\nu) = \int g_\rho \, d\nu$.

The map Φ is continuous, because its coordinates are evaluations. Let $r \in \overline{\mathcal{R}_C}$ be the pointwise limit of a net (r_{ρ_i}). Then the functions $g_{\rho_i} = \Phi(r_{\rho_i})$ converge pointwise to $g = \Phi(r) \in M_\Xi$, and, by Lemma 7.6, $r(\nu) = \lim_i \int g_{\rho_i} \, d\nu = \int g \, d\nu$. Thus the response r is determined by $\Phi(r)$, the map Φ is injective, and, as a continuous injection from a compact space, it is a homeomorphism onto its image. The image is compact and contains the functions g_ρ.

It remains to be shown that the functions g_ρ are dense in M_Ξ. Let $g \in M_\Xi$ and a finite set $F \subseteq \Xi$ containing 0 and 1 be given. In the continuous case, the piecewise linear interpolation $\tilde{g}$ of g on F is continuous and nondecreasing, with values 0 at 0 and 1 at 1; hence $\tilde{g} = g_\rho$ for a state ρ whose distribution has $\tilde{g}$ as its distribution function, which is absolutely continuous, and $\tilde{g}$ agrees with g on F. In the discrete case, write $F = \{\xi_0 < \xi_1 < \cdots < \xi_m\}$, with $\xi_0 = 0$ and $\xi_m = 1$. The probability measure with mass $g(\xi_{j+1}) - g(\xi_j)$ at ξ_j for $0 \le j < m$ is the distribution of a state ρ with finitely many photons, and $g_\rho(\xi_j) = g(\xi_j) - g(\xi_0) = g(\xi_j)$ for every j, so the function g_ρ agrees with g on F. Every neighborhood of g therefore contains some g_ρ. □

7.4. **Location in the classification.** We can now place both comparisons in the classification of Theorem 6.9.

Theorem 7.8. *Let $C = [A < B]$ be the comparison of Alice's and Bob's photon numbers, in the discrete case, or of the positions of their particles in a box, in the continuous case.*

(a) *In the discrete case, the observable C is unstable and NIP$_3$.*

(b) *In the continuous case, the observable C is unstable and NIP$_1$, but not NIP$_2$. Hence, for every trace-norm dense sequence $(\rho_n)_{n\in\mathbb{N}}$ of states of Alice's particle, there is an increasing injection $\iota\colon 2^{<\mathbb{N}} \to \mathbb{N}$ such that the responses $\{r_{\rho_{\iota(t)}} : t \in 2^{<\mathbb{N}}\}$ are equivalent to D_6 or to D_7, the two minimal families of the heptachotomy whose closures are first countable but not hereditarily separable.*

Proof. (a) By Proposition 7.7, the compactum of idealized responses is homeomorphic to a subspace of $[0,1]^\Xi$ with Ξ countable, which is metrizable.

(b) By Proposition 7.7, the compactum of idealized responses is homeomorphic to $M_{[0,1]}$, a copy of the Helly space. It is first countable. Given $g \in M_{[0,1]}$, let D_g be the countable set consisting of the rational points, the jump points of g and the endpoints; we claim that the sets $\{h : |h(p) - g(p)| < 1/n \text{ for } p \in F\}$, with $F \subseteq D_g$ finite and $n \geq 1$, form a neighborhood base at g. It suffices to show that each set $\{h : |h(y) - g(y)| < \varepsilon\}$, with $y \in [0,1]$ and $\varepsilon > 0$, contains one of them. If $y \in D_g$, this is clear. If $y \notin D_g$, then g is continuous at y, and there are rationals $p < y < q$ with $g(q) - g(p) < \varepsilon/2$. Every $h \in M_{[0,1]}$ with $|h - g| < \varepsilon/4$ at p and q satisfies $h(p) \leq h(y) \leq h(q)$, because h is nondecreasing, and hence $|h(y) - g(y)| < \varepsilon$.

The compactum is not hereditarily separable. For $0 < a < 1$, let h_a be 0 on $[0,a)$, $\frac{1}{2}$ at a and 1 on $(a,1]$. The set $\{h \in M_{[0,1]} : \frac{1}{4} < h(a) < \frac{3}{4}\}$ is a neighborhood of h_a that contains no other h_b, so the set $\{h_a : 0 < a < 1\}$ is an uncountable discrete subspace.

The observable is therefore NIP$_1$ but not NIP$_2$, and the last statement is Theorem 6.9(ii). □

The idealized responses of the continuous comparison have a direct physical reading, which is also a probabilistic one. Alice's idealized preparations are described by arbitrary functions g in $M_{[0,1]}$, where $g(y)$ is the probability that Alice's particle lies to the left of y, and the value of g at a jump records how the particle approaches the jump point: from the left, from the right, or both. The function h_a describes a half-and-half mixture of a particle concentrating at a from the left and one concentrating at a from the right, and Bob's idealized probe concentrated at a reads the value $\frac{1}{2}$. In the discrete case no such splitting is possible, because the eigenvalues are isolated, and the only idealization is the escape to infinity.

7.5. **Unconfined quantities.** Confinement to a box is not what produces the Helly space: a comparison depends only on the order of the two outcomes, so it is unchanged when both quantities are replaced by the same increasing function of themselves, and a suitable function turns unconfined quantities into positions in a box (see Proposition 7.9 below).

Proposition 7.9. *Let A and B be self-adjoint operators, each unitarily equivalent to multiplication by x on $L^2(\mathbb{R})$; for instance, the positions of two particles on a line, or quadratures of two modes of light. Then the comparison $[A < B]$ is unstable, and its idealized responses form a copy of the Helly space of nondecreasing functions $[0,1] \to [0,1]$ taking the value 0 at 0 and 1 at 1. Hence the comparison is NIP$_1$ but not NIP$_2$.*

Proof. We may assume that A and B are multiplication by x on $L^2(\mathbb{R})$. Let $\theta\colon \mathbb{R} \to (0,1)$ be the increasing diffeomorphism $\theta(x) = \frac{1}{2}(1 + \tanh x)$. The operator U given by $(U\psi)(u) = \psi(\theta^{-1}(u))\,|(\theta^{-1})'(u)|^{1/2}$ is a unitary from $L^2(\mathbb{R})$ onto $L^2[0,1]$ that carries multiplication by $\theta(x)$ to multiplication by u. Since θ is strictly increasing, $x < y$ exactly when $\theta(x) <$

$\theta(y)$; hence the spectral projection $[A < B]$ equals $[\theta(A) < \theta(B)]$, and $U \otimes U$ carries it to the comparison of the continuous case. The unitary $U \otimes U$ preserves expectation values and carries actual and idealized states to actual and idealized states; it therefore induces a homeomorphism between the two compacta of idealized responses. The statement now follows from Proposition 7.7 and Theorem 7.8(b). □

What matters, then, is that the quantities are continuous, not that they are confined. For quantities whose expectation values matter, such as energies, the rescaling by θ is not available, because it is not linear (see the discussion of the logical setting in the introduction).

7.6. **The causal order is not first countable.** The compactum of idealized responses of a comparison is first countable because its members are monotone, and a monotone function has at most countably many jumps, each with two sides (see the proof of Theorem 7.8(b)). A relation between Bob's value and two values of Alice's, one on each side, removes this restriction (see Theorem 7.10 below), and the natural example is the causal order of spacetime. We work in a $(1+1)$-dimensional spacetime, with one dimension of time and one of space, in units in which the speed of light is 1. An event (t, ξ), at time t and position ξ, can influence a detection by Bob at the point y at time 0 exactly when $|y - \xi| \le -t$, that is, when y lies between the points $a = \xi + t$ and $b = \xi - t$ at which the two light rays through the event cross Bob's time slice. We describe Alice's event by these null coordinates and confine everything to the causal diamond over Bob's interval $[0, 1]$, the set of events whose two light rays cross Bob's slice inside that interval: Alice's Hilbert space is $H_1 = L^2([0,1]^2)$, with variables (a, b), Bob's is $H_2 = L^2[0, 1]$, with variable y, and the *causal-order observable* C is multiplication by the indicator of

$$R = \{(a, b, y) : a \le y \le b\}.$$

Events with $a > b$ lie in the future of Bob's slice and influence none of his detections.

For a state ρ of Alice's system, let μ_ρ be the distribution of (a, b) in ρ, an absolutely continuous probability measure on $[0,1]^2$, and similarly for Bob. The observable C is the relation observable C_R for this relation, so, by Lemma 7.1, its expectation value in states ρ and σ of the two systems is $\varphi_C(\rho, \sigma) = (\mu_\rho \otimes \mu_\sigma)(R) = \int g_\rho \, d\mu_\sigma$, where $g_\rho(y) = \mu_\rho(\{(a, b) : a \le y \le b\})$ is the probability that Alice's event can influence a detection at y. The function g_ρ is continuous: the sections of R at y and at y' differ by a set contained in two strips, each of area at most $|y - y'|$, and the μ_ρ-measure of such strips tends to 0 with $|y - y'|$, because μ_ρ is absolutely continuous. The effective observable $T_C(\rho)$ is multiplication by g_ρ.

Theorem 7.10. *The causal-order observable is NIP but not NIP$_1$. Hence, for every trace-norm dense sequence $(\rho_n)_{n\in\mathbb{N}}$ of states of Alice's system, there is an increasing injection $\iota\colon 2^{<\mathbb{N}} \to \mathbb{N}$ such that the responses $\{r_{\rho_{\iota(t)}} : t \in 2^{<\mathbb{N}}\}$ are equivalent to D_5, the minimal family of the heptachotomy whose closure, $\hat{A}(2^{\mathbb{N}})$, is not first countable.*

Proof. For every state ρ, the event $\{a \le y \le b\}$ is the event $\{a \le y\}$ with the event $\{a \le y,\ b < y\}$ removed, so

$$g_\rho(y) = \mu_\rho\big(\{(a, b) : a \le y\}\big) - \mu_\rho\big(\{(a, b) : a \le y,\ b < y\}\big),$$

a difference of two nondecreasing functions with values in $[0, 1]$. By Proposition 7.3, whose constant is here 1, the observable C is uniformly NIP, and in particular NIP. By Lemma 6.3 the compactum of idealized responses is separable, so its level is defined.

The probe algebra consists of operators M_f of multiplication by continuous functions f; it is therefore a C*-subalgebra of $\{M_f : f \in C[0,1]\}$, which is isometrically isomorphic to $C[0,1]$, because $\|M_f\| = \sup|f|$. Every state of $\mathfrak{A}_C$ extends to a state of this larger algebra (see [KR83, Theorem 4.3.13]), and, by the Riesz representation theorem, every state τ of $\mathfrak{A}_C$ is thus represented by a probability measure ν on $[0,1]$, in the sense that $\tau(M_f) = \int f\,d\nu$ for $M_f \in \mathfrak{A}_C$; in particular, $r_\rho(\tau) = \int g_\rho\,d\nu$. For each point $c \in (0,1)$, the evaluation $\tau_c(M_f) = f(c)$ is a state of $\mathfrak{A}_C$, Bob's probe localized at c.

Let ρ_0 be a state whose distribution is uniform on $[\frac{1}{2},1] \times [0,\frac{1}{2}]$, an event after Bob's slice. Then $g_{\rho_0} = 0$, and the zero function is a response. For $c \in (0,1)$ and large n, let $\rho_{c,n}$ be a state whose distribution is uniform on $[c-\frac{2}{n}, c-\frac{1}{n}] \times [c+\frac{1}{n}, c+\frac{2}{n}]$, an event at position near c shortly before Bob's slice. Then $g_{\rho_{c,n}}(c) = 1$, and $g_{\rho_{c,n}}(y) = 0$ when $|y-c| > \frac{2}{n}$. Hence the function $g_{\rho_{c,n}}$ converges pointwise to the indicator of $\{c\}$, and, by dominated convergence, $r_{\rho_{c,n}}(\tau) \to \nu(\{c\})$ for every probe τ and every measure ν representing it. The limit r_c is an idealized response, with $r_c(\tau_c) = 1$.

Suppose that the zero function had a countable base of neighborhoods (U_k) in the compactum of idealized responses. Each U_k contains a set $\{r : |r(\tau)| < \varepsilon_k \text{ for } \tau \in F_k\}$ with F_k finite. Choose a representing measure for each probe in $F = \bigcup_k F_k$; together these measures have countably many atoms, so some $c \in (0,1)$ is an atom of none of them. Then $r_c(\tau) = 0$ for $\tau \in F$, so the response r_c lies in every U_k. Since the space is Hausdorff, the intersection of a neighborhood base at a point is that point; hence $r_c = 0$, contrary to $r_c(\tau_c) = 1$. The compactum is therefore not first countable, and the last statement is Theorem 6.9(iii). □

Physically, the proof says the following. The idealized response r_c is that of an event on Bob's time slice at the point c: its light cone meets the slice in the single point c, so only probes that assign positive weight to the point c itself detect it. There is one such event for each point c, and countably many of Bob's probes cannot tell all of them apart from an event that influences nothing; this is the failure of first countability. The relation R is the intersection of two comparisons, $a \le y$ and $y \le b$, one for each light ray. A single comparison has monotone idealized responses; the two rays together allow idealized responses supported on a single point. Here Todorčević's trichotomy is used in full: it turns the failure of first countability into a dyadic tree of actual events whose responses are equivalent to the canonical family D_5. The proof suggests a tree of this kind, namely events whose light cones meet Bob's slice in the intervals of a Cantor construction, each emitted later than its predecessor, although we do not claim that the trichotomy produces this one.

7.7. **The comparison of phases is not first countable.** Orders on a line give first countable compacta of idealized responses (see Theorem 7.8 and Proposition 7.9). In Theorem 7.12 below, we show that on a circle first countability can fail. Identify the circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ with $[0,1)$. Alice and Bob each hold a particle on a ring, a planar rotor, with Hilbert spaces $H_1 = H_2 = L^2(\mathbb{T})$, the angle acting by multiplication. The *phase comparison* is the relation observable $C = C_R$ for

$$R = \big\{(x,y) \in [0,1)^2 :\ y - x \in (0,\tfrac{1}{2}) \bmod 1\big\},$$

the event "Bob's angle is ahead of Alice's by less than half a turn". For a state ρ of Alice's rotor, the distribution μ_ρ of her angle is an absolutely continuous probability measure on $\mathbb{T}$, and every such measure occurs. By Lemma 7.1, the phase comparison has, in states ρ and σ of the two rotors, the expectation value $\varphi_C(\rho,\sigma) = \int g_\rho\,d\mu_\sigma$, where $g_\rho(y) = \mu_\rho\big((y-\frac{1}{2}, y)\big)$ is the probability that Alice's angle lies in the open half-circle ending at y; the function g_ρ is continuous on $\mathbb{T}$, because μ_ρ has no atoms.

Lemma 7.11. *For every probability measure μ on $\mathbb{T} = [0,1)$, the function $g(y) = \mu((y-\frac{1}{2}, y))$ on $[0,1)$ is $2(u-v)$ for two nondecreasing functions $u, v\colon [0,1) \to [0,1]$.*

Proof. Let $G(y) = \mu([0,y))$. For $y \in [\frac{1}{2}, 1)$ the half-circle is the interval $(y-\frac{1}{2}, y)$, and $g(y) = G(y) - \mu([0, y-\frac{1}{2}])$; for $y \in [0, \frac{1}{2})$ it is $(y+\frac{1}{2}, 1) \cup [0, y)$, and $g(y) = G(y) + 1 - \mu([0, y+\frac{1}{2}])$. Let $V(y) = \mu([0, y+\frac{1}{2}]) - 1$ for $0 \le y < \frac{1}{2}$ and $V(y) = \mu([0, y-\frac{1}{2}])$ for $\frac{1}{2} \le y < 1$. Then $g = G - V$. The functions G and V are nondecreasing, with values in $[0,1]$ and $[-1,1]$, respectively; indeed, V is nondecreasing on each half, and at $\frac{1}{2}$ it passes from $\lim_{y\uparrow 1/2} V(y) = 0$ to $V(\frac{1}{2}) = \mu(\{0\}) \ge 0$. Put $u = \frac{1}{2}(G+1)$ and $v = \frac{1}{2}(V+1)$; then $g = 2(u-v)$. □

Theorem 7.12. *The phase comparison is uniformly NIP, with bounds that depend only on the margin, and it is not NIP$_1$: its compactum of idealized responses is not first countable. Hence, for every trace-norm dense sequence $(\rho_n)_{n\in\mathbb{N}}$ of states of Alice's rotor, there is an increasing injection $\iota\colon 2^{<\mathbb{N}} \to \mathbb{N}$ such that the responses $\{r_{\rho_{\iota(t)}} : t \in 2^{<\mathbb{N}}\}$ are equivalent to D_5.*

Proof. By Lemma 7.11, with u and v extended constantly outside $[0,1)$, Proposition 7.3 applies with the constant 2, and the observable is uniformly NIP.

As in the proof of Theorem 7.10, every probe τ is represented by a probability measure ν on $\mathbb{T}$, with $r_\rho(\tau) = \int g_\rho \, d\nu$, and for $z \in \mathbb{T}$ the evaluation $\tau_z(M_f) = f(z)$ is a probe, Bob's rotor localized at z. Let ρ_0 be the uniform state. Every open half-circle has measure $\frac{1}{2}$, so $r_{\rho_0}(\tau) = \frac{1}{2}$ for every probe τ.

For $a \in \mathbb{T}$ and large n, let $\rho_{a,n}$ be a state whose distribution is uniform, with weight $\frac{1}{2}$ each, on the arcs $B_1 = (a - \frac{1}{n}, a)$ and $B_2 = (a + \frac{1}{2}, a + \frac{1}{2} + \frac{1}{n})$: half of the angle just behind a, half just ahead of the antipode. Write $A_y = (y - \frac{1}{2}, y)$. If $y = a$, then the half-circle A_y contains B_1 and B_2, so $g_{\rho_{a,n}}(a) = 1$. If $y = a + \frac{1}{2}$, then the half-circle $A_y = (a, a + \frac{1}{2})$ meets neither, and $g_{\rho_{a,n}}(a + \frac{1}{2}) = 0$. Otherwise the endpoints of A_y are at some positive distance d from a and from $a + \frac{1}{2}$; an open half-circle whose endpoints differ from these two antipodal points contains exactly one of them. Once $\frac{1}{n} < d$, the half-circle A_y therefore contains the arc B_1 just behind a if it contains a, and the arc B_2 just ahead of $a + \frac{1}{2}$ if it contains $a + \frac{1}{2}$, but not both, and $g_{\rho_{a,n}}(y) = \frac{1}{2}$. Hence the function $g_{\rho_{a,n}}$ converges pointwise to $g_a = \frac{1}{2} + \frac{1}{2}\mathbf{1}_{\{a\}} - \frac{1}{2}\mathbf{1}_{\{a+1/2\}}$, and, by dominated convergence, $r_{\rho_{a,n}}(\tau) \to \frac{1}{2} + \frac{1}{2}\nu(\{a\}) - \frac{1}{2}\nu(\{a + \frac{1}{2}\})$ for every probe τ and every measure ν representing it. The limit r_a is an idealized response, with $r_a(\tau_a) = 1$.

Suppose that r_{ρ_0} had a countable base of neighborhoods (U_k) in the compactum of idealized responses. Each U_k contains a set $\{r : |r(\tau) - \frac{1}{2}| < \varepsilon_k \text{ for } \tau \in F_k\}$ with F_k finite. The measures representing the probes in $F = \bigcup_k F_k$ have countably many atoms, so there is $a \in \mathbb{T}$ such that neither a nor $a + \frac{1}{2}$ is one of them. Then $r_a(\tau) = \frac{1}{2}$ for every $\tau \in F$, so r_a lies in every U_k, and, the space being Hausdorff, the response r_a equals r_{ρ_0}, contrary to $r_a(\tau_a) = 1$. The compactum is therefore not first countable, and the last statement is Theorem 6.9(iii). □

Here, too, the proof has a physical reading. The uniform state responds $\frac{1}{2}$ to every probe. For an antipodal pair of points a and $a + \frac{1}{2}$, the idealized limit of preparations that put half of the angle's distribution just behind a and half just ahead of $a + \frac{1}{2}$ also responds $\frac{1}{2}$, except that a probe at a reads 1 and a probe at the antipode reads 0: the jumps at the two ends of the half-turn point in opposite directions and cancel everywhere else. There is one such pair for each point of the circle, and countably many probes cannot distinguish all of them from the uniform state. On a line this cannot happen, because a monotone response

cannot hide a jump (see the proof of Theorem 7.8(b)); on a circle there is no first or last point. The same contrast appears in topological dynamics: Glasner and Megrelishvili showed that compact metric linearly ordered systems have first countable enveloping semigroups, while the circular analogue of the Helly space is not first countable [GM22, Theorem 8.8, Proposition 8.10 and Remark 8.11]. Their spaces consist of order-preserving maps of the circle, whereas the responses here are real-valued; neither result implies the other.

7.8. **Interference fringes are not NIP.** Orders are tame. Oscillation at ever finer scales is not, as we show in Proposition 7.13 below; its simplest physical instance is an interference pattern. Alice prepares a mode of light in one of the states e_n, $n \in \mathbb{N}$, whose interference pattern on Bob's screen has wavenumber $k_n = 8^n$, and Bob's detector, at position $x \in [0, 2\pi]$, clicks with probability $f_n(x) = \frac{1}{2}(1 + \cos k_n x)$: it asks whether the photon lands on a bright fringe. The *fringe observable* is $C = \sum_n |e_n\rangle\langle e_n| \otimes M_{f_n}$ on $\ell^2(\mathbb{N}) \otimes L^2[0, 2\pi]$: the label n of Alice's state selects the test f_n applied to Bob's detector, and C is thus a label-controlled observable in the sense of Definition 8.1 below.

Proposition 7.13. *The fringe observable is not NIP: Alice's states e_n form an independent sequence, and the closure of their responses contains a copy of $\beta\mathbb{N}$.*

Proof. Let $E, F \subseteq \mathbb{N}$ be finite and disjoint. Choose $y \in [0, 1)$ whose base-8 digit in place $n + 1$ is 0 for $n \in F$ and 4 for $n \in E$, and put $x = 2\pi y$. Then $\cos k_n x = \cos(2\pi\{8^n y\})$, where $\{8^n y\}$, the fractional part of $8^n y$, lies in $[0, \frac{1}{8})$ for $n \in F$ and in $[\frac{1}{2}, \frac{5}{8})$ for $n \in E$; hence $f_n(x) \geq \frac{3}{4}$ for $n \in F$ and $f_n(x) \leq \frac{1}{4}$ for $n \in E$. By continuity of the finitely many functions f_n involved, a state σ of Bob's detector concentrated near x has expectation values at least $\frac{5}{8}$ for $n \in F$ and at most $\frac{3}{8}$ for $n \in E$. By Lemma 5.2(c), the restrictions of such states are probes; the sequence (e_n) is therefore independent. As in the proof of Theorem 6.1(c), no subsequence of the responses converges pointwise, and Fact 3.4 yields a subsequence whose closure is homeomorphic to $\beta\mathbb{N}$. □

The contrast between the fringes and the orders of this section is the physical content of NIP. A comparison asks on which side of a threshold a value lies, and finitely many probes can cut an ordered line only into finitely many pieces; a fringe of wavenumber 8^n cuts Bob's screen into 8^n pieces, and the fringes of different wavenumbers are independent of one another, like the digits of a number.

8. The seven minimal families in physics

This section is devoted to proving that each of the seven minimal families $D_1, \ldots, D_7$ of the heptachotomy is realized by a tree of preparations of a physical observable, and that each realization displays what is characteristic of its family. For whole compacta of idealized responses, mixtures merge the hereditarily separable class with the metrizable one (see Corollary 6.8); in Theorem 8.5 below, we show that for trees of preparations all seven families occur, because a tree may have a closure of any of the seven kinds inside a larger convex compactum, as a copy of the split interval sits inside the Helly space of Proposition 7.9, the space of nondecreasing functions from $[0, 1]$ to $[0, 1]$ that take the value 0 at 0 and 1 at 1.

8.1. **Label-controlled observables.** Several of our observables are of the following kind. Alice's system is a register that holds a label, and the label selects the test applied to Bob's system: a detector that accepts Bob's system at each point with a probability that depends on the label and varies continuously with the point.

Definition 8.1. Let L be a compact metrizable space, λ a Borel probability measure on L with full support, I a countable set and $(f_i)_{i\in I}$ a family of continuous functions $L \to [0,1]$. The *label-controlled observable* is

$$C = \sum_{i\in I} |e_i\rangle\langle e_i| \otimes M_{f_i} \qquad \text{on} \quad \ell^2(I) \otimes L^2(L,\lambda),$$

where (e_i) is the standard basis of $\ell^2(I)$ and M_f is multiplication by f.

The sum converges strongly, that is, on each vector, because its terms act on orthogonal subspaces, and C lies between 0 and the identity operator. When L contains a countable dense set D of isolated points, we take $\lambda = \sum_{x\in D} w_x \delta_x$, a combination of the point masses δ_x with positive weights w_x of sum 1; then the space $L^2(L,\lambda)$ is a weighted copy of $\ell^2(D)$, and Bob's system is located at one of the points of D.

Lemma 8.2. *Let C be as in Definition 8.1, and suppose that the functions f_i separate the points of L. Then the probe algebra $\mathfrak{A}_C$, generated by the effective observables, is $\{M_f : f \in C(L)\}$, the probe space Y_C, its state space, is the space $P(L)$ of Borel probability measures on L with the weak* topology, and the response of the label state e_i is $r_{e_i}(\nu) = \int f_i \, d\nu$.*

Proof. For a density operator ρ, tracing out Alice's system yields $T_C(\rho) = \sum_i \langle e_i, \rho e_i\rangle M_{f_i}$, a norm-convergent sum, so the slice map takes values in $\{M_f : f \in C(L)\}$. Since λ has full support and f is continuous, $\|M_f\| = \sup|f|$; hence the map $f \mapsto M_f$ is an isometric $*$-isomorphism from $C(L)$ onto its image. The unital C*-subalgebra of $C(L)$ generated by the functions f_i is $C(L)$, by the Stone–Weierstrass theorem, because they separate points. It follows that $\mathfrak{A}_C = \{M_f : f \in C(L)\}$, and, by the Riesz representation theorem, its states are the measures $\nu \in P(L)$, acting by $M_f \mapsto \int f \, d\nu$. □

8.2. **Transfer from points to measures.** The responses of our observables are integrals $\nu \mapsto \int g \, d\nu$ of functions g on a compact space against Bob's probes. In Lemma 8.3 below, we show that a family of such responses has a closure homeomorphic to that of the family of integrands, by a homeomorphism that matches the two families term by term.

Lemma 8.3. *Let L be a compact metrizable space and $(f_t)_{t\in T}$ a countable family of continuous functions $L \to [0,1]$ that is relatively compact in the space $B_1(L)$ of Baire-1 functions on L. For $t \in T$ and $\nu \in P(L)$, put $F_t(\nu) = \int f_t \, d\nu$. Then the families $\{F_t\}$, of functions on $P(L)$, and $\{f_t\}$, of functions on L, are equivalent in the sense of Definition 4.1, and $\{F_t\}$ is NIP.*

Proof. Every sequence (f_{t_n}) has a subsequence that converges pointwise on L. Indeed, otherwise, by Fact 3.4, some subsequence would have a closure homeomorphic to $\beta\mathbb{N}$, which is not Fréchet–Urysohn (a space is Fréchet–Urysohn if every point in the closure of a subset is the limit of a sequence from that subset), whereas the closure of $\{f_t\}$ is a Rosenthal compactum and therefore Fréchet–Urysohn (see [BFT78, Theorem 3F]). By the dominated convergence theorem, the corresponding subsequence of (F_{t_n}) converges pointwise on $P(L)$.

The family $\{F_t\}$ is thus NIP. Indeed, if a sequence (F_{t_n}) and reals $a < b$ witnessed the independence property, then, since $P(L)$ is compact and the functions F_t are continuous, for each $E \subseteq \mathbb{N}$ there would be ν with $F_{t_n}(\nu) \le a$ for $n \in E$ and $F_{t_n}(\nu) \ge b$ for $n \notin E$ (see the proof of Theorem 3.6); taking for E every other index of a pointwise convergent subsequence would yield a ν at which that subsequence does not converge, contrary to the previous paragraph. By Theorem 3.6, the closure of $\{F_t\}$ is contained in $B_1(P(L))$; it is a Rosenthal compactum, hence Fréchet–Urysohn.

Let Φ send a function G on $P(L)$ to the function $x \mapsto G(\delta_x)$ on L. The map Φ is continuous for the pointwise topologies, and $\Phi(F_t) = f_t$. The map Φ is injective on the closure of $\{F_t\}$: if G lies in that closure, then, the closure being Fréchet–Urysohn, the function G is the pointwise limit of a sequence (F_{t_n}); the sequence (f_{t_n}) then converges pointwise to $\Phi(G)$, which is therefore a bounded Borel function, and, by dominated convergence, the value $G(\nu)$ equals $\lim_n \int f_{t_n}\, d\nu = \int \Phi(G)\, d\nu$ for every ν. Thus G is determined by $\Phi(G)$. As a continuous injection on a compact space, the map Φ is a homeomorphism of the closure of $\{F_t\}$ onto its image; this image is compact, contains $\{f_t\}$ and lies in the closure of $\{f_t\}$, so it is that closure. □

We also need to discard points that carry no information about limits.

Lemma 8.4. *Let L be a set, $L_0 \subseteq L$, and $(g_t)_{t\in T}$ a countable pointwise bounded family of real functions on L whose pointwise closure K, a compact space, is Fréchet–Urysohn. Suppose that*

(i) *the restrictions $g_t|_{L_0}$ are pairwise distinct, and none of them lies in the closure of the others;*
(ii) *if two limits of sequences $(g_{t_k})_{k\in\mathbb{N}}$ with pairwise distinct indices agree on L_0, then they are equal.*

Then the families (g_t) and $(g_t|_{L_0})$ are equivalent.

Proof. Restriction $\pi\colon K \to \mathbb{R}^{L_0}$ is continuous, and its image is the closure of $(g_t|_{L_0})$. Let $f \in K$ be different from every g_t. Since K is Fréchet–Urysohn, the function f is the limit of a sequence (g_{t_k}); since K is Hausdorff and f is not a g_t, no index occurs infinitely often, and, after passage to a subsequence, the indices are pairwise distinct. By (i), the map π is injective on the family; moreover, $\pi(g_t) \neq \pi(f)$ for every such limit f and every t, because $\pi(f)$ lies in the closure of $\{g_s|_{L_0} : s \neq t\}$, whereas $g_t|_{L_0}$ does not. By (ii), the map π is also injective on the limits. Hence π is a continuous injection of the compact space K, and therefore a homeomorphism onto its image. □

8.3. **Three settings.** The realizations use three physical settings.

Photon numbers. This is the discrete comparison of Section 7, in which Alice's and Bob's values lie in $\Xi = \{x_l : l \in \mathbb{N}\} \cup \{1\}$, where $x_l = 1 - \frac{1}{l+1}$. For Alice's number state e_l, the state with exactly l photons, the function $g_l = g_{e_l}$, whose value at a point y is the probability that Alice's value lies below y, is the indicator of $\{x_j : j > l\} \cup \{1\}$; the probe space is the space $P(\Xi)$ of Borel probability measures on Ξ, and $r_{e_l}(\nu) = \int g_l\, d\nu$ (see Proposition 7.7).

Positions in a box. This is the continuous comparison of Section 7: the probe space is $P([0,1])$, and a state ρ of Alice's particle with distribution μ_ρ has response $\nu \mapsto \int g_\rho\, d\nu$, where $g_\rho(y) = \mu_\rho([0,y))$. We place a Cantor set in the box. Let $\psi(\sigma) = \frac{3}{4} - \sum_n \sigma(n) 3^{-n-1}$ for $\sigma \in 2^{\mathbb{N}}$; this is an order-reversing homeomorphism of the Cantor space $2^{\mathbb{N}}$, with its lexicographic order $\prec$, onto a Cantor set $C_0 \subseteq [\frac{1}{4}, \frac{3}{4}]$. For a finite binary sequence $s \in 2^{<\mathbb{N}}$, let I_s be the smallest interval containing the image $\psi(V_s)$ of the clopen set V_s of infinite extensions of s; it has length $\ell_s = \frac{1}{2}3^{-|s|}$, left endpoint $\psi(s^\frown 1^\infty)$ and right endpoint $\psi(s^\frown 0^\infty)$, the images of the extensions of s by ones and by zeros. If τ is the immediate successor of σ in $\prec$, then the interval $(\psi(\tau), \psi(\sigma))$ is a component of $[0,1] \setminus C_0$; the other components are $[0, \frac{1}{4})$ and $(\frac{3}{4}, 1]$. The components adjacent to I_s have length at least $\ell_s/2$: each of them is either $[0, \frac{1}{4})$ or $(\frac{3}{4}, 1]$, of length $\frac{1}{4} \geq \ell_s/2$, or the gap between $I_{u^\frown 0}$ and $I_{u^\frown 1}$ for some proper initial segment u of s, of length $\ell_u/3 \geq \ell_s$.

Registers. Alice's system is a chain of spins of variable finite length, with Hilbert space $\ell^2(2^{<\mathbb{N}})$ over the set $2^{<\mathbb{N}}$ of finite binary sequences; for $t \in 2^{<\mathbb{N}}$, the basis vector e_t is the chain of length $|t|$ whose spin at site n is up exactly when $t(n) = 1$. Bob probes it in two ways. In *scanning*, Bob's probe sits at a site $n \in \mathbb{N}$ and asks whether Alice's spin at that site is up: the test is $a_t(n) = t(n)$ for $n < |t|$ and $a_t(n) = 0$ otherwise. In *template matching*, Bob holds a reference configuration u, finite or infinite, and asks whether Alice's chain coincides with the beginning of his reference: the test is $b_t(u) = 1$ if $t \sqsubseteq u$ and $b_t(u) = 0$ otherwise. We extend a_t to the one-point compactification $L_a = \mathbb{N} \cup \{\infty\}$ by $a_t(\infty) = 0$, and we let L_b be the space of finite and infinite binary sequences, with the ultrametric $d(u, u') = 2^{-|u \wedge u'|}$ for $u \neq u'$, where $u \wedge u'$ is the longest common initial segment; it is compact, the finite sequences are isolated and dense, and the functions b_t are continuous, since $\{u : t \sqsubseteq u\}$ is clopen. The *scanning*, *template* and *combined* observables are the label-controlled observables with labels $2^{<\mathbb{N}}$, on L_a, L_b and their disjoint sum $L_a \oplus L_b$, with the functions a_t, b_t and (a_t, b_t), the last equal to a_t on L_a and to b_t on L_b, and with measures concentrated on $\mathbb{N}$, on the finite sequences, and on both, respectively. These functions separate the points of L_a, of L_b and of $L_a \oplus L_b$; for instance, the functions of the empty label, $a_\emptyset = 0$ and $b_\emptyset = 1$, separate L_a from L_b. Hence Lemma 8.2 applies to all three observables.

8.4. **The realization.** Together, the three settings realize all seven minimal families.

Theorem 8.5. *Each minimal family is realized by a tree of actual preparations of a physically natural observable: for each $i = 1, \dots, 7$, the observable and the pure states $(\rho_t)_{t \in 2^{<\mathbb{N}}}$ of Alice's system listed below, given by pairwise orthogonal vectors, have responses $\{r_{\rho_t}\}$ equivalent to D_i. Here $(s_t)_{t \in 2^{<\mathbb{N}}}$ is the canonical enumeration of the regular dyadic subtree Θ fixed in Section 4.*

D_1: the comparison of photon numbers, with Alice's number states in any order;
D_2: scanning, with Alice's chains e_{s_t}, whose scanning functions are $s_t^\frown 0^\infty$, for $s_t \in \Theta$;
D_3: the comparison of positions, with Alice's particle spread over the gap of C_0 immediately to the right of $\psi(s_t^\frown 0^\infty)$;
D_4: the comparison of positions, with Alice's particle spread over the gap of C_0 immediately to the left of $\psi(s_t^\frown 1^\infty)$;
D_5: template matching, with Alice's chains e_t;
D_6: the combined observable, with Alice's chains e_t;
D_7: the comparison of positions, with Alice's particle spread half over a short interval just to the left of I_{s_t} and half over a short interval just to the right of it.

Moreover, the causal order of spacetime realizes D_5 in the sense of Theorem 7.10.

Proof. In each case the probe space is $P(L)$ for a compact metrizable space L, the responses are $r_{\rho_t}(\nu) = \int g_t \, d\nu$ with g_t continuous, and the closure of $\{g_t\}$ consists of Baire-1 functions; by Lemma 8.3, it suffices to show that $\{g_t\}$ is equivalent to D_i. We use the notation of Section 4: the function v_t is the characteristic function of the set V_t of infinite binary sequences that extend t, and the functions x_σ^+ and x_σ^- are the characteristic functions of $\{\tau \in 2^{\mathbb{N}} : \sigma \preceq \tau\}$ and $\{\tau \in 2^{\mathbb{N}} : \sigma \prec \tau\}$, for the lexicographic order $\prec$.

D_1. Index the functions g_l, $l \in \mathbb{N}$, by $2^{<\mathbb{N}}$ through any bijection, and write g_t for the function of index t. Every sequence of pairwise distinct members converges to the indicator of $\{1\}$, which is not a member, and each member is isolated. Hence the closure is a countable compact space with a single non-isolated point, and so is the closure $D_1 \cup \{0\}$ of D_1; the bijection $g_t \mapsto \frac{1}{|t|+1} v_t$, extended by sending the indicator of $\{1\}$ to 0, is a homeomorphism.

The space Ξ is countable, so every function on it is Baire-1, every subset of a countable space being an F_σ.

D_2. Here $g_t = a_{s_t}$ is $s_t^\frown 0^\infty$ on $\mathbb{N}$ and 0 at ∞. This is the family D_2, extended continuously to $\mathbb{N}\cup\{\infty\}$; every function in its closure vanishes at ∞, so the extension does not change the closure, and every function on the countable space L_a is Baire-1.

D_3 and D_4. Let $\sigma_t = s_t^\frown 0^\infty$. If $\sigma_t \neq 0^\infty$, it has an immediate predecessor σ_t^-, and $G_t = (\psi(\sigma_t), \psi(\sigma_t^-))$ is a gap; if $\sigma_t = 0^\infty$, let $G_t = (\frac{3}{4}, 1]$. Let ρ_t be uniform on G_t. Then $g_t = 0$ on $[0, \psi(\sigma_t)]$, $g_t = 1$ to the right of G_t, and g_t is linear on G_t. On C_0, the function g_t satisfies $g_t(\psi(\tau)) = 0$ exactly when $\tau \succeq \sigma_t$; that is, $g_t \circ \psi = 1 - x^+_{\sigma_t}$. The sequences σ_t are pairwise distinct, and so are the gaps G_t. We apply Lemma 8.4 with $L_0 = C_0$. The closure of $\{g_t\}$ consists of nondecreasing functions; it is therefore a Rosenthal compactum, and hence Fréchet–Urysohn. Condition (i) holds because D_3 is discrete in its closure. For (ii), let J be a component of $[0,1]\setminus C_0$. For pairwise distinct t_k, eventually $G_{t_k} \neq J$, and then the function g_{t_k} is constant on $\overline{J}$; every such limit f is thus constant on $\overline{J}$, and so is determined on J by its value at an endpoint of J in C_0. Consequently, the family $\{g_t\}$ is equivalent to $\{g_t|_{C_0}\}$, which is carried to D_3 by the homeomorphism ψ followed by the map $f \mapsto 1-f$, a homeomorphism for the pointwise topology. The case of D_4 is symmetric: with $\sigma_t = s_t^\frown 1^\infty$ and its immediate successor σ_t^+, the gap is $(\psi(\sigma_t^+), \psi(\sigma_t))$, or $[0, \frac{1}{4})$ if $\sigma_t = 1^\infty$, and $g_t \circ \psi = 1 - x^-_{\sigma_t}$.

D_5. Here $g_t = b_t$ on L_b; its restriction to the infinite sequences is v_t. For pairwise distinct t_k, eventually t_k is longer than any given finite u, so every such limit vanishes on the finite sequences. Hence Lemma 8.4, with $L_0 = 2^{\mathbb{N}}$, applies, condition (i) holding because D_5 is discrete in its closure $\hat{A}(2^{\mathbb{N}})$. The closure consists of Baire-1 functions: the restrictions of its members to the closed set $2^{\mathbb{N}}$ lie in $\hat{A}(2^{\mathbb{N}}) \subseteq B_1(2^{\mathbb{N}})$, and the remaining points are isolated.

D_6. Here $g_t = (a_t, b_t)$; its restriction to $\mathbb{N}\oplus 2^{\mathbb{N}}$ is $(t^\frown 0^\infty, v_t)$, which is the member $(v_t, t^\frown 0^\infty)$ of D_6 with the two summands written in the other order. As above, every limit vanishes at ∞ and on the finite sequences; the closure consists of Baire-1 functions, by the arguments for D_2 and D_5; and Lemma 8.4 applies with $L_0 = \mathbb{N} \oplus 2^{\mathbb{N}}$.

D_7. Let $\ell = \ell_{s_t}$, and let ρ_t be the state whose distribution is half uniform on $A_t = [\min I_{s_t} - \frac{\ell}{2}, \min I_{s_t} - \frac{\ell}{4}]$ and half uniform on $B_t = [\max I_{s_t} + \frac{\ell}{4}, \max I_{s_t} + \frac{\ell}{2}]$; these intervals lie in the components adjacent to I_{s_t}. On C_0, the function g_t equals 0 to the left of I_{s_t}, $\frac{1}{2}$ on I_{s_t} and 1 to the right; that is, $g_t \circ \psi = 1 - x^+_{\sigma_t} + \frac{1}{2}v_{s_t}$ with $\sigma_t = s_t^\frown 0^\infty$, since $\psi(\sigma_t) = \max I_{s_t}$. Each member of the closure of D_7 has the form (v, x), for functions v and x on $2^{\mathbb{N}}$ with $v \leq x$ that take only the values 0 and 1; hence the map $(v, x) \mapsto (1 - x + \frac{1}{2}v) \circ \psi^{-1}$ is a continuous injection, and it carries D_7 to $\{g_t|_{C_0}\}$; as D_7 is discrete in its closure, condition (i) of Lemma 8.4 holds. For (ii), note that A_t lies in the component whose right endpoint is $\psi(s_t^\frown 1^\infty)$, and B_t in the component whose left endpoint is $\psi(s_t^\frown 0^\infty)$. These sequences are pairwise distinct for $s_t \in \Theta$, so each component of $[0,1]\setminus C_0$ contains at most one interval A_t and at most one interval B_t. For pairwise distinct t_k, eventually neither A_{t_k} nor B_{t_k} lies in a given component J, and then the function g_{t_k} is constant on $\overline{J}$. Every such limit is therefore constant on $\overline{J}$ and is determined by its values on C_0.

Finally, in the photon and register cases the states are distinct basis vectors, and in the box we take the vectors $\sqrt{d\mu_{\rho_t}/dy}$, whose supports meet in at most finitely many points, so they are orthogonal. □

The realizations display what distinguishes the families.

- D_1, *escape to infinity*: an oscillator excited ever higher has a single idealized limit, and each actual preparation is isolated from it.

- D_2, *infinite configurations*: the idealized limits of the chains are the infinite spin configurations, a Cantor set of them, and countably many site probes tell them all apart, so the closure is metrizable; but any two configurations differ by 1 at some site, so the closure is not separable in the supremum norm.
- D_3 and D_4, *one-sided localization*: a particle that approaches a point from only one side produces one half of a split pair, and the split Cantor set appears; the closure is hereditarily separable but not metrizable.
- D_5, *point-supported limits*: light cones that shrink to points on Bob's slice, or templates that become infinitely long, yield idealized responses supported on a single point, and the closure is not first countable.
- D_6, *twins over a metrizable base*: each infinite configuration has two idealized versions, one that every template misses and one that the template of the configuration itself detects; the scanning probes alone cannot tell the twins apart, and, as seen by these probes, the limits form a Cantor set.
- D_7, *twins over a split base*: a particle that approaches a point from both sides at once produces the midpoint of a split pair, as the symmetric state does in the Ising model (see Section 9); its twin structure sits over the split Cantor set.

In Proposition 8.6 and Corollary 8.7 below, we show that these observables are NIP, which places them in the classification of Theorem 6.9; the proof uses a theorem of Stegall. Recall that a subset of a Banach space is *weakly precompact* if it is bounded and every sequence in it has a weakly Cauchy subsequence, one whose values under each continuous linear functional form a Cauchy sequence, and that the *absolutely convex hull* of a set consists of the finite combinations $\sum_k a_k x_k$ of its elements with $\sum_k |a_k| \leq 1$. Stegall's theorem states that the absolutely convex hull of a weakly precompact set is weakly precompact (see [Rod21], where the result is attributed to Stegall).

Proposition 8.6. *Let C be a label-controlled observable whose functions f_i separate the points of L and have the property that every sequence of them has a subsequence converging pointwise on L. Then the observable C is NIP for Alice's responses.*

Proof. For a density operator ρ, put $p_i = \langle e_i, \rho e_i \rangle$, a probability distribution on I, and $g_\rho = \sum_i p_i f_i$, a norm-convergent sum in $C(L)$; then $r_\rho(\nu) = \int g_\rho \, d\nu$. By dominated convergence, a sequence of functions f_i that converges pointwise on L is weakly Cauchy in $C(L)$; hence the set $\{f_i\}$ is weakly precompact. By Stegall's theorem, so is its absolutely convex hull, and so is the norm closure of that hull: given y_n in the closure, choose x_n in the hull with $\|x_n - y_n\| < 1/n$ and a weakly Cauchy subsequence of (x_n); the corresponding subsequence of (y_n) is weakly Cauchy. The partial sums of g_ρ lie in the hull, and every function g_ρ therefore lies in this closure. Consequently, every sequence (g_{ρ_n}) has a weakly Cauchy subsequence, and, since every $\nu \in P(L)$ is a continuous linear functional on $C(L)$, the corresponding subsequence of (r_{ρ_n}) converges pointwise on $P(L) = Y_C$. As in the proof of Lemma 8.3, an independent sequence has no such subsequence. □

Corollary 8.7.

(a) *The scanning observable is unstable and NIP$_3$, but not uniformly NIP.*
(b) *The template observable is NIP but not NIP$_1$.*
(c) *The scanning observable is not NIP for Bob's responses, that is, the observable obtained by exchanging the two tensor factors is not NIP for Alice's responses. Hence NIP for Alice's responses does not imply NIP for Bob's.*

Proof. (a) Every sequence of functions a_t on the countable set L_a has a pointwise convergent subsequence, by a diagonal argument, so the observable is NIP by Proposition 8.6. For a trace-norm dense sequence $(\rho_n)_{n\in\mathbb{N}}$, the closure of the responses is that of $\{r_{\rho_n}\}$ (see Lemma 6.3), which by Lemma 8.3 is homeomorphic to the closure of $\{g_{\rho_n}\}$ in $\mathbb{R}^{L_a}$; since L_a is countable, this space is metrizable. For each positive integer m, the chains $t_1,\dots,t_m$ of length 2^m with $t_i(n)$ the i-th binary digit of n are shattered, with values 0 and 1, by the site probes $n<2^m$; hence the fat-shattering dimension is infinite at every margin $\delta\le 1$. Finally, the observable is unstable: the functions a_{1^k} of the chains 1^k of k up spins, for $k\in\mathbb{N}$, converge pointwise to the indicator of $\mathbb{N}$, so the idealized response $\nu\mapsto\nu(\mathbb{N})$, which is not continuous on $P(L_a)$, is a limit of responses (see Theorem 6.1(a)).

(b) Let (t_k) be a sequence of labels. By compactness of L_b, a subsequence converges in L_b; if the limit is finite, the subsequence is eventually constant. If the limit is an infinite sequence σ, then $b_{t_k}(u)\to 0$ for finite u, and $b_{t_k}(\tau)\to 0$ for infinite $\tau\ne\sigma$, since eventually t_k agrees with σ beyond the first place where σ and τ differ; and, along a further subsequence, the value $b_{t_k}(\sigma)$ is constant. Hence the observable is NIP by Proposition 8.6. By Theorem 8.5, the closure of its responses contains a closed copy of $\hat{A}(2^{\mathbb{N}})$, which is not first countable.

(c) The states e_n of Bob's system in which his probe sits at the site n, for $n\in\mathbb{N}$, form an independent sequence: for disjoint finite sets $E,F\subseteq\mathbb{N}$, Alice's chain t of length $1+\max(E\cup F)$ with $t(n)=1$ exactly for $n\in F$ has $a_t(n)=1$ for $n\in F$ and $a_t(n)=0$ for $n\in E$. Alice's actual states are points of the probe space of the exchanged observable (see Lemma 5.2(c)), so the family of Bob's responses has the independence property. This is the relation $n\in u$ of Remark 3.8, realized by a physical observable. □

The scanning observable shows the difference between NIP and uniform NIP (see Remark 6.2): its idealized responses form a metrizable compactum, yet its responses are not learnable with a number of probes bounded independently of the distribution of the probes [ABDCBH97], because Bob's site probes can read off every finite pattern of sufficiently long chains.

9. Spontaneous symmetry breaking is a split pair

In this section and the next we show that the levels of Section 4 have a physical meaning beyond bipartite observables. Several notions of physics rest on an order of limits of a family of responses: a preparation, such as a finite sample, has a response, such as its magnetization as a function of an external field, and the notion depends on whether the sample becomes infinite before or after the field is switched off.

To make these orders of limits precise, we use the definitions and results of Sections 3 and 4, which apply verbatim to a pointwise bounded family $\mathcal{R}$ of continuous functions on a compact metrizable space Y of probes. The family is *stable* if, for all sequences $(r_k)_{k\in\mathbb{N}}$ in $\mathcal{R}$ and $(y_l)_{l\in\mathbb{N}}$ in Y, the iterated limits $\lim_l\lim_k r_k(y_l)$ and $\lim_k\lim_l r_k(y_l)$ are equal whenever both exist, that is, if it satisfies Grothendieck's double limit exchange condition; by Grothendieck's theorem [Gro52], this holds exactly when every idealized response, that is, every function in the pointwise closure of $\mathcal{R}$, is continuous, and then the family is NIP_3, by the argument of Proposition 6.5. The family is NIP exactly when every idealized response is of the first Baire class (see Theorem 3.6), and its level is given by Definition 4.4. The families below need not be convex, so Theorem 6.6 does not apply to them; their closures are computed by hand. The physical hypotheses are stated explicitly, and the physics enters only through standard facts, which we cite.

The standard example of an order of limits is spontaneous symmetry breaking: for a ferromagnet below its critical temperature, letting the external field vanish after the volume becomes infinite yields a nonzero magnetization, whereas the other order yields zero [FV17]. In Theorem 9.2 below, we show that this is an instability in our sense, absent above the critical temperature, and that, with an offset field that moves the transition point, the idealized responses below the critical temperature are NIP_1 but not NIP_2: the idealized responses of the two pure phases, the equilibrium states magnetized in opposite directions, form a split pair, two functions that differ only at the transition point, and the idealized response of the symmetric state, the equal mixture of the two phases, is their midpoint, as in the proof of Theorem 6.6.

9.1. **The Ising model.** We recall the facts we need; all of them can be found in the book of Friedli and Velenik [FV17]. Fix a dimension $d \geq 2$ and an inverse temperature $\beta > 0$. For $N \geq 1$, let $\Lambda_N = \{-N, \dots, N\}^d$ with periodic boundary conditions, and for a real field h let

$$Z_N(h) = \sum_{\sigma \in \{-1,1\}^{\Lambda_N}} \exp\Big(\beta \sum_{x \sim y} \sigma_x \sigma_y + h \sum_x \sigma_x\Big), \qquad p_N(h) = \frac{1}{|\Lambda_N|} \log Z_N(h),$$

where $x \sim y$ ranges over the nearest-neighbor pairs of the torus; in this convention the field h already includes the factor β. The magnetization per site in the Gibbs state, the probability measure on configurations σ proportional to the corresponding term of $Z_N(h)$, is $m_N(h) = p_N'(h)$.

(F1) The function p_N is smooth, convex and even. Hence the function m_N is continuous, nondecreasing and odd, with $|m_N| \leq 1$ and $m_N(0) = 0$.

(F2) The limit $p(h) = \lim_N p_N(h)$ exists for every h; it is convex and even, and it is differentiable at every $h \neq 0$, by the Lee–Yang theorem [LY52]. Since p_N and p are convex, $m_N(h) \to p'(h)$ at every $h \neq 0$, by Griffiths' lemma on derivatives of convex functions.

(F3) The spontaneous magnetization $m^*(\beta) = p'(0^+)$ is the right derivative of p at 0, and $p'(0^-) = -m^*(\beta)$. The critical inverse temperature $\beta_c = \inf\{\beta : m^*(\beta) > 0\}$ is positive, and it is finite by Peierls's argument [Pei36, Gri64]; since $m^*(\beta)$ is the magnetization of the plus state (the infinite-volume limit of the Gibbs states with all boundary spins fixed up), it is nondecreasing in β, by the Griffiths–Kelly–Sherman inequalities, and $m^*(\beta) > 0$ for $\beta > \beta_c$ (see [FV17, Chapter 3]).

As p' is nondecreasing and odd, (F3) implies that $p'(t) \geq m^*$ for $t > 0$ and $p'(t) \leq -m^*$ for $t < 0$.

9.2. **Preparations and probes.** A preparation consists of a finite system Λ_N and an offset field $a \in [-1, 1]$, that is, an idealized uniform field built into the sample; a probe is an applied field $h \in Y = [-2, 2]$. The system then feels the field $h - a$, and the response of the preparation (N, a) is the function

$$r_{N,a}(h) = m_N(h - a) \qquad (h \in Y).$$

The offset is a device that moves the transition point; without it, the family $\{m_N\}$ has a single transition point, at $h = 0$. Let $\mathcal{R} = \{r_{N,a} : N \geq 1,\ a \in [-1, 1]\}$. Each $r_{N,a}$ is continuous and nondecreasing, and $r_{N,a'} \to r_{N,a}$ uniformly as $a' \to a$, because m_N is uniformly continuous on compact sets. The responses with rational a are therefore dense in $\mathcal{R}$, and the closure $\overline{\mathcal{R}}$ is separable. Every function in $\overline{\mathcal{R}}$ is nondecreasing, and so of the first Baire class; thus $\overline{\mathcal{R}}$ is a Rosenthal compactum, and, by Theorem 3.6, the family $\mathcal{R}$ is NIP.

We write $m^* = m^*(\beta)$, and for $a \in [-1,1]$ and $c \in [-m^*, m^*]$ we let $g_{a,c}\colon Y \to \mathbb{R}$ be given by $g_{a,c}(h) = p'(h-a)$ for $h \neq a$ and $g_{a,c}(a) = c$. These functions describe the idealized responses.

Lemma 9.1. *The following hold:*

(i) *every $g_{a,c}$ with $-1 < a < 1$ belongs to $\overline{\mathcal{R}}$;*
(ii) *every function in $\overline{\mathcal{R}} \setminus \mathcal{R}$ equals $g_{a,c}$ for some $a \in [-1,1]$ and $c \in [-m^*, m^*]$.*

Proof. (i) Let $-1 < a < 1$. If $m^* > 0$, then, since $\overline{\mathcal{R}}$ is closed and $g_{a,c} \to g_{a,\pm m^*}$ pointwise as $c \to \pm m^*$, it suffices to take $|c| < m^*$, and by symmetry $0 \le c < m^*$; if $m^* = 0$, then $c = 0$. If $c = 0$, then $r_{N,a} \to g_{a,0}$, by (F1) at $h = a$ and by (F2) elsewhere. If $0 < c < m^*$, fix $\eta > 0$; by (F2), $m_N(\eta) \to p'(\eta) \ge m^* > c$, so for all large N, since $m_N(0) = 0$, the intermediate value theorem yields $\varepsilon_N \in (0, \eta)$ with $m_N(\varepsilon_N) = c$. As η is arbitrary, we may choose $\varepsilon_N \to 0$. Put $a_N = a - \varepsilon_N$. Then $r_{N,a_N}(a) = m_N(\varepsilon_N) = c$. For $h \neq a$, let $t = h - a$ and fix η' with $0 < \eta' < |t|$; for large N, we have $t - \eta' \le h - a_N \le t + \eta'$, and hence, by monotonicity, $m_N(t - \eta') \le r_{N,a_N}(h) \le m_N(t + \eta')$. By (F2), the outer terms converge to $p'(t - \eta')$ and $p'(t + \eta')$, which tend to $p'(t)$ as $\eta' \to 0$, because p' is continuous at $t \neq 0$. Finally, as $a > -1$, the offsets a_N lie in $[-1,1]$ for large N, and $r_{N,a_N} \to g_{a,c}$.

(ii) Let $g \in \overline{\mathcal{R}} \setminus \mathcal{R}$. Since $\overline{\mathcal{R}}$ is a Rosenthal compactum, it is Fréchet–Urysohn [BFT78, Theorem 3F], so the function g is the limit of a sequence r_{N_k,a_k}. Passing to a subsequence, we may assume that $a_k \to a$ and that either N_k is constant or $N_k \to \infty$. In the first case, uniform continuity forces $g = r_{N,a} \in \mathcal{R}$, which is excluded. In the second case, the argument of the previous paragraph shows that $g(h) = p'(h-a)$ for $h \neq a$. Finally, given $\eta > 0$, we have $|a - a_k| < \eta$ for all large k, and then $|r_{N_k,a_k}(a)| = |m_{N_k}(a - a_k)| \le m_{N_k}(\eta)$, because m_{N_k} is odd and nondecreasing; the right side tends to $p'(\eta)$, and $p'(\eta) \to m^*$ as $\eta \to 0$. Hence $|g(a)| \le m^*$, and $g = g_{a,g(a)}$. □

9.3. **The classification detects the phase transition.** By Lemma 9.1, the classification of the family $\mathcal{R}$ of responses with offsets reduces to the behavior of p' at 0.

Theorem 9.2. *Let $d \ge 2$ and $\beta > 0$ be given.*

(a) *If $m^*(\beta) = 0$, in particular if $\beta < \beta_c$, then every idealized response is continuous: the family $\mathcal{R}$ is stable, and $\overline{\mathcal{R}}$ is metrizable.*
(b) *If $m^*(\beta) > 0$, in particular if $\beta > \beta_c$, then the family $\mathcal{R}$ is unstable, and $\overline{\mathcal{R}}$ is first countable but not hereditarily separable: the family is NIP$_1$ but not NIP$_2$.*

Already the family $\{m_N\}$ without offsets is stable exactly when $m^*(\beta) = 0$; it is NIP$_3$ for every β.

Proof. (a) If $m^* = 0$, then the pressure p is differentiable everywhere, and its derivative p', being the derivative of a convex function, is continuous. By Lemma 9.1(ii), every idealized response outside $\mathcal{R}$ has the form $g_{a,0}(h) = p'(h-a)$ and is therefore continuous; so are the responses in $\mathcal{R}$. By Grothendieck's theorem [Gro52], a family of continuous functions on a compact space whose pointwise closure consists of continuous functions satisfies the double limit condition, so the family $\mathcal{R}$ is stable. The closure is metrizable by the argument of Proposition 6.5: evaluation at the rational points of Y is a continuous injection on continuous functions.

(b) Let $m^* > 0$. For the family without offsets, take $h_l = 1/l$ for $l \ge 1$ and a sequence $(N_k)_{k \in \mathbb{N}}$ of positive integers with $N_k \to \infty$. By (F2) and (F3), $\lim_l \lim_k m_{N_k}(h_l) =$

$\lim_l p'(1/l) = m^*$, whereas, by continuity of each m_N and (F1), $\lim_k \lim_l m_{N_k}(h_l) = \lim_k m_{N_k}(0) = 0$. The family $\{m_N\}$, and with it the family $\mathcal{R}$, is therefore unstable.

Every function in $\overline{\mathcal{R}}$ is nondecreasing, and the argument in the proof of Theorem 7.8(b) shows that the space of nondecreasing functions $Y \to [-1,1]$ is first countable; hence so is $\overline{\mathcal{R}}$. For $a \in (-1,1)$, the neighborhood $\{g : |g(a)| < m^*/2\}$ of $g_{a,0}$ contains no $g_{b,0}$ with $b \neq a$, because $|g_{b,0}(a)| = |p'(a-b)| \geq m^*$. The set $\{g_{a,0} : -1 < a < 1\}$ is thus an uncountable discrete subspace, and $\overline{\mathcal{R}}$ is not hereditarily separable.

The first part of the last statement follows from part (a), since $\{m_N\} = \{r_{N,0}\}$ is a subfamily of $\mathcal{R}$, and from the computation at the beginning of (b). For the second part, the sequence (m_N) converges pointwise, by (F1) and (F2), so its closure is countable, and a countable compact Hausdorff space is metrizable. □

Physically, the proof tells a familiar story. At the transition point $h = a$, the idealized responses take every value c between $-m^*$ and m^*. The extreme values $\pm m^*$ are the magnetizations of the two pure phases, obtained by letting the field reach the transition from one side after the volume has become infinite; they are the values at a of the idealized responses g_{a,m^*} and $g_{a,-m^*}$, which form a split pair. The value $c = (2\lambda - 1)m^*$ is the magnetization of the mixture of the two phases with weights λ and $1-\lambda$, for $0 \leq \lambda \leq 1$, and it is reached through finite-size rounding of the transition, the smoothing of the jump of the magnetization in a finite sample [BK90], by tuning the offset as the volume grows. The uncountable discrete set that destroys hereditary separability consists of the symmetric responses $g_{a,0}$, the midpoints of the split pairs, exactly as in the proof of Theorem 6.6. Above the critical temperature there is no split pair, and the idealized responses are continuous. In this sense the classification detects the phase transition: crossing β_c moves the family with offsets from the stable, metrizable class to the class of NIP_1 families that are not NIP_2, the class of the Helly space, the compactum of idealized responses of the comparison of positions in a box. Without offsets, the family stays NIP_3, and only stability registers the transition.

9.4. **The quantum Ising model.** In Theorem 9.6 below, we show that Theorem 9.2 holds verbatim for a Hamiltonian whose terms do not commute: the Ising model in a transverse field. The only ingredient of the classical proof that requires an argument for this model is the differentiability of the pressure away from zero field. It can be derived from the Griffiths–Hurst–Sherman inequality, which holds for this model through its space-time representation, a description of the quantum model by random configurations in one more dimension, that of imaginary time [BG09, Lemma 4.6]; we derive it instead from a Lee–Yang theorem, which yields more (see Propositions 9.4 and 9.5 below): the partition function has no zeros off the imaginary axis of the complex longitudinal field, and the pressure is real analytic away from zero field.

Fix $d \geq 1$, a coupling $J > 0$, a transverse field $\Gamma \geq 0$ and an inverse temperature $\beta > 0$. On the torus Λ_N of Section 9, with Hilbert space $(\mathbb{C}^2)^{\otimes \Lambda_N}$, the Pauli matrices $\sigma^x = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $\sigma^z = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ acting at each site i as σ_i^x and σ_i^z, and $S_N = \sum_i \sigma_i^z$, let

$$H_N = -J \sum_{i \sim j} \sigma_i^z \sigma_j^z - \Gamma \sum_i \sigma_i^x, \qquad Z_N(h) = \operatorname{Tr} \exp\bigl(-\beta H_N + h S_N\bigr),$$

for complex h; as in Section 9, the longitudinal field h includes the factor β. For real h put $p_N(h) = |\Lambda_N|^{-1} \log Z_N(h)$ and $m_N = p_N'$. For $\Gamma = 0$ this is the classical model with βJ in place of β.

(Q1) The function p_N is smooth, convex and even, and $m_N(h)$ is the expectation of $S_N/|\Lambda_N|$ in the Gibbs state, the density matrix $e^{-\beta H_N + hS_N}/Z_N(h)$; hence the function m_N is continuous, nondecreasing and odd, with $|m_N| \le 1$, because $\|S_N\| = |\Lambda_N|$, and $m_N(0) = 0$. Indeed, by cyclicity of the trace, $\frac{d}{dh} \operatorname{Tr} e^{X+hS} = \operatorname{Tr}(S e^{X+hS})$ for self-adjoint matrices X and S, and $h \mapsto \log \operatorname{Tr} e^{X+hS} = \sup_\omega \{\operatorname{Tr} \omega(X + hS) - \operatorname{Tr} \omega \log \omega\}$, the supremum over density matrices ω, is convex as a supremum of affine functions; the unitary $U = \prod_i \sigma_i^x$ commutes with H_N and satisfies $U S_N U^* = -S_N$, so $Z_N(-h) = Z_N(h)$ for all complex h.

(Q2) The limit $p(h) = \lim_N p_N(h)$ exists for every real h; it is convex and even. For free boundary conditions, in which the bonds that wrap around the torus are omitted, the existence is standard [Sim93], and the periodic Hamiltonian differs from the free one by at most J times the number of these bonds, which is at most $d(2N+1)^{d-1}$, so the two pressures differ by $O(1/N)$, because $|\log \operatorname{Tr} e^A - \log \operatorname{Tr} e^B| \le \|A - B\|$ for self-adjoint matrices A and B.

In the proof of Proposition 9.4 below, we show, by Suzuki's correspondence [Suz76] between quantum spin systems and classical Ising systems in one more dimension, that the partition function of the quantum model is a limit of partition functions of classical Ising models on a larger graph, in which each site of the torus is repeated M times, for a positive integer M that tends to infinity, one copy for each slice of "imaginary time". The copies of a site are coupled ferromagnetically, so the classical Lee–Yang theorem applies to each of these classical models: none of their partition functions vanishes off the imaginary axis. A limit of holomorphic functions without zeros has no zeros either, unless it vanishes identically (see Fact 9.3(b)), and so the quantum partition function inherits the zero-free half-planes. The analyticity of the pressure then follows, in Proposition 9.5, from a uniform bound and two classical theorems of complex analysis. The two proofs use only the following standard facts.

Fact 9.3.

(a) Lee–Yang theorem [LY52] *(see* [FV17, Chapter 3]*). Let G be a finite graph with vertex set V and edge set E, let $J_e \ge 0$ for $e \in E$, and let h_v be complex numbers with $\operatorname{Re} h_v > 0$ for $v \in V$. Then $\sum_{s \in \{-1,1\}^V} \exp\bigl(\sum_{e=\{u,w\} \in E} J_e s_u s_w + \sum_{v \in V} h_v s_v\bigr) \ne 0$.*

(b) Hurwitz's theorem [Tit39]. *If holomorphic functions without zeros on a connected open set $U \subseteq \mathbb{C}$ converge uniformly on compact subsets of U to f, then either f has no zeros in U or f vanishes identically.*

(c) Borel–Carathéodory inequality [Tit39]. *If f is holomorphic on a neighborhood of the closed disc $|z - z_0| \le R$ and $\operatorname{Re} f \le A$ there, then $|f(z) - f(z_0)| \le \frac{2r}{R-r}\bigl(A - \operatorname{Re} f(z_0)\bigr)$ for $|z - z_0| \le r < R$.*

(d) Vitali's theorem [Tit39]. *A sequence of holomorphic functions on a connected open set U that is uniformly bounded on each compact subset of U and converges at every point of a set with an accumulation point in U converges uniformly on compact subsets of U; by Weierstrass's theorem, so do the derivatives.*

Proposition 9.4 (Lee–Yang theorem for the quantum Ising model). *For every N, the function Z_N is entire, $Z_N(h) \ne 0$ whenever $\operatorname{Re} h \ne 0$, and $|Z_N(h)| \le Z_N(\operatorname{Re} h)$.*

Proof. Write $X = \beta J \sum_{i \sim j} \sigma_i^z \sigma_j^z + hS_N$, which is diagonal in the basis $\{|s\rangle : s \in \{-1,1\}^{\Lambda_N}\}$ of eigenvectors of the operators σ_i^z, and $W = \beta\Gamma \sum_i \sigma_i^x$. The function $Z_N(h) = \operatorname{Tr} e^{X+W}$ is entire, because the exponential series converges locally uniformly in h.

Step 1: the Lie–Trotter formula. For matrices A and B, we have $e^{A/M} e^{B/M} = I + (A+B)/M + O(M^{-2})$, uniformly for A and B in a bounded set; by taking the principal

logarithm, we obtain $e^{A/M}e^{B/M} = \exp\big((A+B)/M + R_M\big)$ with $\|R_M\| = O(M^{-2})$, and hence $(e^{A/M}e^{B/M})^M = \exp(A+B+MR_M) \to e^{A+B}$, uniformly for A and B in a bounded set. Applied to X and W, this yields $Z_{N,M}(h) = \mathrm{Tr}(e^{X/M}e^{W/M})^M \to Z_N(h)$ as $M \to \infty$, uniformly for h in compact subsets of $\mathbb{C}$.

Step 2: a classical ferromagnet. Suppose first that $\Gamma > 0$, and put $\varepsilon = \beta\Gamma/M$. Since $e^{\varepsilon\sigma^x} = \cosh\varepsilon\, I + \sinh\varepsilon\,\sigma^x$, the matrix element $\langle s|e^{\varepsilon\sigma^x}|s'\rangle$ of a single spin is $\cosh\varepsilon$ if $s = s'$ and $\sinh\varepsilon$ if $s \neq s'$, that is, $\langle s|e^{\varepsilon\sigma^x}|s'\rangle = c_\varepsilon e^{J_\varepsilon ss'}$ with $c_\varepsilon = (\sinh\varepsilon\cosh\varepsilon)^{1/2} > 0$ and $J_\varepsilon = \frac{1}{2}\log\coth\varepsilon > 0$. Insertion of the basis between the factors of $(e^{X/M}e^{W/M})^M$ shows that, for $M \geq 3$,

$$Z_{N,M}(h) = c_\varepsilon^{M|\Lambda_N|} \sum_{s^1,\dots,s^M} \exp \sum_{k=1}^{M} \Big(\frac{\beta J}{M} \sum_{i\sim j} s_i^k s_j^k + J_\varepsilon \sum_i s_i^k s_i^{k+1} + \frac{h}{M} \sum_i s_i^k \Big),$$

where $s^1, \dots, s^M$ range over $\{-1,1\}^{\Lambda_N}$ and $s^{M+1} = s^1$. This is $c_\varepsilon^{M|\Lambda_N|}$ times the partition function of a classical Ising model on the finite graph $\Lambda_N \times \{1,\dots,M\}$, with nonnegative couplings and the same field h/M at every site. By Fact 9.3(a), it does not vanish when $\mathrm{Re}\, h > 0$. Moreover, by the triangle inequality applied to the sum, $|Z_{N,M}(h)| \leq Z_{N,M}(\mathrm{Re}\, h)$. If $\Gamma = 0$, then the function Z_N itself is a classical Ising partition function, and the same two statements hold directly.

Step 3: Hurwitz's theorem. The functions $Z_{N,M}$ have no zeros in the connected open set $\{\mathrm{Re}\, h > 0\}$ and converge to Z_N uniformly on its compact subsets, and Z_N is not identically zero there, since $Z_N(1) > 0$. By Fact 9.3(b), the function Z_N has no zeros in $\{\mathrm{Re}\, h > 0\}$, and by the evenness in (Q1) none in $\{\mathrm{Re}\, h < 0\}$. The inequality $|Z_N(h)| \leq Z_N(\mathrm{Re}\, h)$ follows from Step 2 in the limit. □

Proposition 9.5. *The pressure p is real analytic on $\mathbb{R}\setminus\{0\}$, and $m_N(h) \to p'(h)$ for every $h \neq 0$.*

Proof. By evenness it suffices to treat $h > 0$. The half-plane $\{\mathrm{Re}\, h > 0\}$ is simply connected and Z_N does not vanish there, so the function Z_N has a holomorphic logarithm on it, and we let f_N be $|\Lambda_N|^{-1}$ times the logarithm that is real on $(0,\infty)$; thus $f_N = p_N$ on $(0,\infty)$ and $\mathrm{Re}\, f_N(h) = |\Lambda_N|^{-1}\log|Z_N(h)| \leq p_N(\mathrm{Re}\, h)$, by Proposition 9.4.

We bound p_N on the real line uniformly in N. The torus has $d|\Lambda_N|$ bonds, whence $\|\beta H_N\| \leq \beta(dJ+\Gamma)|\Lambda_N|$. Since the trace of the exponential of a self-adjoint matrix of size $2^{|\Lambda_N|}$ and norm at most c lies between $2^{|\Lambda_N|}e^{-c}$ and $2^{|\Lambda_N|}e^{c}$, it follows that $|p_N(0) - \log 2| \leq \beta(dJ+\Gamma) =: b$; as $|p_N'| \leq 1$, also $|p_N(x) - \log 2| \leq b + |x|$ for real x.

Fix $x_0 > 0$, and put $R = x_0/2$ and $r = x_0/4$. On the disc $|h - x_0| \leq R$ we have $\mathrm{Re}\, f_N \leq \log 2 + b + \frac{3}{2}x_0$, and $f_N(x_0) = p_N(x_0) \geq \log 2 - b - x_0$. By the Borel–Carathéodory inequality (see Fact 9.3(c)),

$$|f_N(h) - f_N(x_0)| \leq \frac{2r}{R-r}\Big(\sup_{|z-x_0|\leq R} \mathrm{Re}\, f_N(z) - \mathrm{Re}\, f_N(x_0)\Big) \leq 2\big(2b + \tfrac{5}{2}x_0\big)$$

for $|h - x_0| \leq r$. Since also $|f_N(x_0)| \leq \log 2 + b + x_0$, the functions f_N are uniformly bounded on each disc $\{|h - x_0| < x_0/4\}$, and therefore locally uniformly bounded on the connected open set U that is the union of these discs, which contains $(0,\infty)$. They converge on $(0,\infty)$, by (Q2). By Vitali's theorem (see Fact 9.3(d)), they converge uniformly on compact subsets of U to a holomorphic function F, which equals p on $(0,\infty)$; hence the pressure p is real analytic there, and $m_N = f_N' \to F' = p'$ on $(0,\infty)$. □

Put $m^*(\beta,\Gamma) = p'(0^+)$, the spontaneous magnetization; by evenness, $p'(0^-) = -m^*(\beta,\Gamma)$. With offsets as in Section 9, let $r_{N,a}(h) = m_N(h-a)$ for $N \geq 1$, $a \in [-1,1]$ and $h \in Y = [-2,2]$, and let $\mathcal{R}_\Gamma$ be the family of these functions.

Theorem 9.6. *Let $d \geq 1$, $J > 0$, $\Gamma \geq 0$ and $\beta > 0$ be given.*

(a) *If $m^*(\beta,\Gamma) = 0$, then every idealized response is continuous: the family $\mathcal{R}_\Gamma$ is stable, and its closure is metrizable.*

(b) *If $m^*(\beta,\Gamma) > 0$, then the family $\mathcal{R}_\Gamma$ is unstable, and its closure is first countable but not hereditarily separable: the family is NIP$_1$ but not NIP$_2$. For each $a \in (-1,1)$, the functions g_{a,m^*} and $g_{a,-m^*}$, defined as before Lemma 9.1 with $m^* = m^*(\beta,\Gamma)$ and with the pressure p of this model, lie in the closure and differ only at a, and their midpoints $g_{a,0}$ form an uncountable discrete set.*

The family $\{m_N\}$ without offsets is stable exactly when $m^*(\beta,\Gamma) = 0$, and it is NIP$_3$ in all cases.

Proof. The discussion preceding Lemma 9.1, the lemma and the proof of Theorem 9.2 use only the following properties: m_N is continuous, nondecreasing and odd, with $|m_N| \leq 1$ and $m_N(0) = 0$; the limit p exists, is convex and even, and is differentiable at every $h \neq 0$, with $m_N(h) \to p'(h)$; and $p'(0^\pm) = \pm m^*$. These hold by (Q1), (Q2) and Proposition 9.5, and the proofs apply verbatim with $m^*(\beta,\Gamma)$ in place of $m^*(\beta)$. □

In physical terms, the functions $g_{a,\pm m^*}$ are the magnetizations of the two pure phases at the transition point $h = a$, and $g_{a,0}$ is that of the symmetric state; the classification thus reads off the quantum phase diagram. Both cases of the theorem occur: Björnberg and Grimmett [BG09] write the Hamiltonian as $-\frac{1}{2}\lambda\sum\sigma^z\sigma^z - \delta\sum\sigma^x - \gamma\sum\sigma^z$, in which the parameters are $\lambda = 2J$, $\delta = \Gamma$ and $h = \beta\gamma$, and they define the spontaneous magnetization as the limit, as $\gamma \downarrow 0$, of the infinite-volume magnetization at field γ on the same periodic boxes Λ_N, through the space-time representation. By translation invariance and Proposition 9.5, that magnetization is $p'(\beta\gamma)$, and since the right derivative of a convex function is the right limit of its derivative, their spontaneous magnetization is $p'(0^+) = m^*(\beta,\Gamma)$. For $d = 1$ and $\beta < \infty$ it vanishes, and for $d \geq 2$ and $\Gamma > 0$ there is a critical value of $2J/\Gamma$, depending on $\beta\Gamma$, below which $m^* = 0$ and above which $m^* > 0$ (see [BG09, Section 1 and Theorems 1.1–1.2], where the existence of the critical value is attributed to standard methods). Order at low temperature and small Γ/J goes back to Ginibre [Gin69].

The Lee–Yang theorem [LY52] was extended to general classical ferromagnets by Newman [New74] and by Lieb and Sokal [LS81], and Lee–Yang theorems for quantum spin systems go back to Asano [Asa70] and to Suzuki and Fisher [SF71]. Applied to H_N, these results, in the form given by Fröhlich and Rodriguez [FR12, Theorem 5], exclude zeros only for $\operatorname{Re} h > \beta\Gamma$; the point of Proposition 9.4 is the whole half-plane. We have not found Proposition 9.4 stated in the literature; its proof, an approximation of the quantum model by classical ferromagnets in the spirit of Simon and Griffiths [SG73] and of Suzuki [Suz76], combines standard tools, and the result may well be known to experts.

10. Chiral symmetry breaking and metallic conduction are instabilities

Finally, we show that two orders of limits of quantum theory, the Banks–Casher relation for the chiral condensate and the Scalapino–White–Zhang criterion for superconductivity, are statements about the stability of a family of responses, that is, about the agreement of its two iterated limits whenever both exist (see Section 9). In each of them the response is an

integral against a family of spectral measures, and stability fails exactly when spectral weight accumulates at zero energy (see Theorems 10.3 and 10.4). In both, the transition occurs at a single value of the probe, and the family is then NIP_3, that is, its closure is metrizable, whether or not it is stable, by Lemma 10.1 below.

Lemma 10.1. *Let Q be a compact metrizable space, q_0 a point of Q that is not isolated, and $q \mapsto g_q$ a map from Q into $\mathbb{R}^{[0,1]}$ that is continuous for the topology of pointwise convergence. Suppose that g_q is continuous for $q \neq q_0$, and that $G = g_{q_0}$ is continuous on $(0,1]$. Then the family $\mathcal{R} = \{g_q : q \neq q_0\}$ is NIP$_3$, and it is stable if and only if G is continuous at 0. If $G(0^+) = \lim_{s\to 0^+} G(s)$ exists, then for $q_k \to q_0$ with $q_k \neq q_0$ and $s_l \to 0$ with $s_l > 0$,*

$$\lim_l \lim_k g_{q_k}(s_l) = G(0^+), \qquad \lim_k \lim_l g_{q_k}(s_l) = G(0).$$

Proof. The image of Q is compact, contains $\mathcal{R}$ and is contained in $\overline{\mathcal{R}}$, because q_0 is not isolated; it is thus $\overline{\mathcal{R}}$. Its members are continuous on $(0,1]$, so two distinct members differ at a rational point of $[0,1]$, and $\overline{\mathcal{R}}$ is metrizable by Proposition 4.5. The function G is of the first Baire class, as the pointwise limit of the continuous functions that equal G on $[1/n, 1]$ and are linear on $[0, 1/n]$ with value $G(0)$ at 0; hence, by Theorem 3.6, the family is NIP, and, its closure being metrizable, it is NIP_3. By Grothendieck's theorem, it is stable exactly when every member of $\overline{\mathcal{R}}$ is continuous, that is, exactly when G is continuous at 0. The iterated limits follow from the continuity of $q \mapsto g_q$ and of each g_{q_k}. □

10.1. **The chiral condensate.** We work with a lattice regularization, in which Euclidean spacetime (spacetime with imaginary time) is replaced by a lattice of points in a box of volume V. The quarks are described by an anti-self-adjoint Dirac operator D_V, as for staggered fermions, a standard discretization of quarks on the lattice; the Dirac operator governs the motion of the quarks in a given gauge field, the gluon field with which they interact. The eigenvalues of D_V come in pairs $\pm i\lambda_n$, indexed by n in a finite set, with $\lambda_n > 0$, and we restrict to the sector of trivial topology, in which there are no exact zero modes, that is, no eigenvectors with eigenvalue 0; in a sector of topological charge $\nu \neq 0$, an integer invariant of the gauge field, the zero modes add $|\nu|/(mV)$ to the condensate at quark mass m, defined below, and make it unbounded as $m \to 0$. The spectral density is averaged over gauge fields with a measure that does not depend on the mass m: either the quenched theory, in which the quarks do not act back on the gauge field, or the valence condensate at a fixed sea quark mass, which is the setting of the Banks–Casher relation; here the valence quarks are those of the condensate, whose mass m varies, and the sea quarks are those that act back on the gauge field. The chiral condensate at quark mass $m > 0$ is

$$\Sigma_V(m) = \frac{1}{V}\operatorname{Tr}(D_V + m)^{-1} = \int \frac{2m}{m^2 + \lambda^2}\, d\kappa_V(\lambda), \qquad \kappa_V = \frac{1}{V}\sum_n \delta_{\lambda_n},$$

where δ_{λ_n} is the point mass at λ_n, and we put $\Sigma_V(0) = 0$, the value given by continuity. We assume:

(C1) each κ_V is a finitely supported positive measure on $(0, \Lambda]$, of total mass at most M_0, where the cutoff Λ and the bound M_0 do not depend on V;
(C2) as $V \to \infty$ along a given sequence of volumes, the measure κ_V converges weakly on $[0, \Lambda]$ to a measure κ that, on some interval $[0, \delta]$, has a density f continuous at 0.

Hypothesis (C1) holds on the lattice. Hypothesis (C2) is the physical assumption at issue, the existence of a limiting density of near-zero eigenvalues per unit volume; its value at 0 is

$f(0)$. A preparation is a volume V, a probe is a mass $m \in [0,1]$, and the response is Σ_V. Each Σ_V is continuous on $[0,1]$, because κ_V has finitely many atoms, all at positive λ.

Lemma 10.2. *Assume* (C1) *and* (C2). *Then* $\Sigma_V(m) \to \Sigma(m) = \int \frac{2m}{m^2+\lambda^2}\, d\kappa(\lambda)$ *for every* $m > 0$, *the function* Σ *is continuous on* $(0,1]$, *and* $\Sigma(m) \to \pi f(0)$ *as* $m \to 0^+$.

Proof. For fixed $m > 0$, the function $\lambda \mapsto 2m/(m^2+\lambda^2)$ is continuous and bounded on $[0,\Lambda]$, so the first statement is weak convergence, and the second follows from dominated convergence. For the third, choose δ so small that f is bounded on $[0,\delta]$; since κ has the density f there, it has no atom at 0. On $(\delta,\Lambda]$ the integrand is at most $2m/\delta^2$, and the contribution of this interval therefore tends to 0. On $[0,\delta]$, the substitution $\lambda = mu$ shows that

$$\int_0^\delta \frac{2m}{m^2+\lambda^2} f(\lambda)\, d\lambda = \int_0^{\delta/m} \frac{2}{1+u^2} f(mu)\, du,$$

and the right side tends to $f(0)\int_0^\infty \frac{2\,du}{1+u^2} = \pi f(0)$, by dominated convergence. □

Theorem 10.3. *Assume* (C1) *and* (C2).

(a) *The iterated limits of the condensate are*

$$\lim_{m\to 0}\lim_{V\to\infty} \Sigma_V(m) = \pi f(0), \qquad \lim_{V\to\infty}\lim_{m\to 0} \Sigma_V(m) = 0.$$

(b) *The family* $\{\Sigma_V\}$ *is stable if and only if* $f(0) = 0$, *that is, if and only if chiral symmetry is not spontaneously broken* [BC80].

(c) *In all cases the family is NIP*$_3$.

Proof. Let Q consist of the numbers $1/V$ along the sequence of volumes, together with $q_0 = 0$, put $g_{1/V} = \Sigma_V$, and let $G(m) = \Sigma(m)$ for $m > 0$ and $G(0) = 0$. By Lemma 10.2 and $\Sigma_V(0) = 0$, the map $q \mapsto g_q$ is continuous, the function G is continuous on $(0,1]$, and $G(0^+) = \pi f(0)$. The theorem now follows from Lemma 10.1. □

Letting the volume grow before the mass vanishes keeps a density of near-zero modes, and the condensate is $\pi f(0)$; the other order removes every mode from the neighborhood of 0 before the mass vanishes, and the condensate is 0. When the mass is tied to the volume, as in the Leutwyler–Smilga microscopic regime, where $\mu = mV\Sigma_0$, with $\Sigma_0 = \pi f(0)$, is held fixed [LS92, VW00], the condensate takes values strictly between 0 and $\pi f(0)$. As for the Ising model, these values become idealized responses only when there is a continuum of transition points (see Section 11).

10.2. **The current response.** The Scalapino–White–Zhang criterion concerns the current response of a clean many-body system, one free of disorder, at zero temperature. Let H be its Hamiltonian, with ground state Ω of energy E_0, and let j_q be the transverse current operator at wave vector q, the component perpendicular to q of the Fourier transform of the current density, normalized per unit volume. For a finite system these are operators and a vector; for an infinite system, the objects below are the infinite-volume limits of the corresponding finite-volume spectral measures, and Theorem 10.4 below is a statement about these measures. The paramagnetic current response at imaginary frequency $i\omega$ has the spectral (Lehmann) representation

$$\Pi(q, i\omega) = \int \frac{2\nu}{\nu^2+\omega^2}\, d\mu_q(\nu),$$

where μ_q is the spectral measure of $H - E_0$ in the vector $j_q\Omega$, restricted to $(0,\infty)$; for a finite system, $\mu_q = \sum_k |\langle \psi_k, j_q\Omega\rangle|^2 \delta_{E_k - E_0}$, the sum running over the k with $E_k > E_0$, where the vectors ψ_k, for k in a countable index set, form an orthonormal basis of eigenvectors of H, with $H\psi_k = E_k\psi_k$. The total response is $\mathcal{K}(q, i\omega) = D_0 - \Pi(q, i\omega)$, where D_0 is the diamagnetic term, the expectation of minus the kinetic energy along the field on a lattice, and, in the continuum, the ratio n/m of the density of the charge carriers to their mass. Following Scalapino, White and Zhang [SWZ93], we define the *superfluid weight* and the *Drude weight*, up to their factor πe^2, where e is the charge of the carriers, by

$$D_s = \lim_{q\to 0} \mathcal{K}(q, 0), \qquad D = \lim_{\omega\to 0} \lim_{q\to 0} \mathcal{K}(q, i\omega).$$

The superfluid weight is the transverse static limit of Scalapino, White and Zhang. Their Drude weight, and that of Kohn [Koh64], is taken at $q = 0$ and continued from imaginary to real frequency; it agrees with D when the measures below depend continuously on q at $q = 0$ away from zero energy and when, in addition, the imaginary-frequency limit agrees with the real-frequency one. In an insulator $D_s = D = 0$; in a clean normal metal $D > 0 = D_s$; in a clean superconductor $D_s = D > 0$ [SWZ93]. In a disordered metal the Drude weight vanishes, and at positive temperature the normal fluid, the thermally excited carriers that do not condense, yields $D > D_s$ even in a superconductor; both cases lie outside the hypotheses below.

It is convenient to absorb a factor into the measure: put $d\zeta_q(\nu) = \frac{2}{\nu}\, d\mu_q(\nu)$, whence

$$\Pi(q, i\omega) = \int \frac{\nu^2}{\nu^2 + \omega^2}\, d\zeta_q(\nu), \qquad \Pi(q, 0) = \zeta_q\big((0,\infty)\big).$$

We assume:

(S1) each ζ_q, for $0 < q \le 1$, is a positive measure on $(0, \Lambda]$ of total mass at most D_0, where the bandwidth Λ does not depend on q;

(S2) there is a measure ζ_0 on $[0, \Lambda]$ such that, for every $h \in C[0, \Lambda]$, the map $q \mapsto \int h\, d\zeta_q$, with each ζ_q regarded as a measure on $[0, \Lambda]$, is continuous on $(0, 1]$ and converges to $\int h\, d\zeta_0$ as $q \to 0$.

The bound in (S1) is the statement that $\mathcal{K}(q, 0) \ge 0$, that is, that the ground state is a local minimum of the energy as a function of a static vector potential. This is an assumption: for lattice fermions it can fail, as Lieb's flux-phase theorem shows [Lie94]. The compact support in (S1) is a convenience: since $\nu^2/(\nu^2 + \omega^2)$ extends continuously to $[0,\infty]$, the proof below works for tight families of measures on $(0,\infty)$. The limit ζ_0 may have an atom at 0, which carries the weight of current-carrying excitations whose energy tends to 0 with q. A preparation is a wave vector $q \in (0, 1]$, a probe is a frequency $\omega \in [0, 1]$, and the response is $g_q(\omega) = \Pi(q, i\omega)$.

Theorem 10.4. *Assume* (S1) *and* (S2).

(a) *We have* $D_s = D_0 - \zeta_0\big([0,\Lambda]\big)$ *and* $D = D_0 - \zeta_0\big((0,\Lambda]\big)$, *whence* $D - D_s = \zeta_0(\{0\})$.

(b) *The family of responses* $\{g_q : 0 < q \le 1\}$ *is stable if and only if* $D = D_s$. *In particular, the responses of a clean superconductor and of an insulator are stable, and those of a clean normal metal are unstable.*

(c) *In all cases the family is NIP*$_3$.

Proof. Write $\kappa_\omega(\nu) = \nu^2/(\nu^2 + \omega^2)$. For $\omega > 0$ the function κ_ω is continuous on $[0, \Lambda]$ with $\kappa_\omega(0) = 0$, while $\kappa_0 = 1$ on $(0, \Lambda]$. Each g_q is continuous on $[0, 1]$, by dominated convergence.

By (S2), applied to the continuous function 1, we have $g_q(0) = \zeta_q((0,\Lambda]) \to \zeta_0([0,\Lambda])$, and the formula for D_s follows; applied to κ_ω with $\omega > 0$, (S2) shows that $g_q(\omega) \to G(\omega) = \int_{(0,\Lambda]} \kappa_\omega \, d\zeta_0$, and, by dominated convergence, $G(\omega) \to \zeta_0((0,\Lambda])$ as $\omega \to 0$, which yields the formula for D. This proves (a). Put $g_0 = G$, with G extended to 0 by $G(0) = \zeta_0([0,\Lambda])$. The map $q \mapsto g_q$ is continuous on $[0,1]$, by (S2) and the computation just made, and G is continuous on $(0,1]$ with $G(0) - G(0^+) = \zeta_0(\{0\}) = D - D_s$. Parts (b) and (c) now follow from Lemma 10.1. □

The difference $D - D_s$ is the weight of current-carrying excitations whose energy tends to zero with the wave vector, the gapless quasiparticles of a metal, and the presence of such weight is exactly the failure of stability. In a superconductor the energy gap, a positive lower bound for the energies of the excitations, keeps this weight away from zero energy, and in an insulator there is no current-carrying weight at low energy.

11. Questions

(1) *Light cones.* In Theorem 7.10 Bob detects at a fixed time. Locate the causal order in the classification when Bob's detection time is also uncertain, and in spacetimes of dimension $1 + d$, with $d \geq 2$ dimensions of space.

(2) *The minimal families of the box.* By Theorem 8.5, the comparison of positions in a box realizes the minimal families D_3, D_4 and D_7 of the heptachotomy. Does it realize D_6? Which of D_6 and D_7 does Theorem 7.8(b) produce from a given dense sequence?

(3) *Noncommutative probes.* Find a natural observable that is NIP but unstable and whose probe algebra is not commutative, and locate it in the classification. Can noncommutativity produce a phenomenon in the classification that observables with commutative probe algebras cannot produce, for instance in the minimal families of the heptachotomy that arise from a given dense sequence of preparations?

(4) *Orientations.* By Theorem 7.12, the comparison of phases on a circle is NIP but not NIP_1. Locate in the classification the comparison of directions on a sphere, "Bob's direction lies in the open hemisphere centered at Alice's", and relative orientations in higher dimensions.

(5) *Finitary levels.* Is there a finitary, combinatorial characterization of NIP_1, NIP_2 and NIP_3, in the spirit of the fat-shattering dimension for NIP, sharpening Proposition 4.6?

(6) *Quantum phase transitions.* Extend Theorem 9.6 to the ground state of the Ising model in a transverse field, where the transition is driven by the transverse field Γ at zero temperature.

(7) *A continuum of transitions.* Is there a natural family of chiral condensates with a continuum of transition points, obtained for instance from twisted boundary conditions or a chemical potential, that is NIP_1 but not NIP_2?

(8) *Positive temperature.* Extend Theorem 10.4 to positive temperature, where the Matsubara frequencies, the imaginary frequencies at which thermal response functions are evaluated, form a discrete set, and the Drude weight is obtained by extrapolation.

(9) *Unbounded formulas.* In the setting of Keisler, classify the energy functional $\rho \mapsto \mathrm{Tr}(\rho H)$ of a Hamiltonian H, with the energy constraints $\mathrm{Tr}(\rho H) \leq E$, for real numbers E, as shards (the parts of the state space on which every predicate stays within prescribed bounds).

(10) *Forcing.* Can Todorčević's trichotomy, or its consequences for observables, be proved without forcing?

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Declarations

Competing interests. The authors have no competing interests to declare.
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